

偏微分方程

袁浩然 数学强基 2301

1 方程的导出和定解条件

1.1 基本概念

在本节，我们首先将介绍关于偏微分方程的一些基本概念。对偏微分方程，我们将其分为先行与非线性两种：

定义 1.1.1. 对某个偏微分方程，若其关于所有的未知函数及其偏导数都是线性的，则称其为一个线性偏微分方程。

例如对向量 $\mathbf{a} = (x, y, z)$ 及关于 $\mathbf{a}$ 的函数 $u(\mathbf{a})$ ，Laplace 方程

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

就是线性偏微分方程。

而对非线性偏微分方程，我们又将其分为三种：

定义 1.1.2. 对某偏微分方程，若其中函数的最高阶导数项系数 δ 与未知函数及其导数无关，则称其为半线性偏微分方程 (*semi-linear partial differential equation*) .

例如下面的偏微分方程最高阶导数项 u_{xx} , u_{yy} 系数均为常数项，却又不是线性偏微分方程：

$$u_{xx} + u_{yy} + u^3 u_x = u^2$$

定义 1.1.3. 对某偏微分方程，若其最高阶导数项系数 δ 与未知函数的最高阶导数无关，则称其为拟线性偏微分方程 (*quasi-linear partial differential equation*)。

例如下面的方程仅与一阶导有关：

$$|\nabla \mathbf{u}|^2(u_{xx} + u_{yy}) = f(x, y)$$

他就是拟线性偏微分方程。而其他的情形就是完全非线性偏微分方程 (*fully-nonlinear partial differential equation*)。

求解偏微分方程本质上是求解由其本身与初值条件和边界条件所构成的定解问题，下面我们介绍初值条件与边界条件的概念：

定义 1.1.4. 初值条件是指在给定初始时刻下，已知系统状态的条件。

另外，我们有三种边界条件：

定义 1.1.5. 对某个偏微分方程中的未知函数 $u(x, t)$ ，如果我们规定了其在边界上的值，即

$$u(x, t) = \phi(t), \quad x \in \partial\Omega$$

则称这种边界条件为 *Dirichlet* 边界条件。

定义 1.1.6. 对某个偏微分方程的未知函数 $u(x, t)$ ，如果我们规定了函数在边界上的法向导数：

$$\frac{\partial u}{\partial n}(x, t) = \psi(t), \quad x \in \partial\Omega$$

则称这种边界条件为 *Neumann* 边界条件。

定义 1.1.7. 对某个偏微分方程的未知函数 $u(x, t)$ 如果我们知道了函数值与法向导数的线性组合：

$$\alpha u + \beta \frac{\partial u}{\partial n} = \gamma, \quad x \in \partial\Omega$$

则称这种边界条件为 *Robin* 边界条件。

1.2 变分原理

本节我们需要解决的问题就是求某一特定泛函在定义域内的极值，在此之前我们来明确一些基本定义。

定义 1.2.1. 设 Ω 为 $\mathbb{R}^2$ 中的区域，定义在 Ω 上的无穷次可微且在 Ω 的边界附近为零的函数的全体，记为 $C_0^\infty(\Omega)$ 。

进而我们有

引理 1.2.1. 设 Ω 为 $\mathbb{R}^2$ 中有界区域 $f(x, y)$ 在 Ω 上连续，如果 $\forall \varphi(x, y) \in C_0^\infty(\Omega)$,

$$\iint_{\Omega} f(x, y) \varphi(x, y) dx dy = 0$$

则 $f(x, y)$ 在 Ω 上恒为零。

解答 对 $\rho(x, y) = \begin{cases} k \exp \left\{ \frac{-1}{1-x^2-y^2} \right\}, & x^2 + y^2 < 1 \\ 0, & x^2 + y^2 \geq 1 \end{cases}$, 其中 k 是常数.

我们可以调整 k s.t. $\iint_{\mathbb{R}^2} \rho(x, y) dx dy = 1$, 再定义 $\rho_n(x, y) = n^2 \rho(nx, ny)$, 则 $\rho_n(x, y) \in C_0^\infty(\mathbb{R}^2)$ 且 $\iint_{\mathbb{R}^2} \rho_n(x, y) dx dy = 1$

反设 $\exists (x_0, y_0) \in \Omega$ s.t. $f(x_0, y_0) > 0$, 则结合 $f \in C(\Omega)$, 则 $\exists M > 0$ s.t. $\exists \delta > 0$ 当 $(x, y) \in B((x_0, y_0), \delta)$ 有 $f(x, y) > M$, 取 $\varphi(x, y) = \rho_n(x - x_0, y - y_0)$, 有 (其中 $\frac{1}{n} < \delta$) .

$$\iint_{\Omega} f(x, y) \varphi(x, y) dx dy > M \iint_{B(x_0, \frac{1}{n})} \varphi(x, y) dx dy > 0 \quad \text{矛盾!}$$

□

则我们是这样定义变分问题的：

定义 1.2.2. 对函数集合 M_φ 求 $u \in M_\varphi$ 使得泛函 J 最小，即

$$j(u) = \min_{v \in M_\varphi} J(v)$$

我们称这样的问题为变分问题，函数集合称为变分问题的允许函数类，或称为泛函 $J(v)$ 的定义域， u 称为变分问题的解。

我们求解的思路是这样的，如果 u 是变分问题的解，我们构造函数 $j(\varepsilon) = J(u + \varepsilon v)$ 其中 $v \in M_\varphi$ 则 $j(\varepsilon)$ 在 $\varepsilon = 0$ 时取最小值，即 $j'(0) = 0, j''(0) > 0$.

1.2.1 最小曲面方程

重要! 新增内容 若考虑平面上有界区域 Ω , 在边界 $\partial\Omega$ 上给定一条空间闭曲线 L :
$$\begin{cases} x = x(s) \\ y = y(s) \\ z = \varphi(s) \end{cases} \quad 0 \leq s \leq s_0, \text{ 其中 } (x(0), y(0)) =$$

$(x(s_0), y(s_0)), \varphi(0) = \varphi(s_0)$ 。我们需要找定义在 $\bar{\Omega}$ 上的曲面 S , s.t. 1) S 以 L 为边界 2) S 的表面积最小。

我们将问题转化为给定函数集合

$$M_\varphi = \{v \mid v \in C^1(\bar{\Omega}), v|_{\partial\Omega} = \varphi\}$$

求 $u \in M_\varphi$, 使得 $J(u) = \min_{v \in M_\varphi} J(v)$, 其中 $J(v) = \iint_{\Omega} \sqrt{1 + v_x^2 + v_y^2} dx dy$

补充知识 Green 公式诱导的分部积分公式

$$\iint_{\Omega} (Pv_x + Qv_y) dx dy = \oint_{\partial\Omega} v(Pn_x + Qn_y) ds - \iint_{\Omega} v\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dx dy$$

解答 设 $M_0 = \{v \mid v \in C^1(\bar{\Omega}), v|_{\partial\Omega} = 0\}$ 。则对 $u + \varepsilon v \in M_\varphi$, 其中 $u \in M_\varphi, v \in M_0$, 我们要使 u 是 $\arg \min_u J(u + \varepsilon v)$, 即令 $j(\varepsilon) = J(u + \varepsilon v)$ 中 $j'(\varepsilon) = 0, j''(\varepsilon) \geq 0$

$$j(\varepsilon) = J(u + \varepsilon v) = \iint_{\Omega} \sqrt{1 + (u + \varepsilon v)_x^2 + (u + \varepsilon v)_y^2} dx dy$$

$$j'(\varepsilon) = \iint_{\Omega} \frac{(u_x + \varepsilon v_x)v_x + (u_y + \varepsilon v_y)v_y}{\sqrt{1 + (u_x + \varepsilon v_x)^2 + (u_y + \varepsilon v_y)^2}} dx dy \Rightarrow j'(0) = \iint_{\Omega} \frac{u_x v_x + u_y v_y}{\sqrt{1 + u_x^2 + u_y^2}} dx dy = 0$$

利用 Green 公式诱导的分部积分公式, 我们有

$$\oint_{\partial\Omega} v\left(\frac{u_x n_x + u_y n_y}{\sqrt{1 + u_x^2 + u_y^2}}\right) ds - \iint_{\Omega} v\left(\frac{\partial}{\partial x} \frac{u_x}{\sqrt{1 + u_x^2 + u_y^2}} + \frac{\partial}{\partial y} \frac{u_y}{\sqrt{1 + u_x^2 + u_y^2}}\right) dx dy$$

结合 $v|_{\partial\Omega} = 0$, 及 v 的连续性

$$\frac{\partial}{\partial x} \left(\frac{u_x}{\sqrt{1 + u_x^2 + u_y^2}} \right) + \frac{\partial}{\partial y} \left(\frac{u_y}{\sqrt{1 + u_x^2 + u_y^2}} \right) = 0$$

□

在这样的思路之下, 通过推导我们可以确定对某个偏微分方程求解问题, 变分问题的解与边值问题的等价性。

例 1.2.1. 求解变分问题: 求 $u \in M = \{y(x) \mid y \in C^1[0, 1], y(1) = 0\}$ 使得 $J(u) = \min_{y \in M} J(y)$, 其中

$$J(y) = \frac{1}{2} \int_0^1 (y'(x))^2 dx - 2 \int_0^1 y(x) dx - y(0).$$

解答 设 $u \in M$ 是变分问题的最优解. 任取 $v \in M$, 构造函数 $j(\varepsilon) = J(u + \varepsilon v)$. 则求 $J(u) = \min_{y \in M} J(y)$ 转化为求使 $j'(0) = 0$ 且 $j''(0) \geq 0$ 成立的函数 u .

$$\begin{aligned} j(\varepsilon) &= J(u + \varepsilon v) = \frac{1}{2} \int_0^1 (u'(x) + \varepsilon v'(x))^2 dx - 2 \int_0^1 (u(x) + \varepsilon v(x)) dx - u(0) - \varepsilon v(0) \\ &= \frac{1}{2} \varepsilon^2 \int_0^1 (v'(x))^2 dx + \varepsilon \left(\int_0^1 (u'(x)v'(x) - 2v(x)) dx - v(0) \right) + \int_0^1 \left(\frac{1}{2} u'(x)^2 - 2u(x) \right) dx - u(0) \\ j'(\varepsilon) &= \varepsilon \int_0^1 (v'(x))^2 dx + \left(\int_0^1 (u'(x)v'(x) - 2v(x)) dx - v(0) \right). \end{aligned}$$

则取 $j'(0) = 0$, 即 $\int_0^1 (u'(x)v'(x) - 2v(x)) dx - v(0) = 0$.

$$\begin{aligned} \int_0^1 u'(x)v'(x) dx - 2 \int_0^1 v(x) dx - v(0) &= u'(x)v(x) \Big|_0^1 - \int_0^1 v(x)(u''(x) + 2) dx - v(0) \\ &= u'(1)v(1) - (u'(0) + 1)v(0) - \int_0^1 v(x)(u''(x) + 2) dx = 0 \end{aligned}$$

结合 $u, v \in M$, 我们需要保证 $u(0) = v(0) = 0$, 则有

$$\begin{cases} u''(x) + 2 = 0 \\ u'(0) + 1 = 0 \\ u(1) = 0 \end{cases} \Rightarrow u(x) = -x^2 - x$$

此时 $j''(\varepsilon) = \int_0^1 (v'(x))^2 dx \geq 0$ 恒成立; 则 $u = -x^2 - x$

□

例 1.2.2. 求 $u \in M = C^1[0, 1]$ 使得 $J(u) = \min_{y \in M} J(y)$, 其中 $J(y) = \frac{1}{2} \int_0^1 (y'^2 + y^2) dx + \frac{1}{2} [y^2(0) + y^2(1)] - 2y(0)$

解答 构造函数 $j(\varepsilon) = J(u + \varepsilon v)$ 其中 $u, v \in M$. 当 $\varepsilon = 0$ 时 $j(\varepsilon)$ 取 $\min$ 时 u 即为 $\arg \min_{y \in M} J(y)$.

$$\begin{aligned} j(\varepsilon) &= J(u + \varepsilon v) = \frac{1}{2} \int_0^1 ((u'(x) + \varepsilon v'(x))^2 + (u(x) + \varepsilon v(x))^2) dx + \frac{1}{2} [(u(0) + \varepsilon v(0))^2 + (u(1) + \varepsilon v(1))^2] - 2(u(0) + \varepsilon v(0)). \\ &= \varepsilon^2 \left(\int_0^1 \left(\frac{1}{2} (v'(x))^2 + \frac{1}{2} (v(x))^2 \right) dx + \frac{1}{2} (v^2(0) + v^2(1)) \right) + \varepsilon \left(\int_0^1 (u'(x)v'(x) + u(x)v(x)) dx + u(0)v(0) + u(1)v(1) - 2v(0) \right) \\ &\quad + \frac{1}{2} \int_0^1 ((u'(x))^2 + (u(x))^2) dx + \frac{1}{2} (u^2(0) + u^2(1)) - 2u(0) \\ j'(\varepsilon) &= \varepsilon \left(\int_0^1 (v'(x))^2 + (v(x))^2 dx + v^2(0) + v^2(1) \right) + \int_0^1 u'(x)v'(x) + u(x)v(x) dx + u(0)v(0) + u(1)v(1) - 2v(0). \end{aligned}$$

取 $j'(0) = 0$, 有 $\int_0^1 u'(x)v'(x) + u(x)v(x) dx + u(0)v(0) + u(1)v(1) - 2v(0) = 0$

结合

$$\begin{aligned} &\int_0^1 u'(x)v'(x) dx + \int_0^1 u(x)v(x) dx + u(0)v(0) + u(1)v(1) - 2v(0) \\ &= u'(x)v(x) \Big|_0^1 + \int_0^1 (u(x) - u''(x))v(x) dx + u(0)v(0) + u(1)v(1) - 2v(0). \\ &= (u'(1) + u(1))v(1) + (u(0) - u'(0) - 2)v(0) + \int_0^1 (u(x) - u''(x))v(x) dx = 0 \end{aligned}$$

则

$$\begin{cases} u''(x) - u(x) = 0 \\ u(1) + u'(1) = 0 \\ u(0) - u'(0) - 2 = 0 \end{cases} \quad u(x) = e^{-x}$$

结合 $j''(\varepsilon) = \int_0^1 (v'(x))^2 + (v(x))^2 dx + v^2(0) + v^2(1) \geq 0$, 则 $u(x) = e^{-x}$ 构成 $\arg \min_{y \in M} J(y)$

□

1.2.2 膜平衡问题

重要! 新增内容 我们取 $M_\varphi = \{v \mid v \in C^1(\bar{\Omega}), v|_\Gamma = \varphi\}$, 寻找 $u \in M_\varphi$ s.t $u = \arg \min_{v \in M_\varphi} J(v)$ 其中

$$J(v) = \frac{T}{2} \iint_{\Omega} (v_x^2 + v_y^2) dx dy - \iint_{\Omega} f(x, y) v(x, y) dx dy - \int_{\Gamma} p(s) v(s) ds$$

补充知识 $\int_{\Omega} \nabla u \cdot \nabla v dx = - \int_{\Omega} (\Delta u) v dx + \int_{\partial\Omega} \frac{\partial u}{\partial n} v ds$ (Green 第一恒等式).

解答 对 $u \in M_\varphi, v \in M_0$, 并取 $j(\varepsilon) = J(u + \varepsilon v)$.

$$j(\varepsilon) = J(u + \varepsilon v) = \frac{T}{2} \iint_{\Omega} (u_x + \varepsilon v_x)^2 + (u_y + \varepsilon v_y)^2 dx dy - \iint_{\Omega} f(x, y) (u(x, y) + \varepsilon v(x, y)) dx dy - \int_{\Gamma} p(s) (u(s) + \varepsilon v(s)) ds$$

$$j'(\varepsilon) = T \iint_{\Omega} (u_x + \varepsilon v_x) v_x + (u_y + \varepsilon v_y) v_y dx dy - \iint_{\Omega} f(x, y) v(x, y) dx dy - \int_{\Gamma} p(s) v(s) ds$$

$$j'(0) = T \iint_{\Omega} u_x v_x + u_y v_y dx dy - \iint_{\Omega} f(x, y) v(x, y) dx dy - \int_{\Gamma} p(s) v(s) ds = \iint_{\Omega} (T \nabla u \cdot \nabla v - f v) dx dy - \int_{\Gamma} p v ds = 0 \quad (*)$$

则对 $w, u \in M_\varphi, w - u \in M_0$

$$\begin{aligned} J(w) - J(u) &= \iint_{\Omega} \frac{T}{2} |\nabla w|^2 - f w dx dy - \int_{\Gamma} p w ds - \iint_{\Omega} \frac{T}{2} |\nabla u|^2 - f u dx dy + \int_{\Gamma} p u ds \\ &= \iint_{\Omega} \frac{T}{2} (|\nabla w|^2 - 2 \nabla w \cdot \nabla u + |\nabla u|^2 - 2 |\nabla u|^2 + 2 \nabla w \cdot \nabla u) - f(w - u) dx dy - \int_{\Gamma} p(w - u) ds \\ &= \iint_{\Omega} \frac{T}{2} |\nabla(w - u)|^2 dx dy + T \iint_{\Omega} \nabla u \cdot \nabla(w - u) - f(w - u) dx dy - \int_{\Gamma} p(w - u) ds \end{aligned}$$

在划线处, 由于 $w - u \in M_0, u \in M_\varphi$, 利用 (*) 式, 有

$$J(w) - J(u) = \iint_{\Omega} \frac{T}{2} |\nabla(w - u)|^2 dx dy \geq 0$$

再结合 $w - u|_\Gamma = 0$, 我们有 $w = u$ 成立, 则原问题是变分。利用 Green 公式, 我们有

$$j'(0) = - \iint_{\Omega} (T \Delta u + f) v dx dy + \int_{\Gamma} (T \frac{\partial u}{\partial n} - p) v ds = 0$$

对 $v \in M_0$ 的任意性, 取 $v \in C_0^\infty(\Omega)$, 则 $-(T\Delta u + f)$ 在 Ω 内连续, 因此就有

$$-T\Delta u = f \quad \forall x \in \Omega$$

则代入后有 $\int_{\Omega} (T \frac{\partial u}{\partial n} - p) v ds = 0 \quad \forall v \in M_0$, 则立即就有

$$T \frac{\partial u}{\partial n} = p \quad \forall x \in \Gamma$$

□

例 1.2.3. (重要!) 设 $J(v) = \frac{1}{2} \int_{\Omega} (|\nabla v|^2 + v^2) dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x) v^2 ds - \int_{\Omega} f v dx - \int_{\partial\Omega} g v ds$, 其中 $\alpha(x) \geq 0$, 考虑以下三个问题

问题 I (变分问题): 求 $u \in M = C^1(\bar{\Omega})$, 使得

$$J(u) = \min_{v \in M} J(v)$$

问题 II: 求 $u \in M = C^1(\bar{\Omega})$, 使得对于任意 $v \in M$, 都满足

$$\int_{\Omega} (\nabla u \cdot \nabla v + u \cdot v - f v) dx + \int_{\partial\Omega} (\alpha(x) u v - g v) ds = 0$$

问题 III (第三边值问题): 求 $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$, 满足下列边值问题

$$\begin{cases} -\Delta u + u = f, & x \in \Omega \\ \frac{\partial u}{\partial n} + \alpha(x) u = g, & x \in \partial\Omega \end{cases}$$

(1) 证明问题 I 与问题 II 等价

(2) 当 $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ 时, 证明问题 I, II, III 等价

解答 本题将第一问充分性的证明做了修改, 以保证更直观的看出关系

(1) 取 $u = \arg \min_{v \in M} J(v)$ 且 $u \in M$, $\forall v \in M$, 有 $j(\varepsilon) = J(u + \varepsilon v)$, 则我们应保证 $j'(0) = 0, j''(0) \geq 0$

$$j(\varepsilon) = \frac{1}{2} \int_{\Omega} |\nabla(u + \varepsilon v)|^2 + (u + \varepsilon v)^2 dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x) (u + \varepsilon v)^2 ds - \int_{\Omega} f(u + \varepsilon v) dx - \int_{\partial\Omega} g(u + \varepsilon v) ds$$

$$j'(\varepsilon) = \varepsilon \left(\int_{\Omega} |\nabla v|^2 + v^2 dx + \int_{\partial\Omega} v^2 \alpha(x) ds \right) - \int_{\Omega} f v dx - \int_{\partial\Omega} g v ds + \int_{\Omega} (\nabla u \cdot \nabla v + uv) dx$$

则 $j'(0) = 0$ 即 $\int_{\Omega} (\nabla u \cdot \nabla v + uv - f v) dx + \int_{\partial\Omega} (\alpha(x) uv - g v) ds = 0$

结合 3.1) 的 $\Leftarrow$ 部分, 结合 $\Rightarrow$ 我们已经证明 $\forall u \in M_{\varphi}, v \in M_0$

$$j'(0) = \int_{\Omega} (\nabla u \cdot \nabla v + uv - f v) dx + \int_{\partial\Omega} (\alpha(x) uv - g v) ds = 0$$

对 $\forall u, w \in M_{\varphi}$

$$\begin{aligned} J(w) - J(u) &= \frac{1}{2} \int_{\Omega} (|\nabla w|^2 - |\nabla u|^2 + w^2 - u^2) dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x) (w^2 - u^2) ds - \int_{\Omega} f(w - u) dx - \int_{\partial\Omega} g(w - u) ds \\ &= \frac{1}{2} \int_{\Omega} (|\nabla(w - u)|^2 - 2|\nabla u|^2 + 2\nabla w \cdot \nabla u + (w - u)^2 - 2u^2 + 2wu) dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x) ((w - u)^2 - 2u^2 + 2wu) ds \\ &\quad - \int_{\Omega} f(w - u) dx - \int_{\partial\Omega} g(w - u) ds \\ &= \frac{1}{2} \int_{\Omega} |\nabla(w - u)|^2 + (w - u)^2 dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x) (w - u)^2 ds + \\ &\quad \int_{\Omega} (\nabla u (\nabla w - \nabla u) + u(w - u)) dx + \int_{\partial\Omega} \alpha(x) u(w - u) ds - \int_{\Omega} f(w - u) dx - \int_{\partial\Omega} g(w - u) ds \end{aligned}$$

结合 $w - u \in M_0$, 利用 $j'(0) = 0$, 有

$$\int_{\Omega} |\nabla(w - u)|^2 + (w - u)^2 dx + \int_{\partial\Omega} \alpha(x) (w - u)^2 ds = 0$$

则必有 $w = u$ 成立, 则方程的解唯一。

(2) 利用证明 II $\Leftrightarrow$ III

“ $\Rightarrow$ ” 结合 Green 第一恒等式, 我们有

$$\int_{\Omega} (-\Delta u + u - f) v dx + \int_{\partial\Omega} \left(\frac{\partial u}{\partial n} + \alpha u - g \right) v ds = 0$$

结合 v 的任意性, 我们有

$$\begin{cases} -\Delta u + u = f, & x \in \Omega \\ \frac{\partial u}{\partial n} + \alpha u = g, & x \in \partial\Omega \end{cases}$$

“ $\Leftarrow$ ” 结合 $-\Delta u + u = f, x \in \Omega$, 则对其做积分, 有 $\int_{\Omega} (-\Delta u + u - f) dx = 0$. 同理, $\int_{\partial\Omega} (\frac{\partial u}{\partial n} + \alpha u - g) ds = 0$ 两者相加并使用 Green 第一恒等式有

$$\int_{\Omega} (\nabla u \cdot \nabla v + u \cdot v - fv) dx + \int_{\partial\Omega} (\alpha(x)uv - gv) ds = 0$$

□

2 波动方程

2.1 一阶线性方程的特征线解法

首先我们来明确什么是特征线：

定义 2.1.1. 我们称下列常微分方程初值问题

$$\begin{cases} \frac{dx}{dt} = a \\ x(0) = c \end{cases}$$

的解 $x(t, c) = at + c$ 为方程 (Cauchy 问题)

$$\frac{\partial \rho}{\partial t} + a \frac{\partial \rho}{\partial x} = 0$$

的特征线，其中 c 为常数。

我们直接来考虑最复杂的情形：

$$\begin{cases} \frac{\partial u}{\partial t} + f(x, t) \frac{\partial u}{\partial x} = g(u; x, t), & x \in \mathbb{R}^1, t > 0 \\ u(x, 0) = h(x). \end{cases}$$

解：我们构造特征线

$$\begin{cases} \frac{dx}{dt} = f(x, t) \\ x(0) = c \end{cases} \quad \text{得到方程的解为 } x(t, c), \text{ 其中 } c \text{ 可以由 } x, t \text{ 表示: } c = c_0(x, t).$$

取 $U(t) = u(x(t, c), t)$ ，则我们有

$$\begin{cases} \frac{dU}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial t} = \frac{\partial u}{\partial t} + f(x, t) \frac{\partial u}{\partial x} = g(U; x(t, c), t) \\ U(0) = u(x(0, c), 0) = u(c, 0) = h(c). \end{cases}$$

此时我们将问题转化为求解上述 ODE 问题，得到方程的解为 $U(t, c) = U(t, c_0(x, t))$. 此时把 $c = c_0(x, t)$ ，也就得到了原方程的解。 □

另外, 我们也可以考虑如下问题, 它们是原问题的简化形式:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0 \quad / \quad \frac{\partial u}{\partial t} + \frac{\partial}{\partial x}(q(u(x))) = 0$$

例 2.1.1. 求解一阶 PDE 问题

$$\begin{cases} u_t + au_x = f(x, t), & t > 0, x \in \mathbb{R} \\ u(x, 0) = \varphi(x). \end{cases}$$

解答 对特征线

$$\begin{cases} \frac{dx}{dt} = a \\ x(0) = c \end{cases} \quad \text{有 } x(t, c) = at + c, \text{ 则 } c = x - at$$

取 $U(t) = u(x(t, c), t)$, 则有

$$\begin{cases} \frac{dU}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{dx}{dt} = \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = f(x(t, c), t) \\ U(0) = u(x(0, c), 0) = u(c, 0) = \varphi(c) \end{cases}$$

则

$$\begin{aligned} U(t) &= \int_0^t U'(\tau) d\tau + U(0) = U(0) + \int_0^t f(x(\tau, c), \tau) d\tau = \varphi(c) + \int_0^t f(x(\tau, c), \tau) d\tau \\ &= \varphi(c) + \int_0^t f(a\tau + c, \tau) d\tau = \varphi(x - at) + \int_0^t f(x + a(\tau - t), \tau) d\tau. \end{aligned}$$

□

从这个问题中, 我们指出, 当我们求出 $c = c_0(x, t)$ 之后, 对它的替换应当放在整个题目计算的最后一步。

例 2.1.2. 用特征线法求解下述 Cauchy 问题

$$(1) \begin{cases} u_t + 2u_x = 0, & t > 0, -\infty < x < \infty \\ u|_{t=0} = x^2, & -\infty < x < \infty \end{cases} \quad (2) \begin{cases} u_t + (1 + x^2)u_x - u = 0, & t > 0, -\infty < x < \infty \\ u|_{t=0} = \arctan x, & -\infty < x < \infty \end{cases}$$

解答 (1) 取特征线问题

$$\begin{cases} \frac{dx}{dt} = 2 \\ x(0) = c \end{cases} \Rightarrow \begin{cases} x(t) = 2t + c \\ c = x - 2t \end{cases}$$

则对 $U(t) = u(x(t, c), t)$, 我们有 $\frac{dU}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial t} = 0$. 则 $u(x(t, c), t) = u(x(0, c), 0) = u(c, 0) = c^2 = (x - 2t)^2$

(2) 取特征线问题

$$\begin{cases} \frac{dx}{dt} = 1 + x^2 \\ x(0) = c \end{cases} \Rightarrow \begin{cases} x(t, c) = \tan(t + \arctan c) \\ c = \tan(\arctan x - t) \end{cases}$$

则对 $U(t) = u(x(t, c), t)$, 我们有 $\frac{dU}{dt} = U$ 则

$$\begin{cases} \frac{dU}{dt} = U \\ U(0) = u(x(0, c), 0) = u(c, 0) = \arctan c \end{cases}$$

有 $U(x(t, c), t) = \arctan c e^t = (\arctan x - t)e^t$. □

例 2.1.3. 遗漏了答案 求下列 *Cauchy* 问题的解

$$\begin{cases} \frac{\partial u}{\partial t} + (x + t) \frac{\partial u}{\partial x} + u = x, & x \in \mathbb{R}, t > 0 \\ u(x, 0) = x \end{cases}$$

解答

$$\begin{cases} \frac{dx}{dt} = x + t \\ x(0) = c \end{cases}$$

则有 $(x'(t) - x(t))e^{-t} = te^{-t}$, 解得 $x(t) = (c + 1)e^t - (t + 1)$, $c = (x + t + 1)e^{-t} - 1$

令 $U(t) = u(x(t, c), t)$

$$\begin{cases} \frac{dU}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial t} = x - U = (c + 1)e^t - (t + 1) - U \\ U(0) = u(x(0, c), 0) = c \end{cases}$$

则 $(U(t) + U'(t))e^t = (c+1)e^{2t} - (t+1)e^t$

$$\begin{aligned} U(t)e^t &= U(0) + \int_0^t ((c+1)e^{2\tau} - (\tau+1)e^\tau) d\tau \\ &= c + \frac{1}{2}(c+1)(e^{2t} - 1) - (t+1)e^t + 1 + e^t - 1 \\ &= \frac{1}{2}(c+1)e^{2t} - te^t + \frac{1}{2}(c-1) \\ &= \frac{1}{2}e^t(x+t+1) - te^t + \frac{1}{2}((x+t+1)e^{-t} - 2) \end{aligned}$$

则 $u(x, t) \equiv \frac{1}{2}(x-t+1) + \frac{1}{2}(x+t+1)e^{-2t} - e^{-t}$

□

(相对不重要)知识回顾: 隐函数定理

定理 2.1.1. 设 E 是 $\mathbb{R}^{n+m}$ 中开集, $F = (f_1, \dots, f_m)^T : E \rightarrow \mathbb{R}^m$ 连续可微, 又设 $\mathbf{a} \in \mathbb{R}^n$ 及 $\mathbf{b} \in \mathbb{R}^m$, 使得 $(\mathbf{a}, \mathbf{b}) \in E$ 且 $F(\mathbf{a}, \mathbf{b}) = \mathbf{0}$, 现将 F 的 Jacobi 矩阵写作为如下形状

$$\begin{bmatrix} \frac{\partial F}{\partial \mathbf{x}}, \frac{\partial F}{\partial \mathbf{y}} \end{bmatrix}$$

其中 $\frac{\partial F}{\partial \mathbf{x}} = (\frac{\partial f_i}{\partial x_j})_{1 \leq i \leq m, 1 \leq j \leq n}$, $\frac{\partial F}{\partial \mathbf{y}} = (\frac{\partial f_i}{\partial y_j})_{1 \leq i \leq m, 1 \leq j \leq n}$, 则当 $\det\left(\frac{\partial F}{\partial \mathbf{y}}\right)(\mathbf{a}, \mathbf{b}) \neq 0$ 时, 存在 $\mathbf{a}$ 在 $\mathbb{R}^n$ 中心邻域 U 以及唯一连续可微映射 $g : U \rightarrow \mathbb{R}^m$, 使得 (1) $g(\mathbf{a}) = \mathbf{b}$ (2) 对任意 $\mathbf{x} \in U$ 有 $F(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ (3) 对任意 $\mathbf{x} \in U$ 有 $\det \frac{\partial F}{\partial \mathbf{y}}(\mathbf{x}, g(\mathbf{x})) \neq 0$ 且 $g'(\mathbf{x}) = -\left(\frac{\partial F}{\partial \mathbf{y}}(\mathbf{x}, g(\mathbf{x}))\right)^{-1} \frac{\partial F}{\partial \mathbf{x}}(\mathbf{x}, g(\mathbf{x})).$

隐函数定理可以用来解决下面这个问题:

定理 2.1.2. 一阶拟线性方程问题

$$\begin{cases} u_t + uu_x = 0 \\ u(x, 0) = \varphi(x) \end{cases}$$

存在整体 C^1 解且仅当 $\varphi'(x) \geq 0$

解答

对特征线问题

$$\begin{cases} \frac{dx}{dt} = u \\ x(0) = c \end{cases} \quad \text{得到特征线 } x(t, c)$$

则对 $U(t) = u(x(t, c), t)$

$$\frac{dU}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{dx}{dt} = u_t + uu_x = 0, \quad \text{则 } u(x(0, c), 0) = \varphi(c).$$

则有 $x(t, c) = \varphi(c)t + c$ 结合隐函数定理为由式解出 c , 我们应保证 $1 + \varphi'(c)t > 0$

必要性: 当 $\varphi'(x) \geq 0$ 时, 有 $1 + \varphi'(c)t \geq 1 > 0$, 有必要性保证

充分性: 反设 $\exists x_0 \in \mathbb{R}$ s.t. $\varphi'(x_0) < 0$, 则对 $c_1 < c_2$, 有 $\varphi(c_1) > \varphi(c_2)$ 在某一区间内则对两条特征线 $x_1(t) = \varphi(c_1)t + c_1$, $x_2(t) = \varphi(c_2)t + c_2$, 我们来找两者相交的部分即 $x_1(t) = x_2(t) = \varphi(c_1)t + c_1 = \varphi(c_2)t + c_2$. 则有 $t = \frac{c_2 - c_1}{\varphi(c_1) - \varphi(c_2)} > 0$. 但此时有

$$u(x_1(t, c), t) = \varphi(c_1) \quad u(x_2(t, c), t) = \varphi(c_2)$$

$\varphi(c_1) \neq \varphi(c_2)$ 两者矛盾, 则有充分性保证. □

我们给出两个例子:

实例 1

$$\begin{cases} u_t + uu_x = 0 \\ u(x, 0) = \sin x \end{cases}$$

结合 $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial c} \frac{\partial c}{\partial x} = \varphi'(c) \frac{1}{1 + \varphi'(c)t}$ $\varphi'(c) = \cos c$, 当 $c = \pi$ 时 $\varphi'(c) = -1$ 则取 $t \in (0, 1)$, 我们有 $\varphi'(c) \rightarrow \infty$, 有 C^1 解

实例 2

$$\begin{cases} u_t + uu_x = 0 \\ u(x, 0) = \tan x \end{cases}$$

结合 $(\tan x)' = \frac{1}{\cos^2 x} \geq 0$ 则方程有整体解

2.2 初值问题 (一维情形)

在这里我们重点解决这样的偏微分方程问题

$$\begin{cases} \square u = \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t), \mathbb{R} \times (0, \infty) \\ u(x, 0) = \varphi(x), x \in \mathbb{R} \\ u_t(x, 0) = \psi(x), x \in \mathbb{R} \end{cases}$$

事实上上面偏微分方程是一个线性偏微分方程, 我们可以将问题转化为下面的三个方程

$$\begin{cases} \square u_1 = 0, \mathbb{R} \times (0, \infty) \\ u_1(x, 0) = \varphi(x), x \in \mathbb{R} \\ (u_1)_t(x, 0) = 0, x \in \mathbb{R} \end{cases}, \begin{cases} \square u_2 = 0, \mathbb{R} \times (0, \infty) \\ u_2(x, 0) = 0, x \in \mathbb{R} \\ (u_2)_t(x, 0) = \psi(x), x \in \mathbb{R} \end{cases}, \begin{cases} \square u_3 = f(x, t), \mathbb{R} \times (0, \infty) \\ u_3(x, 0) = 0, x \in \mathbb{R} \\ (u_3)_t(x, 0) = 0, x \in \mathbb{R} \end{cases}$$

此时我们只需要加和三个解

$$u = u_1 + u_2 + u_3$$

便可以得到最终解, 在此之中, 我们称问题 2 是基本的两外两个解可以通过其表出, 即如下定理:

定理 2.2.1. 设 $u_2 = M_\psi(x, t)$ 是定解问题 2 的解 (这里 M_ψ 表示以 ψ 为初值的定解问题的解), 则定解问题 1, 3 的解 u_1, u_3 可以分别表示为

$$u_1 = \frac{\partial}{\partial t} M_\varphi(x, t)$$

$$u_3 = \int_0^t M_{f_\tau}(x, t - \tau) d\tau$$

其中 $f_\tau = f(x, \tau)$, 并且假定 $M_\varphi(x, t)$ 和 $M_{f_\tau}(x, t - \tau)$ 分别在区域 $\{x \in \mathbb{R}, 0 \leq t < \infty\}$ 和 $\{x \in \mathbb{R}, 0 \leq \tau \leq t < \infty\}$ 上对变量 x, t 和 τ 充分光滑。

解答 (问题 3 又称为 Duhamel 原理, 我们来证明之, 证明过程进行了修改, 原本的写法是有问题的) 取 $u(x, t) = \int_0^t W(x, t - \tau) d\tau$

$\tau; \tau) d\tau$, 其中辅助函数 $W(x, t; \tau)$ 是依赖于参数 τ 的齐次波动方程的解, 满足如下定解条件:

$$\begin{cases} W_{tt} - a^2 W_{xx} = 0, & x \in \mathbb{R}, t > 0 \\ W(x, 0; \tau) = 0, & x \in \mathbb{R} \\ W_t(x, 0; \tau) = f(x, \tau), & x \in \mathbb{R} \end{cases} \quad (1)$$

这里 $M_{f_\tau}(x, t - \tau)$ 即对应于上述 $W(x, t - \tau; \tau)$ 。利用含参变量积分的求导法则 (Leibniz 公式), 我们直接计算 $u(x, t)$ 的导数。首先对 t 求一阶导数:

$$u_t(x, t) = W(x, t - \tau; \tau) \Big|_{\tau=t} + \int_0^t W_t(x, t - \tau; \tau) d\tau$$

注意到由辅助方程 (1) 的初始条件知 $W(x, 0; \tau) = 0$, 故上式第一项为零, 即有 $u_t(x, t) = \int_0^t W_t(x, t - \tau; \tau) d\tau$ 。由此立即可得 $u_t(x, 0) = 0$, 且显然有 $u(x, 0) = 0$, 即满足零初始条件。

接下来求二阶导数:

$$u_{tt}(x, t) = W_t(x, t - \tau; \tau) \Big|_{\tau=t} + \int_0^t W_{tt}(x, t - \tau; \tau) d\tau$$

根据辅助方程 (1) 的初始速度条件, 边界项 $W_t(x, 0; \tau)$ 在 $\tau = t$ 时恰为 $f(x, t)$ 。同时, 对 x 求二阶导数可得 $u_{xx}(x, t) = \int_0^t W_{xx}(x, t - \tau; \tau) d\tau$ 。

将 u_{tt} 和 u_{xx} 代入波动方程左端, 并利用 W 满足齐次方程 $W_{tt} = a^2 W_{xx}$ 的性质, 有:

$$\begin{aligned} u_{tt} - a^2 u_{xx} &= f(x, t) + \int_0^t [W_{tt}(x, t - \tau; \tau) - a^2 W_{xx}(x, t - \tau; \tau)] d\tau \\ &= f(x, t) + \int_0^t 0 \cdot d\tau = f(x, t) \end{aligned}$$

综上所述, 由积分公式定义的 $u(x, t)$ 满足非齐次波动方程及零初始条件, 故原命题成立。□

结合上面的讨论, 我们仅需解出问题 2 的解即可, 我们介绍的第一种方法称为特征线法。结合式

$$\square u = \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = \left(\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} \right) \left(\frac{\partial u}{\partial t} - a \frac{\partial u}{\partial x} \right) = 0$$

我们将问题分解为

$$\frac{\partial v}{\partial t} + a \frac{\partial v}{\partial x} = 0, \quad \frac{\partial u}{\partial t} - a \frac{\partial u}{\partial x} = v$$

对初值条件

$$u(x, 0) = 0, \quad v(x, 0) = \left. \frac{\partial u}{\partial t} \right|_{t=0} - a \left. \frac{\partial u}{\partial x} \right|_{t=0} = \psi(x)$$

我们就将问题转化为了两个一阶线性方程原始问题的求解问题，即

$$\begin{cases} \frac{\partial u}{\partial t} - a \frac{\partial u}{\partial x} = v \\ v(x, 0) = \psi(x) \end{cases} \quad \begin{cases} \frac{\partial u}{\partial t} - a \frac{\partial u}{\partial x} = v \\ u(x, 0) = 0 \end{cases}$$

利用第一个方程我们可以得到 $v(x, t) = \psi(x - at)$ ，带入第二个方程，使用第一节的方法，最终我们得到了波动方程的解：

$$\begin{aligned} u(x, t) &= \frac{\partial}{\partial t} \left[\frac{1}{2a} \int_{x-at}^{x+at} \varphi(\xi) d\xi \right] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi + \int_0^t \left[\frac{1}{2a} \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi \right] d\tau \\ &= \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi + \frac{1}{2a} \int_0^t d\tau \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi \end{aligned}$$

当 $f \equiv 0$ 时，上面的式子被称为 D'Alembert 公式。

2.3 求解波动方程的行波法

求解波动方程，我们有另一种方法：行波法。我们来回顾一下问题：

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u(x, 0) = \varphi(x) \\ u_t(x, 0) = \psi(x) \end{cases}$$

此时我们记 $\xi = x - at$, $\eta = x + at$ ，得到

$$\frac{\partial \xi}{\partial x} = 1, \quad \frac{\partial \xi}{\partial t} = -a, \quad \frac{\partial \eta}{\partial x} = 1, \quad \frac{\partial \eta}{\partial t} = a$$

因此我们来计算 $\frac{\partial^2 u}{\partial t^2}$ 和 $\frac{\partial^2 u}{\partial x^2}$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial t} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial t} = -a \frac{\partial u}{\partial \xi} + a \frac{\partial u}{\partial \eta} \quad \frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}$$

则我们可以求二阶导:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial t} \right) = \frac{\partial}{\partial \xi} \left(\frac{\partial u}{\partial t} \right) \frac{\partial \xi}{\partial t} + \frac{\partial}{\partial \eta} \left(\frac{\partial u}{\partial t} \right) \frac{\partial \eta}{\partial t} = a^2 u_{\xi\xi} - 2a^2 u_{\xi\eta} + a^2 u_{\eta\eta}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial \xi} \left(\frac{\partial u}{\partial x} \right) \frac{\partial \xi}{\partial x} + \frac{\partial}{\partial \eta} \left(\frac{\partial u}{\partial x} \right) \frac{\partial \eta}{\partial x} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

代入, 我们就可以得到 $u_{\xi\eta} = 0$, 则方程的解我们一定可以写作

$$F(\xi, \eta) = f(\xi) + g(\eta)$$

即

$$u(x, t) = f(x - at) + g(x + at)$$

结合初值条件 $u(x, 0) = f(x) + g(x) = \varphi(x)$, $u_t(x, 0) = -af'(x - at) + ag'(x + at) = -cf'(x) + cg'(x) = \psi(x)$, 我们对两侧求积分, 得到

$$g(x) - f(x) = \frac{1}{a} \int_{x_0}^x \psi(s) ds + g(x_0) - f(x_0) \triangleq \frac{1}{a} \int_{x_0}^x \psi(s) ds + K$$

通过化简, 我们就可以得到最终的 D'Alembert 公式:

$$u(x, t) = \frac{1}{2}(\varphi(x - at) + \varphi(x + at)) + \frac{1}{2a} \int_{x-at}^{x+at} \psi(s) ds$$

例 2.3.1. 试求解初值问题

$$\begin{cases} u_{tt} - u_{xx} = 0, & -\infty < x < \infty, t > ax \\ u|_{t=ax} = u_0(x), & -\infty < x < \infty \\ u_t|_{t=ax} = u_1(x), & -\infty < x < \infty \end{cases}, \text{ 其中 } a \neq \pm 1, \text{ 若初值只给定在 } c \leq x \leq b \text{ 上, 试问需在什么区域上确定解.}$$

解答 取 $u(x, t) = v(x, t - ax) \triangleq v(x, m)$.

则

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial v}{\partial x} + \frac{\partial v}{\partial m} \frac{\partial m}{\partial x} = \frac{\partial v}{\partial x} - a \frac{\partial v}{\partial m} & \frac{\partial^2 u}{\partial x^2} &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial m} \frac{\partial m}{\partial x} \right) - a \left(\frac{\partial v}{\partial x} + \frac{\partial v}{\partial m} \frac{\partial m}{\partial x} \right) = \frac{\partial^2 v}{\partial x^2} - 2a \frac{\partial^2 v}{\partial m \partial x} + a^2 \frac{\partial^2 v}{\partial m^2} \\ \frac{\partial u}{\partial t} &= \frac{\partial v}{\partial m} & \frac{\partial^2 u}{\partial t^2} &= \frac{\partial^2 v}{\partial m^2}\end{aligned}$$

则 $u_{tt} - u_{xx} = 0$ 转化为 $(1 - a^2)v_{mm} + 2av_{mx} - v_{xx} = 0$, $u|_{t=ax} = u_0(x)$ 转化为 $v|_{t=0} = u_0(x)$, $u_t|_{t=ax} = u_1(x)$ 转化为 $v_t|_{t=0} = u_1(x)$
即

$$\begin{cases} v_{tt} + \frac{2a}{1-a^2}v_{tx} - \frac{1}{1-a^2}v_{xx} = 0, & -\infty < x < \infty, t > 0 \\ v|_{t=0} = u_0(x), & -\infty < x < \infty \\ v_t|_{t=0} = u_1(x), & -\infty < x < \infty \end{cases}$$

此时取 $\xi = x - \frac{t}{1-a}$ $\eta = x + \frac{t}{1+a}$

$$\frac{\partial \xi}{\partial x} = \frac{\partial \eta}{\partial x} = 1 \quad \frac{\partial \xi}{\partial t} = -\frac{1}{1-a} \quad \frac{\partial \eta}{\partial t} = \frac{1}{1+a}$$

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial v}{\partial \eta} \frac{\partial \eta}{\partial x} = \frac{\partial v}{\partial \xi} + \frac{\partial v}{\partial \eta} \quad \frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial \xi^2} \frac{\partial \xi}{\partial x} + \frac{\partial^2 v}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial x} + \frac{\partial^2 v}{\partial \eta \partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial^2 v}{\partial \eta^2} \frac{\partial \eta}{\partial x} = v_{\xi\xi} + 2v_{\xi\eta} + v_{\eta\eta}.$$

$$\frac{\partial \xi}{\partial t} + \frac{\partial v}{\partial \eta} \frac{\partial \eta}{\partial t} = -\frac{1}{1-a} \frac{\partial v}{\partial \xi} + \frac{1}{1+a} \frac{\partial v}{\partial \eta} \quad \frac{\partial^2 v}{\partial t^2} = -\frac{1}{1-a} \left(\frac{\partial^2 v}{\partial \xi^2} \frac{\partial \xi}{\partial t} + \frac{\partial^2 v}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial t} \right) + \frac{1}{1+a} \left(\frac{\partial^2 v}{\partial \eta \partial \xi} \frac{\partial \xi}{\partial t} + \frac{\partial^2 v}{\partial \eta^2} \frac{\partial \eta}{\partial t} \right) = \frac{1}{(1-a)^2} v_{\xi\xi} - \frac{2}{1-a^2} v_{\xi\eta} + \frac{v_{\eta\eta}}{(1+a)^2}$$

$$\frac{\partial^2 v}{\partial x \partial t} = \frac{\partial^2 v}{\partial \xi^2} \frac{\partial \xi}{\partial t} + \frac{\partial^2 v}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial t} + \frac{\partial^2 v}{\partial \eta \partial \xi} \frac{\partial \xi}{\partial t} + \frac{\partial^2 v}{\partial \eta^2} \frac{\partial \eta}{\partial t} = -\frac{1}{1-a} v_{\xi\xi} + \frac{-2a}{1-a^2} v_{\xi\eta} + \frac{1}{1+a} v_{\eta\eta}.$$

则

$$\begin{aligned}(1-a^2)v_{tt} + 2av_{tx} + a^2v_{xx} &= (1-a^2) \left(\frac{1}{(1-a)^2} v_{\xi\xi} - \frac{2}{1-a^2} v_{\xi\eta} + \frac{1}{(1+a)^2} v_{\eta\eta} \right) - \frac{2a}{1-a} v_{\xi\xi} \frac{4a^2}{1-a^2} v_{\xi\eta} + \frac{2a}{1+a} v_{\eta\eta} + a^2(v_{\xi\xi} + 2v_{\xi\eta} + v_{\eta\eta}) \\ &= \left(-\frac{4a}{a^2-1} - 4 \right) v_{\xi\eta} = 0\end{aligned}$$

则我们可以将该问题的解写为

$$v(x, t) = f\left(x - \frac{t}{1-a}\right) + g\left(x + \frac{t}{1+a}\right).$$

结合

$$\begin{cases} v(x, 0) = f(x) + g(x) = u_0(x) \\ v_t(x, 0) = -\frac{1}{1-a}f'(x) + \frac{1}{1+a}g'(x) = u_1(x) \end{cases} \quad \text{即} \quad \begin{cases} f(x) + g(x) = u_0(x) \\ \int_0^x \frac{1}{1+a}g'(m) - \frac{1}{1-a}f'(m)dm = \int_0^x u_1(m)dm. \end{cases}$$

我们有

$$\begin{cases} f(x) + g(x) = u_0(x) \\ \frac{1}{1+a}g(x) - \frac{1}{1-a}f(x) = k + \int_0^x u_1(m)dm. \end{cases} \quad \begin{cases} f(x) = \frac{1-a}{2}u_0(x) - k_0 - \frac{1-a^2}{2} \int_0^x u_1(s)ds \\ g(x) = \frac{1+a}{2}u_0(x) + k_0 + \frac{1-a^2}{2} \int_0^x u_1(s)ds \end{cases}$$

则

$$\begin{aligned} v(x, t) &= f\left(x - \frac{t}{1-a}\right) + g\left(x + \frac{t}{1+a}\right) \\ &= \frac{1-a}{2}u_0\left(x - \frac{t}{1-a}\right) - k_0 - \frac{1-a^2}{2} \int_0^{x-\frac{t}{1-a}} u_1(s)ds + \frac{1+a}{2}u_0\left(x + \frac{t}{1+a}\right) + k_0 + \frac{1-a^2}{2} \int_0^{x+\frac{t}{1+a}} u_1(s)ds \\ &= \frac{1-a}{2}u_0\left(x - \frac{t}{1-a}\right) + \frac{1+a}{2}u_0\left(x + \frac{t}{1+a}\right) + \frac{1-a^2}{2} \int_{x-\frac{t}{1-a}}^{x+\frac{t}{1+a}} u_1(s)ds \end{aligned}$$

$$u(x, t) = v(x, t - ax) = \frac{1-a}{2}u_0\left(x - \frac{t-ax}{1-a}\right) + \frac{1+a}{2}u_0\left(x + \frac{t-ax}{1+a}\right) + \frac{1-a^2}{2} \int_{x-\frac{t-ax}{1-a}}^{x+\frac{t-ax}{1+a}} u_1(s)ds$$

□

例 2.3.2. 求解达布问题 $\begin{cases} \frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, & 0 < x < t, \ t > 0 \\ u|_{x=0} = \varphi(t), & t \geq 0 \\ u|_{x=t} = \psi(t), & t \geq 0 \end{cases}$ 其中 $\varphi(0) = \psi(0)$, 如果 $\varphi(t), \psi(t)$ 都给定在 $[0, a]$, 指出求解条件及决定区域.

解答 用行波法方式, 我们可以得到方程应满足如下形式

$$u(x, t) = f(x + t) + g(x - t)$$

结合

$$\begin{cases} u(0, t) = f(t) + g(-t) = \varphi(t), & t \geq 0 \\ u(t, 0) = f(2t) + g(0) = \psi(t), & t \geq 0 \end{cases}$$

我们有 $f(0) + g(0) = \varphi(0)$, $f(t) + g(0) = \psi(\frac{t}{2})$, 则 $f(t) - f(0) = \psi(\frac{t}{2}) - \varphi(0)$

$$g(t) = \varphi(-t) - f(-t) = \varphi(-t) - f(0) - \psi(-\frac{t}{2}) + \varphi(0).$$

则

$$\begin{aligned} u(x, t) = f(x + t) + g(x - t) &= f(0) + \psi(\frac{x+t}{2}) - \varphi(0) + \varphi(t-x) - f(0) - \psi(\frac{t-x}{2}) + \varphi(0) \\ &= \psi(\frac{x+t}{2}) + \varphi(t-x) - \psi(\frac{t-x}{2}). \end{aligned}$$

□

2.4 D'Alembert 公式的性质与推广

2.4.1 依赖区间和影响区域

从 D'Alembert 公式看出对上半平面的任一固定点 (x, t) , 解 $u(x, t)$ 仅由 φ 在 $x - at$ 和 $x + at$ 两点及 ψ 在 $[x - at, x + at]$ 上的值确定, 于是我们称 x 轴上的区间 $[x - at, x + at]$ 为点 (x, t) 的依赖区间, 而对 $x = x_1 + at$, $x = x_2 - at$ 围成的三角形区域, 其依赖区间都落在区间 $[x_1, x_2]$ 上, 我们称 $[x_1, x_2]$ 为三角形区域的决定区域, 此外, 对 $E = \{(x, t) | x_1 - at \leq x \leq x_2 + at, t > 0\}$ 我们称其为区间 $[x_1, x_2]$ 的影响区域。

2.4.2 D'Alembert 公式的稳定性

所谓稳定性, 是指当问题的初始条件发生微小变化时, 其对应的解是否也只产生微小的变化。现假设两组解:

$$(f_1(x), g_1(x)) \longrightarrow u_1(x, t)$$

$$(f_2(x), g_2(x)) \longrightarrow u_2(x, t)$$

两者之间的差异记为 $\sup_{x \in \mathbb{R}} |f_1(x) - f_2(x)|$ 和 $\sup_{x \in \mathbb{R}} |g_1(x) - g_2(x)|$ ，我们需要证明 $\sup_{\substack{x \in \mathbb{R} \\ t \in [0, T]}} |u_1(x, t) - u_2(x, t)|$ 可以被初始条件的扰动所控制，结合下面的讨论：

$$\begin{aligned} |u_1(x, t) - u_2(x, t)| &= \left| \frac{1}{2} \left(f_1(x - at) - f_2(x - at) \right) + \frac{1}{2} \left(f_1(x + at) - f_2(x + at) \right) + \frac{1}{2a} \int_{x-at}^{x+at} g_1(s) - g_2(s) ds \right| \\ &\leq \frac{1}{2} \left| f_1(x - at) - f_2(x - at) \right| + \frac{1}{2} \left| f_1(x + at) - f_2(x + at) \right| + \frac{1}{2a} \int_{x-at}^{x+at} |g_1(s) - g_2(s)| ds \\ &\leq \frac{1}{2} \sup_{x \in \mathbb{R}} |f_1 - f_2| + \frac{1}{2} \sup_{x \in \mathbb{R}} |f_1 - f_2| + \frac{1}{2a} \int_{x-at}^{x+at} \sup_{x \in \mathbb{R}} |g_1 - g_2| ds \\ &= \sup_{x \in \mathbb{R}} |f_1 - f_2| + t \sup_{x \in \mathbb{R}} |g_1 - g_2| \end{aligned}$$

故有 D'Alembert 公式具有稳定性。

2.4.3 D'Alembert 公式的唯一性

假设存在两个不同的函数 $u_1(x, t)$, $u_2(x, t)$ 均满足波动方程

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u(x, 0) = \varphi(x) \\ u_t(x, 0) = \psi(x) \end{cases}$$

则我们构造函数 $w(x, t) = u_1(x, t) - u_2(x, t)$ ，其满足

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w(x, 0) = 0 \\ w_t(x, 0) = 0 \end{cases}$$

定义能量不等式为 $E(t) = \frac{1}{2} \int_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} \left(\left(\frac{\partial w}{\partial t} \right)^2 + a^2 \left(\frac{\partial w}{\partial x} \right)^2 \right) dx$. 则有 $E(0) = \frac{1}{2} \int_{x_0-at_0}^{x_0+at_0} \left(\left(\frac{\partial w}{\partial t} \right)^2 + a^2 \left(\frac{\partial w}{\partial x} \right)^2 \right) dx = 0$ 且

$$\begin{aligned} \frac{d}{dt} E(t) &= -\frac{a}{2} \left(\left(\frac{\partial w}{\partial t} \right)^2 + a^2 \left(\frac{\partial w}{\partial x} \right)^2 \right) \Big|_{x_0+a(t_0-t)} + \left(\left(\frac{\partial w}{\partial t} \right)^2 + a^2 \left(\frac{\partial w}{\partial x} \right)^2 \right) \Big|_{x_0-a(t_0-t)} + \int_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} \frac{\partial w}{\partial t} \frac{\partial^2 w}{\partial t^2} + a^2 \frac{\partial w}{\partial x} \frac{\partial^2 w}{\partial x \partial t} dx \\ &= -\frac{a}{2} \left(w_t^2 + a^2 w_x^2 \right) \Big|_{x_0+a(t_0-t)} + \left(w_t^2 + a^2 w_x^2 \right) \Big|_{x_0-a(t_0-t)} + \int_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} w_t w_{tt} - a^2 w_t w_{xx} dx + a^2 w_t w_x \Big|_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} \\ &= -\frac{a}{2} \left((w_t^2 - 2aw_t w_x + a^2 w_x^2) \Big|_{x_0+a(t_0-t)} + (w_t^2 + 2aw_t w_x + a^2 w_x^2) \Big|_{x_0-a(t_0-t)} \right) \leq 0 \end{aligned}$$

而 $E(t) \geq 0$, $E(0) = 0$, 则有 $E(t) \equiv 0 \quad \forall t \geq 0$ 成立. 则 $\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} = 0$, 则 $w(x, t) = u_1(x, t) - u_2(x, t) = 0$, 则 $u_1(x, t) = u_2(x, t) = 0$ 成立

2.4.4 能量不等式

这里我们研究能量不等式, 我们将问题限制在特征锥中:

定义 2.4.1. 设 (x_0, t_0) 为上半平面 Q 内任一点, 通过这点向下做两条特征线 $x = x_0 \pm a(t_0 - t)$, 这两条特征线与 x 轴围成的三角形区域称为以 (x_0, t_0) 点为顶点的特征锥, 我们记之为 K .

在引出能量不等式之前, 我们先来介绍一个证明能量不等式的重要不等式:

引理 2.4.1 (Gronwall 不等式). 若非负函数 $G(\tau)$ 在 $[0, T]$ 上连续可微, $G(0) = 0$, 且对 $\tau \in [0, T]$, 有

$$\frac{dG(\tau)}{d\tau} \leq CG(\tau) + F(\tau)$$

其中 $C > 0$ 为常数, $F(\tau)$ 为 $[0, T]$ 上不减的非负可积函数, 则

$$\frac{dG(\tau)}{d\tau} \leq e^{C\tau} F(\tau), \quad G(\tau) \leq C^{-1}(e^{C\tau} - 1)F(\tau)$$

解答 $\frac{dG(t)}{dt} \leq CG(t) + F(t)$ 即得 $e^{-Ct} \left(\frac{dG(t)}{dt} - CG(t) \right) \leq F(t)e^{-Ct}$ 即 $\frac{d}{dt}(e^{-Ct}G(t)) \leq F(t)e^{-Ct}$, 对两侧积分

$$\int_0^\tau \frac{d}{dt}(e^{-Ct}G(t))dt = e^{-C\tau}G(\tau) \leq \int_0^\tau F(t)e^{-Ct}dt \leq F(\tau) \int_0^\tau e^{-Ct}dt = C^{-1}(1 - e^{-C\tau})F(\tau).$$

即 $G(\tau) \leq C^{-1}(e^{C\tau} - 1)F(\tau)$. 代入原式 $\frac{dG(\tau)}{d\tau} \leq CG(\tau) + F(\tau) \leq (e^{C\tau} - 1)F(\tau) + F(\tau) = e^{C\tau}F(\tau)$, 原不等式得证 $\square$

定理 2.4.1 (能量不等式). 大概率这次不考 设 $u \in C^1(\bar{Q}) \cap C^2(Q)$ 是定解问题波动方程的解, 则有估计

$$\begin{aligned} \int_{\Omega_\tau} \left[u_t^2(x, \tau) + a^2 u_x^2(x, \tau) \right] dx &\leq M \left[\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right] \\ \iint_{K_\tau} \left[u_t^2(x, t) + a^2 u_x^2(x, t) \right] dx dt &\leq M \left[\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right] \end{aligned}$$

其中

$$0 \leq \tau \leq t_0, \quad K_\tau = K \cap \{0 \leq t \leq \tau\}$$

$$\Omega_\tau = K \cap \{t = \tau\} = (x_0 - a(t_0 - \tau), x_0 + a(t_0 - \tau)), \quad M = e^{t_0}$$

解答 对 $\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$. 对两侧乘以 $\frac{\partial u}{\partial t}$ 后在 K_τ 上求积分

$$\iint_{K_\tau} \frac{\partial u}{\partial t} \left(\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} \right) dx dt = \iint_{K_\tau} \frac{\partial u}{\partial t} f dx dt.$$

其中

$$\frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial t^2} = \frac{1}{2} \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial t} \right)^2 \quad \frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial t} \right) - \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial t} \right) - \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial x \partial t} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial t} \right) - \frac{1}{2} \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial x} \right)^2$$

则

$$\begin{aligned} \iint_{K_\tau} \frac{\partial u}{\partial t} \left(\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} \right) dx dt &= \iint_{K_\tau} \frac{1}{2} \frac{\partial}{\partial t} \left[\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right] - a^2 \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial t} \right) dx dt \\ &= - \oint_{\partial K_\tau} \frac{1}{2} \left[\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right] dx + a^2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} dt = \iint_{K_\tau} \frac{\partial u}{\partial t} f dx dt \end{aligned}$$

记 Γ_1 为 K_τ 的左边界, Γ_2 为 K_τ 的右边界.

$$\begin{aligned}
J &\triangleq \iint_{K_\tau} \frac{\partial u}{\partial t} \left(\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} \right) dx dt = - \oint_{\partial K_\tau} \frac{1}{2} \left[\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right] dx + a^2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} dt \\
&= - \int_{\Omega_0} \frac{1}{2} (\psi^2 + a^2 \varphi_x^2) dx + \int_{\Omega_\tau} \frac{1}{2} \left[\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right] dx - \int_{\Gamma_1 \cup \Gamma_2} \frac{1}{2} \left[\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right] dx + a^2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} dt. \\
&\triangleq J_1 + J_2 + J_3
\end{aligned}$$

$$J_3 = - \int_{\Gamma_1 \cup \Gamma_2} \frac{1}{2} \left[\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right] dx + a^2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} dt., \text{ 结合在 } \Gamma_1 \text{ 上 } dx = a dt, \text{ 在 } \Gamma_2 \text{ 上 } dx = -a dt.$$

$$J_3 = - \int_{\Gamma_1} \frac{a}{2} \left(\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} \right)^2 dt + \int_{\Gamma_2} \frac{a}{2} \left(\frac{\partial u}{\partial t} - a \frac{\partial u}{\partial x} \right)^2 dt \geq 0 \text{ 恒成立.}$$

则

$$\int_{\Omega_\tau} \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \frac{\partial^2 u}{\partial x^2} dx \leq \int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + 2 \iint_{K_\tau} \frac{\partial u}{\partial t} f dx dt \leq \int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} \left(\frac{\partial u}{\partial t} \right)^2 dx dt + \iint_{K_\tau} f^2 dx dt$$

$$\text{取 } G(\tau) = \iint_{K_\tau} \left[\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right] dx dt. = \int_0^\tau \left[\int_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} \left(\left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right) dx \right] dt. \quad F(\tau) = \int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2 dx dt.,$$

结合

$$\begin{aligned}
\frac{dG(\tau)}{d\tau} &= \int_{x_0-a(t_0-\tau)}^{x_0+a(t_0-\tau)} \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 dx = \int_{\Omega_\tau} \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 dx \\
&\leq \int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} \left(\frac{\partial u}{\partial t} \right)^2 dx dt + \iint_{K_\tau} f^2 dx dt \leq \int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 dx dt + \int_{K_\tau} f^2 dx dt \\
&= G(\tau) + F(\tau).
\end{aligned}$$

$$\text{则 } \frac{dG(\tau)}{d\tau} = \int_{\Omega_\tau} \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 dx \leq e^\tau \left(\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2 dx dt \right) \leq e^{t_0} \left(\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2 dx dt \right), \text{ 进一步的:}$$

$$G(\tau) = \iint_{K_\tau} \left[u_t^2(x, t) + a^2 u_x^2(x, t) \right] dx dt \leq (e^\tau - 1) \left(\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2 dx dt \right) \leq e^{t_0} \left(\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2 dx dt \right)$$

注: Green 公式

$$\int_{\partial S} P dx + Q dy = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

□

有了这样的概念, 我们证明前面所谓的 D'Alembert 公式的唯一性事实上就已经显而易见了, 另外, 我们还得到了这样的结论:

$$\begin{aligned} \int_{\Omega_\tau} u^2(x, \tau) dx &\leq M_1 \left[\int_{\Omega_0} (\varphi^2 + \psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right] \\ \iint_{K_\tau} u^2(x, t) dx dt &\leq M_1 \left[\int_{\Omega_0} (\varphi^2 + \psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right] \end{aligned}$$

其中 $M_1 = e^{t_0}(e^{t_0} + 1)$, $\tau \in [0, t_0]$.

解答 (新增了证明) 我们对

$$\begin{aligned} \int_{\Omega_\tau} u^2(x, \tau) - u^2(x, 0) dx &= \int_{\Omega_\tau} dx \int_0^\tau \frac{\partial}{\partial t} u^2(x, t) dt \\ &= \int_{\Omega_\tau} dx \int_0^\tau 2u(x, t) u_t(x, t) dt \\ &\leq \iint_{K_\tau} u^2(x, t) dx dt + \iint_{K_\tau} u_t^2(x, t) dt dx \end{aligned}$$

取 $G(\tau) = \iint_{K_\tau} u^2(x, t) dx dt$ 则 $\frac{dG(\tau)}{d\tau} = \int_{\Omega_\tau} u^2(x, \tau) dx$, 并取 $F(\tau) = \int_{\Omega_0} u^2(x, 0) dx + \iint_{K_\tau} u_t^2(x, t) dt dx$, 则 $G(0) = 0$ 且 $\frac{dG(\tau)}{d\tau} \leq G(\tau) + F(\tau)$, 结合 Gronwall 不等式:

$$\int_{\Omega_\tau} u^2(x, \tau) dx \leq e^\tau \left(\int_{\Omega_0} u^2(x, 0) dx + \iint_{K_\tau} u_t^2(x, t) dt dx \right)$$

又结合

$$\iint_{K_\tau} u_t^2(x, t) dt dx \leq \iint_{K_\tau} (u_t^2(x, t) + a^2 u_x^2(x, t)) dx dt \leq e^{t_0} \left[\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right]$$

则

$$\begin{aligned} \int_{\Omega_\tau} u^2(x, \tau) dx &\leq e^{t_0} \left(\int_{\Omega_0} \varphi^2 dx + e^{t_0} \left[\int_{\Omega_0} (\psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right] \right) \\ &\leq e^{t_0} (e^{t_0} + 1) \left[\int_{\Omega_0} (\varphi^2 + \psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right] \end{aligned}$$

同理

$$\iint_{K_\tau} u^2(x, t) dx dt \leq e^{t_0} (e^{t_0} + 1) \left[\int_{\Omega_0} (\varphi^2 + \psi^2 + a^2 \varphi_x^2) dx + \iint_{K_\tau} f^2(x, t) dx dt \right]$$

□

2.4.5 半无界问题

现在我们将问题限制在区域

$$\bar{Q} = \{0 \leq x < \infty, 0 \leq t < \infty\}$$

求解波动方程的相关问题, 此时我们求解的方程为

$$\begin{cases} \square u = f(x, t), & 0 < x < \infty, 0 < t < \infty \\ u|_{t=0} = \varphi(x), & 0 < x < \infty \\ u_t|_{t=0} = \psi(x), & 0 < x < \infty \\ u|_{x=0} = g(t), & t > 0 \end{cases}$$

我们分为两种情形来讨论:

1. $g(t) \equiv 0$ 的情形

在这种情况下, 我们只需要将问题紧性延拓即可, 也就是用下面的函数替换原函数:

$$\bar{\varphi}(x) = \begin{cases} \varphi(x), & x \geq 0 \\ -\varphi(-x), & x < 0 \end{cases}, \quad \bar{\psi}(x) = \begin{cases} \psi(x), & x \geq 0 \\ -\psi(-x), & x < 0 \end{cases}$$

$$\bar{f}(x, t) = \begin{cases} f(x, t), & x \geq 0, t \geq 0 \\ -f(x, t), & x < 0, t \geq 0 \end{cases}$$

则原方程的解我们转化为了

$$\bar{u}(x, t) = \frac{1}{2}[\bar{\varphi}(x - at) + \bar{\varphi}(x + at)] + \frac{1}{2a} \int_{x-at}^{x+at} \bar{\psi}(\xi) d\xi + \frac{1}{2a} \int_0^1 dt \int_{x-a(t-\tau)}^{x+a(t-\tau)} \bar{f}(\xi, \tau) d\xi$$

此时我们只要取 $u = \bar{u}|_{\bar{Q}}$ ，我们就得到了原问题的解，当然我们还需要考虑光滑性要求和相容性要求：

定理 2.4.2. 若 $\varphi(x) \in C^2[0, \infty)$, $\psi(x) \in C^1[0, \infty)$, $f(x, t) \in C^1(\bar{Q})$ ，且满足相容性条件

$$\varphi(0) = 0, \psi(0) = g'(0) = 0, a^2\varphi''(0) + f(0, 0) = 0$$

那么半无界问题必有解 $u(x, t) \in C^2(\bar{Q})$ 。

2. $g(t) \neq 0$ 的情形

此时我们令 $u = v + g(t)$ ，则上面的问题我们可以做如下转化：

$$\begin{cases} \square v = \square u - g''(t) = f(x, t) - g''(t), & 0 < x < \infty, 0 < t < \infty \\ v(0, t) = u(0, t) - g(t), & 0 < x < \infty \\ v(x, 0) = u(x, 0) - g(0) = \varphi(x) - g(0), & 0 < x < \infty \\ v_t(x, 0) = u_t(x, 0) - g'(0) = \psi(x) - g'(0), & t > 0 \end{cases}$$

例 2.4.1. 试利用唯一性结果直接证明：当初值 $\varphi(x), \psi(x)$ 是偶函数，非齐次项 $f(x, t)$ 是 x 的偶函数时，非齐次波动方程初值问题的解 $u(x, t)$ 关于 x 也是偶函数

根据以上事实，用延拓法求解半无界问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), & 0 \leq x < +\infty, t > 0 \\ u|_{t=0} = u_t|_{t=0} = 0 & 0 \leq x < +\infty \\ u_x|_{x=0} = 0, & t \geq 0 \end{cases} \quad \text{并说明当 } f(x, t) \text{ 满足什么条件时，导}$$

出的公式确实是问题的解。

解答 设 $u(x, t)$ 为方程

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), & -\infty < x < \infty, t > 0 \\ u|_{t=0} = \varphi(x), & -\infty < x < \infty \\ u_t|_{t=0} = \psi(x), & -\infty < x < \infty \end{cases}$$

的解, 且 $f(x, t), \varphi(x), \psi(x)$ 满足 $f(x, t) = f(-x, t), \varphi(x) = \varphi(-x), \psi(x) = \psi(-x)$.

则对 $v(x, t) = u(-x, t)$, 有 $u_{tt} = v_{tt}, u_{xx} = v_{xx}$, 因此 $v(x, t)$ 满足

$$\begin{cases} v_{tt} - a^2 v_{xx} = f(-x, t) = f(x, t), & -\infty < x < \infty, t > 0 \\ v|_{t=0} = \varphi(-x) = \varphi(x), & -\infty < x < \infty \\ v_t|_{t=0} = \psi(-x) = \psi(x), & -\infty < x < \infty \end{cases}$$

则 $v(x, t) = u(-x, t) \equiv u(x, t)$ 命题成立.

对半无界问题, 当我们有 $f(x, t) = f(-x, t), \forall x \in (-\infty, +\infty), t \in [0, +\infty)$ 时. 取

$$\bar{f}(x, t) = \begin{cases} f(x, t), & x \geq 0 \\ f(-x, t), & x < 0 \end{cases} \quad \bar{\varphi}(x) = \begin{cases} \varphi(x), & x \geq 0 \\ \varphi(-x), & x < 0 \end{cases} \quad \bar{\psi}(x) = \begin{cases} \psi(x), & x \geq 0 \\ \psi(-x), & x < 0 \end{cases}$$

我们有方程

$$\begin{cases} \bar{u}_{tt} - a^2 \bar{u}_{xx} = \bar{f}(x, t), & -\infty < x < \infty, t \geq 0 \\ \bar{u}|_{t=0} = \bar{u}_t|_{t=0} = 0, & -\infty \leq x < \infty \end{cases}$$

的解为

$$\bar{u}(x, t) = \frac{1}{2a} \int_0^t d\tau \int_{x-a(t-\tau)}^{x+a(t-\tau)} \bar{f}(\xi, \tau) d\xi$$

再将其限制在 $x \geq 0$ 即可

□

例 2.4.2. 证明半无界问题 $\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t), & 0 < x < \infty, t > 0 \\ u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x), & 0 \leq x < \infty \\ u|_{x=0} = \mu(t), & t \geq 0 \end{cases}$ 解的唯一性.

解答 设 $u_1(x, t), u_2(x, t)$ 均为该半无界问题的解, 则令 $w(x, t) = u_1(x, t) - u_2(x, t)$, 其满足

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{t=0} = 0, w_t|_{t=0} = 0 \\ w|_{x=0} = 0, t \geq 0 \end{cases} \quad . \text{ 考虑}$$

能量函数 $E(t) = \int_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} \frac{1}{2}(u_t^2 + a^2 u_x^2) dx$, 则有 $E(t_0) = 0$.

$$\begin{aligned} \frac{d}{dt} E(t) &= -\frac{a}{2} \left(u_t^2 + a^2 u_x^2 \Big|_{x_0+a(t_0-t)} + u_t^2 + a^2 u_x^2 \Big|_{x_0-a(t_0-t)} \right) + \int_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} u_{tt} u_t + a^2 u_x u_{xt} dx \\ &= -\frac{a}{2} \left(u_t^2 + a^2 u_x^2 \Big|_{x_0+a(t_0-t)} + u_t^2 + a^2 u_x^2 \Big|_{x_0-a(t_0-t)} \right) + \left(\int_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} \underline{u_t u_{tt} - a^2 u_t u_{xx}} dx + a^2 u_x u_t \Big|_{x_0-a(t_0-t)}^{x_0+a(t_0-t)} \right) \\ &= -\frac{a}{2} \left(u_t^2 - 2a u_t u_x + u_x^2 \Big|_{x_0+a(t_0-t)} + u_t^2 + 2a u_x u_t + a^2 u_x^2 \Big|_{x_0-a(t_0-t)} \right) \leq 0 \end{aligned}$$

则 $E(0) = \int_{x_0-at_0}^{x_0+at_0} \frac{1}{2}(u_t^2(x, 0) + a^2 u_x^2(x, 0)) dx = 0$ 结合 $E(t) \geq 0$, 我们有 $E(t) = 0 \quad \forall t > 0$ 成立. 则 $\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} = 0$

则有 $w(x, t) = u_1(x, t) - u_2(x, t) = 0 \Rightarrow u_1(x, t) = u_2(x, t)$

□

例 2.4.3. 试问下述半无界问题

$$\begin{cases} u_{tt} - u_{xx} + u_t + u_x = 0, & 0 < x < \infty, t > 0 \\ u|_{x=0} = 0, & t \geq 0 \\ u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x), & 0 \leq x < \infty \end{cases}$$

能否直接使用对称开拓法求解? 为什么? 试用特征线法求解.

解答 不可以直接用对称开拓法求解, 因为 $u_t + u_x$ 破坏了原方程的对称性, 结合

$$\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial t} + \frac{\partial}{\partial x} = \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x} + 1 \right) \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x} \right)$$

我们将问题分解为

$$\begin{cases} \frac{\partial v}{\partial t} - \frac{\partial v}{\partial x} + v = 0 \\ v(x, 0) = \varphi'(x) + \psi(x) \end{cases} \quad \begin{cases} \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = v \\ u(x, 0) = \varphi(x) \end{cases} \quad \text{先求解第一个问题}$$

对特征线 $x_1(t, c) = -t + c$, 构造 $V(t) = v(x_1(t, c), t)$

$$\begin{cases} \frac{dV}{dt} = -V \\ V(0) = v(c, 0) = \varphi'(c) + \psi(c) \end{cases} \Rightarrow v(x, t) = (\varphi'(x+t) + \psi(x+t))e^{-t}$$

则

$$\begin{cases} \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = (\varphi'(x+t) + \psi(x+t))e^{-t} \\ u(x, 0) = \varphi(x) \end{cases}$$

对特征线 $x_2(t, c) = t + c$, 构造 $U(t) = u(x_2(t, c), t)$, 有

$$\begin{cases} \frac{dU}{dt} = (\varphi'(2t+c) + \psi(2t+c))e^{-t} \\ U(0) = \varphi(c) \end{cases}$$

$$u(x, t) = U(t) = U(0) + \int_0^t (\varphi'(2\tau+c) + \psi(2\tau+c))e^{-\tau} d\tau = \varphi(x-t) + \int_0^t (\varphi'(2\tau+x-t) + \psi(2\tau+x-t))e^{-\tau} d\tau$$

□

2.5 初值问题 (高维情形)

2.5.1 球面平均法

针对高维情形, 我们介绍球面平均法推导三维波动方程, 我们是这样表述该问题的:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \Delta u = f(x, t), & \mathbb{R}^3 \times (0, \infty) \\ u|_{t=0} = \varphi(x), & x \in \mathbb{R}^3 \\ u_t|_{t=0} = \psi(x), & x \in \mathbb{R}^3 \end{cases} \quad \begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \Delta u = 0, & \mathbb{R}^3 \times (0, \infty) \\ u|_{t=0} = \varphi(x), & x \in \mathbb{R}^3 \\ u_t|_{t=0} = \psi(x), & x \in \mathbb{R}^3 \end{cases}$$

事实上和一维的情形相同, 我们只需要解决右侧的问题, 另外的一般性问题就可以通过这个方程的解解决了, 我们可以首先解出其径向解:

对三维波动方程问题, 我们来做变量替换 $r = \sqrt{x^2 + y^2 + z^2}$, 则

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} = \frac{\partial u}{\partial r} \frac{x}{\sqrt{x^2 + y^2 + z^2}} \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial r^2} \frac{x^2}{x^2 + y^2 + z^2} + \frac{\partial u}{\partial r} \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

则 $\Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r}$. 对初值条件 $u(r, 0) = \phi(r)$ $u_t(r, 0) = \psi(r)$. 我们令 $v(r, t) = ru(r, t)$. 有

$$\frac{\partial^2 u}{\partial t^2} = \frac{1}{r} \frac{\partial^2 v}{\partial t^2}, \quad \Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial^2 v}{\partial r^2}$$

记 $\tilde{\phi}(r) = r\phi(r)$ $\tilde{\psi}(r) = r\psi(r)$, 则我们将原问题转化为

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} - a^2 \frac{\partial^2 v}{\partial r^2} = 0 & , (r, t) \in \mathbb{R}_+ \times \mathbb{R}_+ \\ v(r, 0) = \tilde{\phi}(r) & , r \in \mathbb{R}_+ \\ v_t(r, 0) = \tilde{\psi}(r) & , r \in \mathbb{R}_+ \end{cases}$$

由于 $r > 0$, 我们利用半无界问题的处理方式, 有

$$\begin{aligned} v(r, t) &= \frac{1}{2}[(r - at)\phi(r - at) + (r + at)\phi(r + at)] + \frac{1}{2a} \int_{r-at}^{r+at} s\psi(s)ds \\ u(r, t) &= \frac{1}{2r}[(r - at)\phi(r - at) + (r + at)\phi(r + at)] + \frac{1}{2ar} \int_{r-at}^{r+at} s\psi(s)ds \\ &\quad (r \geq at \text{ 时}) \end{aligned}$$

而当 $r < at$ 时

$$\begin{aligned} v(r, t) &= \frac{1}{2}[(at + r)\phi(at + r) - (at - r)\phi(at - r)] + \frac{1}{2a} \int_{at-r}^{at+r} s\psi(s)ds \\ u(r, t) &= \frac{1}{2r}[(at + r)\phi(at + r) - (at - r)\phi(at - r)] + \frac{1}{2ar} \int_{at-r}^{at+r} s\psi(s)ds \end{aligned}$$

这就是高维波动方程的径向解, 下面我们开始正式研究高维波动方程的一般解, 这里我们主要研究球面平均值公式:

定义 2.5.1. 对三维欧氏空间 $\mathbb{R}^3$, 我们定义函数 u 在以 P 为球心, 半径为 r 的球面上在 t 时刻的平均值为

$$M_u(r, t; P) = \frac{1}{4\pi r^2} \iint_{\partial B(P, r)} u(\xi, \eta, \zeta; t) dS$$

我们介绍这样的定理

引理 2.5.1. (修改了公式记号, 原本是一个识别错误) 对于任何一个足够光滑的函数 u , 他的球面平均 $M_u(r; t)$ 满足

$$\frac{1}{4\pi r^2} \iint_{\partial B(P, r)} \Delta u(\mathbf{x}, t) dS = \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) M_u(r, t, P)$$

其中 $\mathbf{a}$ 是单位向量, $\mathbf{x}$ 是点 P 的坐标。

这里原本的符号有点混乱, 进行了修改 下面我们来求解高维波动方程。结合拉格朗日中值定理, 我们有 $\lim_{r \rightarrow 0} M_u(r, t; P) = u(\mathbf{x}, t)$ 且

$$M_u(r, 0; P) = \frac{1}{4\pi r^2} \iint_{\partial B(P, r)} \varphi(\mathbf{x}) dS, \quad M_\psi(r, t; P) = \frac{\partial M_u(r, t; P)}{\partial t} = \frac{1}{4\pi r^2} \iint_{\partial B(P, r)} \psi(\mathbf{x}) dS$$

则取 $M_u(r, t; P) = \frac{v(\mathbf{x}, t, r)}{r}$, 则 $\lim_{r=0} v(\mathbf{x}, t, r) = 0$, 因此我们有 $u(\mathbf{x}, t) = \lim_{r \rightarrow 0} M_u(r, t; P) = \lim_{r \rightarrow 0} \frac{v(\mathbf{x}, t, r)}{r}$, 结合

$$m(P, t; r) \triangleq M_u(r, t; P) = \frac{1}{4\pi r^2} \iint_{\partial B(P, r)} u(\xi, \eta, \zeta, t) dS$$

$$\frac{\partial^2 m}{\partial t^2} = \frac{1}{4\pi r^2} \iint_{\partial B(P, r)} u_{tt}(\mathbf{x}, t) dS = \frac{1}{4\pi r^2} \iint_{\partial B(P, r)} a^2 \Delta u dS = a^2 \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) M_u(r, t; P)$$

结合 $\frac{\partial m}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial r} - \frac{1}{r^2} v, \quad \frac{\partial^2 m}{\partial t^2} = \frac{1}{r} \frac{\partial^2 v}{\partial t^2}$

$$\frac{\partial^2 m}{\partial r^2} = -\frac{1}{r^2} \frac{\partial v}{\partial r} + \frac{1}{r} \frac{\partial^2 v}{\partial r^2} + \frac{2}{r^3} v - \frac{1}{r^2} \frac{\partial v}{\partial r} = \frac{1}{r} \frac{\partial^2 v}{\partial r^2} - \frac{2}{r^2} \frac{\partial v}{\partial r} + \frac{2v}{r^3}$$

我们有 $\frac{1}{r} \frac{\partial^2 v}{\partial t^2} = \frac{a^2}{r} \frac{\partial^2 v}{\partial r^2}$, 即 $\square v = 0$, 再结合边值条件

$$v(\mathbf{x}, 0, r) = r m(P, 0, r) = r M_u(r, 0; P) = \frac{1}{4\pi r} \iint_{\partial B(P, r)} \varphi(\mathbf{x} + r\mathbf{a}) d\Omega$$

$$v_t(\mathbf{x}, 0, r) = r m_t(P, 0, r) = r M_\psi(r, 0; P) = \frac{1}{4\pi r} \iint_{\partial B(P, r)} \psi(\mathbf{x} + r\mathbf{a}) d\Omega$$

此时 $v(\mathbf{x}, t, r)$ 我们可以使用上面的方式求出径向解, 在上面的径向解求解中, 我们对 $\tilde{\phi}, \tilde{\psi}$ 进行了偶延拓, 这里我们也对 $M_u(r, t; P), M_\psi(r, t; P)$ 进行偶延拓, 得到 $\tilde{M}_u(r, t; P), \tilde{M}_\psi(r, t; P)$, 即

$$v(\mathbf{x}, t, r) = \frac{1}{2} \left((r - at) \tilde{M}_u(r - at, 0; P) + (r + at) \tilde{M}_u(r + at, 0; P) \right) + \frac{1}{2a} \int_{r-at}^{r+at} \rho \tilde{M}_\psi(\rho, 0; P) d\rho$$

进而带入上面的过程有

$$\begin{aligned} u(\mathbf{x}, t) &= \lim_{r \rightarrow 0} \frac{v(\mathbf{x}, r, t)}{r} = \frac{\partial v(\mathbf{x}, r, t)}{\partial r} \Big|_{r=0} \\ &= \frac{1}{2} \left(\tilde{M}_u(r - at, 0; P) + (r - at) \tilde{M}'_u(r - at, 0; P) + \tilde{M}_u(r + at, 0; P) + (r + at) \tilde{M}'_u(r + at, 0; P) \right) \\ &\quad + \frac{1}{2a} \left((r + at) \tilde{M}_\psi(r + at, 0; P) - (r - at) \tilde{M}_\psi(r - at, 0; P) \right) \Big|_{r=0} \\ &= M_u(at, 0; P) + at M'_u(at, 0; P) + t M_\psi(at, 0; P) \end{aligned}$$

另有 $\frac{\partial}{\partial t}[t M_u(at, 0; P)] = M_u(at, 0; P) + at M'_u(at, 0; P)$ 则

$$u(\mathbf{x}, t) = \frac{\partial}{\partial t}[t M_u(at, 0; P)] + t M_\psi(at, 0; P) = \frac{\partial}{\partial t} \left[\frac{1}{4\pi a^2 t} \iint_{\partial B(P, at)} \varphi(\mathbf{x}) d\Omega \right] + \frac{1}{4\pi a^2 t} \iint_{\partial B(P, at)} \psi(\mathbf{x}) d\Omega$$

进一步推广为

$$u(\mathbf{x}, t) = \frac{\partial}{\partial t} \left[\frac{1}{4\pi a^2 t} \iint_{\partial B(P, at)} \varphi(\mathbf{x}) d\Omega \right] + \frac{1}{4\pi a^2 t} \iint_{\partial B(P, at)} \psi(\mathbf{x}) d\Omega + \int_0^t \left[\frac{1}{4\pi a^2 (t - \tau)} \iint_{\partial B(P, (t - \tau))} f(\mathbf{y}, \tau) dS \right] d\tau$$

例 2.5.1. (计算过于复杂) 求解三维波动方程的 *Cauchy* 问题

$$\begin{cases} u_{tt} = a^2(u_{xx} + u_{yy} + u_{zz}) \\ u|_{t=0} = 0 \\ u_t|_{t=0} = x^3 + y^2 z \end{cases}$$

解答 结合 Kirchhoff 公式, 我们有

$$u(\mathbf{x}, t) = \frac{1}{4\pi a^2 t} \iint_{\partial B(\mathbf{x}, at)} (u^3 + v^2 w) dS.$$

取 $u = x + at \sin \varphi \cos \theta$ $v = y + at \sin \varphi \sin \theta$ $w = z + at \cos \varphi$, 则 $dS = a^2 t^2 \sin \varphi d\theta d\varphi$
则

$$\begin{aligned} u(\mathbf{x}, t) &= \frac{1}{4\pi a^2 t} \int_0^{2\pi} d\theta \int_0^\pi ((x + at \sin \varphi \cos \theta)^3 + (y + at \sin \varphi \sin \theta)^2 (z + at \cos \varphi)) a^2 t^2 \sin \varphi d\varphi \\ &= \frac{1}{4\pi a^2 t} \int_0^{2\pi} d\theta \int_0^\pi (\underbrace{x^3 + 3x^2 at \sin \varphi \cos \theta + 3xa^2 t^2 \sin^2 \varphi \cos^2 \theta + a^3 t^3 \sin^3 \varphi \cos^3 \theta}_{\text{Term 1}} + \underbrace{y^2 z + y^2 at \cos \varphi + 2yzat \sin \varphi \sin \theta + 2ya^2 t^2 \sin \varphi \cos \varphi \sin \theta + za^2 t^2 \sin^2 \varphi \sin^2 \theta + a^3 t^3 \sin^2 \varphi \cos \varphi \sin^2 \theta}_{\text{Term 2}}) a^2 t^2 \sin \varphi d\varphi \\ &= \frac{1}{4\pi a^2 t} [2\pi \int_0^\pi (x^3 + y^2 z + y^2 at \cos \varphi) a^2 t^2 \sin \varphi d\varphi + \int_0^{2\pi} \cos^2 \theta d\theta \int_0^\pi 3xa^3 t^4 \sin^3 \varphi d\varphi + \\ &\quad \int_0^{2\pi} \cos^3 \theta d\theta \int_0^\pi a^3 t^3 \sin^3 \varphi \cos \varphi d\varphi + \int_0^{2\pi} \sin^2 \theta d\theta \int_0^\pi (za^4 t^4 \sin^3 \varphi + a^5 t^5 \sin^3 \varphi \cos \varphi) d\varphi] \\ &= \frac{1}{4\pi a^2 t} [2\pi (\int_0^\pi y^2 a^3 t^3 \frac{1}{2} \sin 2\varphi d\varphi + 2a^2 t^2 (x^3 + y^2 z)) + \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} d\theta \int_0^\pi 3xa^4 t^4 \sin \varphi (1 - \cos^2 \varphi) d\varphi + \\ &\quad \int_0^\pi a \cdot 0 (1 - \sin^2 \theta) d\theta \int_0^\pi a^3 t^3 \sin^2 \varphi \cos \varphi d\varphi + \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta \int_0^\pi (za^4 t^4 \sin \varphi (1 - \cos^2 \varphi) + a^5 t^5 \sin^3 \varphi \cos \varphi) d\varphi] \\ &= \frac{1}{4\pi a^2 t} (4\pi a^2 t^2 (x^3 + y^2 z) + \pi (6xa^4 t^4 - 2xa^4 t^4) + \pi \frac{4}{3} za^4 t^4) \\ &= t(x^3 + y^2 z) + t^3(x + \frac{1}{3}z) \end{aligned}$$

□

2.5.2 二维形式柯西问题

在前面一部分, 我们已经推导了三维波动方程的解, 即 Kirchhoff 公式, 既然如此, 我们希望使用三维情形, 将二维问题嵌入其中, 也就是将 Φ 和 Ψ 看成是在 z 方向上恒定不变的三维初始数据

$$\Phi(x, y, z) = \varphi(x, y), \quad \Psi(x, y, z) = \psi(x, y)$$

结合 Kirchhoff 公式，我们知道只要限制了 z 固定不变，结果就是二维情形，我们将

$$\Phi(\mathbf{x} + at\boldsymbol{\xi}) = \varphi(x + ct\xi_1, y + at\xi_2), \quad \Psi(\mathbf{x} + at\boldsymbol{\xi}) = \psi(x + at\xi_1, y + at\xi_2)$$

代入 Kirchhoff 公式即二维情形最原始的形式：

$$u(x, y, t) = U(x, y, 0, t) = \frac{1}{4\pi a^2} \frac{\partial}{\partial t} \left(t \iint_{|\boldsymbol{\xi}|=1} \varphi(x + at\xi_1, y + at\xi_2) dS(\boldsymbol{\xi}) \right) + \frac{t}{4\pi a} \iint_{|\boldsymbol{\xi}|=1} \psi(x + at\xi_1, y + at\xi_2) dS(\boldsymbol{\xi})$$

此时积分值为上半球面积分的 2 倍，且 (y_1, y_2) 满足

$$(y_1 - x_1)^2 + (y_2 - x_2)^2 = a^2 t^2$$

则有 $d\Omega = \frac{at}{\sqrt{a^2 t^2 - (y_1 - x_1)^2 - (y_2 - x_2)^2}} d\mathbf{y}$ ，因此我们就可以将方程转化为

$$\begin{aligned} u(y_1, y_2, t) &= \frac{\partial}{\partial t} \left[\iint_{\partial B(P, at)} \frac{\varphi(\mathbf{y})}{\sqrt{a^2 t^2 - (y_1 - x_1)^2 - (y_2 - x_2)^2}} d\mathbf{y} \right] + \frac{1}{2\pi a} \iint_{\partial B(P, at)} \frac{\psi(\mathbf{y})}{\sqrt{a^2 t^2 - (y_1 - x_1)^2 - (y_2 - x_2)^2}} d\mathbf{y} \\ &\quad + \frac{1}{2\pi a} \int_0^t d\tau \iint_{\partial B(P, a(t-\tau))} \frac{f(\mathbf{y}, \tau)}{\sqrt{a^2 (t-\tau)^2 - (y_1 - x_1)^2 - (y_2 - x_2)^2}} d\mathbf{y} \end{aligned}$$

例 2.5.2. (计算过于复杂) 求解二维波动方程的 *Cauchy* 问题

$$\begin{cases} u_{tt} = a^2(u_{xx} + u_{yy}) \\ u|_{t=0} = x^2(x + y) \\ u_t|_{t=0} = 0 \end{cases}$$

解答 将该问题代入 Poisson 公式

$$u(x, t) = \iint_{\partial B(\vec{x}, at)} \frac{u(u + v)}{\sqrt{a^2 t^2 - (u - x)^2 - (v - y)^2}} du dv$$

取 $u = x + r \cos \theta$ $v = y + r \sin \theta$, 则 $du dv = r dr d\theta$, 则有

$$\begin{aligned}
 I &= \int_0^{at} \frac{r}{\sqrt{a^2 t^2 - r^2}} dr \int_0^{2\pi} (x^2 + 2xr \cos \theta + r^2 \cos^2 \theta)(x + y + r \cos \theta + r \sin \theta) d\theta \\
 &= \int_0^{at} \frac{r}{\sqrt{a^2 t^2 - r^2}} dr \int_0^{2\pi} x^2(x + y) + r(x^2(\cos \theta + \sin \theta) + 2x(x + y) \cos \theta) + r^2((x + y) \cos^2 \theta + 2x \cos \theta(\cos \theta + \sin \theta)) + r^3 \sin \theta \cos^2 \theta d\theta \\
 &= \int_0^{at} \frac{r}{\sqrt{a^2 t^2 - r^2}} dr \int_0^{2\pi} x^2(x + y) + r^2((x + y) \frac{1 + \cos 2\theta}{2} + x(1 + \cos 2\theta) + x \sin 2\theta) + r^3 \sin \theta \cos^2 \theta d\theta \\
 &= \int_0^{at} \frac{r}{\sqrt{a^2 t^2 - r^2}} (2\pi x^2(x + y) + r^2(\pi(x + y) + 2x\pi)) dr
 \end{aligned}$$

取 $r = at \sin \varphi$ $dr = at \cos \varphi d\varphi$

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \frac{1}{\cos \varphi} at \sin \varphi (2\pi x^2(x + y) + a^2 t^2 \sin^2 \varphi \pi(3x + y)) d\varphi \\
 &= 2\pi at(x^3 + x^2 y) + \int_0^{\frac{\pi}{2}} a^3 t^3 (3x + y) \pi (\sin \varphi - \sin \varphi \cos^2 \varphi) d\varphi = 2\pi at(x^3 + x^2 y) + \frac{2}{3} a^3 t^3 \pi (3x + y)
 \end{aligned}$$

$$\text{则 } u(\vec{x}, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} I = x^3 + x^2 y + a^2 t^2 (3x + y)$$

□

关于高维情形的存在性定理, 我们表述如下:

定理 2.5.1. 若 $\varphi \in C^3(\mathbb{R}^n)$, $\psi \in C^2(\mathbb{R}^n)$, $f \in C^2(\bar{Q})$, 这里 $Q = \{(x, t) | x \in \mathbb{R}^n, t > 0\}$, 则由 Kirchhoff 公式和 Poisson 公式给出的函数 $u \in C^2(\bar{Q})$ 是定解问题的解。

2.5.3 特征锥与惠更斯原理

结合物理学规律, 我们知道波动方程有如下几个性质:

1. 解 u 在任一点 $P_0(x_0, y_0, t_0)$ 的值依赖于以此点的顶点的锥体:

$$K_{P_0} = \left\{ (x, y, t) | \sqrt{(x - x_0)^2 + (y - y_0)^2} \leq a(t_0 - t), 0 \leq t \leq t_0 \right\}$$

内, 我们把区域 $\left\{ (x, y, 0) | \sqrt{(x - x_0)^2 + (y - y_0)^2} \leq at_0 \right\}$ 称为点 P_0 对初值的依赖区域, 其本质上表现了扰动的传播速度有限的性质是波动方程的解的最基本的特征。

2. 假设在某一时刻，一般取作为处置时刻，某一小块 D_0 或是一个点 (x_0, y_0) ，什么样的范围受其影响？我们将这样的区域称为影响区域，一般的，我们有：

$$J_{(x_0, y_0)} = \left\{ \rho(x, y, t) \mid \sqrt{(x - x_0)^2 + (y - y_0)^2} \leq at, t \geq 0 \right\}$$

3. 我们定义上半平面中位移 $u(x, y, t)$ 可由 D 中的 φ 和 ψ 的值唯一确定的点集为决定区域，结合上面的讨论，我们有锥体

$$K = \left\{ (x, y, t) \mid (x - x_0)^2 + (y - y_0)^2 \leq a^2(t - t_0)^2 \right\}$$

起了特殊的作用，我们称其为波动方程以 (x_0, y_0, t_0) 为顶点的特征锥。

4. 在波的传播过程中有清晰前阵面和后阵面。

此外，针对高维情形，我们也有能量不等式：

$$\iint_{\Omega_\tau} (u_t^2 + a^2 |\nabla u|^2) dx dy \leq M \left[\iint_{\Omega_0} (\psi^2 + a^2 |\nabla \varphi|^2) dx dy + \iiint_{K_\tau} f^2 dx dy dt \right]$$

2.6 混合问题

在实际的波动问题中，我们可能遇见两端被限制的情形，本节我们就来解决这样的问题：

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0, & Q = \{0 < x < l, t > 0\} \\ u(0, t) = u(l, t) = 0, & t > 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \\ u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

我们将求解该问题分为三步：

1. 找到所有具有变量分离形式的非零特解，即

$$u(x, t) = X(x)T(t)$$

通过化简初始问题，我们本质上将方程转化为了如下的问题：

$$T'' + a^2 \lambda T = 0, \quad X'' + \lambda X = 0, \quad t > 0, 0 < x < l$$

我们称使得常微分方程其次边值条件具有非零解的那些 λ 值称为这个边值问题的特征值，相应的非零解称为对应于这个特征值的特征函数，这样的所有寻找特征值特征函数的问题被称为 Sturm-Liouville 问题。

在介绍下一阶段计算之前我们应当介绍这样的定理，他为我们进一步求解混合问题提供了方法：

定理 2.6.1. 在 $[0, l]$ 上特征值问题

$$\begin{cases} X'' + \lambda X = 0, & 0 < x < l \\ -\alpha_1 X'(0) + \beta_1 X(0) = 0 \\ \alpha_2 X'(l) + \beta_2 X(l) = 0 \end{cases}$$

(其中 $\alpha_i \geq 0, \beta_i \geq 0, \alpha_i + \beta_i \neq 0$) 具有如下性质：

(a) 所有特征值都是非负实数，特别当 $\beta_1 + \beta_2 > 0$ 时，所有特征值都是正数。

(b) 所有特征值组成一个单调递增以无穷远点为凝聚点的序列：

$$0 \leq \lambda_1 < \cdots < \lambda_n < \cdots, \lim_{n \rightarrow \infty} \lambda_n = \infty$$

(c) 不同特征值对应的特征函数必正交；即若 $X_\lambda(x)$ 和 $X_\mu(x)$ 分别对应于特征值 $\lambda, \mu (\lambda \neq \mu)$ 的特征函数，则

$$\int_0^l X_\lambda(x) X_\mu(x) dx = 0$$

(d) 任意函数 $f(x) \in L^2[0, l]$ 可以按特征函数系 $\{X_n(x)\}$ 展开，即

$$f(x) = \sum_{n=1}^{\infty} C_n X_n(x)$$

其中

$$C_n = \frac{\int_0^l f(x) X_n(x) dx}{\int_0^l X_n^2(x) dx}$$

解答 仅证 (1) 和 (3)

(1) 对原方程两侧乘 $X_\lambda(x)$, 并求积分我们有

$$\int_0^L X_\lambda''(x)X_\lambda(x) + \lambda X_\lambda^2(x)dx = 0$$

$$\text{即有 } X_\lambda'(x)X_\lambda(x)\Big|_0^L - \int_0^L (X_\lambda'(x))^2dx + \lambda \int_0^L (X_\lambda(x))^2dx = 0 \text{ 则 } \lambda = \frac{\int_0^L (X_\lambda'(x))^2dx - X_\lambda'(x)X_\lambda(x)\Big|_0^L}{\int_0^L (X_\lambda(x))^2dx}$$

结合

$$\begin{aligned} X_\lambda'(0)X_\lambda(0) &= \frac{1}{2\alpha_1\beta_1}(\alpha_1^2(X_\lambda'(0))^2 + \beta_1^2(X_\lambda(0))^2) \\ X_\lambda'(L)X_\lambda(L) &= -\frac{1}{2\alpha_2\beta_2}(\alpha_2^2(X_\lambda'(L))^2 + \beta_2^2(X_\lambda(L))^2) \end{aligned}$$

有 $\lambda \geq 0$ 恒成立.

当 $\lambda = 0$ 时 $\int_0^L (X_\lambda'(x))^2dx = 0$ 且 $X_\lambda'(0)X_\lambda(0) - X_\lambda'(L)X_\lambda(L) = 0$ 则 $\frac{\beta_1}{2\alpha_1}(X_\lambda(0))^2 + \frac{\beta_2}{2\alpha_2}(X_\lambda(L))^2 = 0$ 且 $X_\lambda(x) = C$ 故当 $\beta_1 + \beta_2 > 0$ 时, 为使上式成立, 有 $X_\lambda(x) = 0$

(3) 对 $X_\mu''(x) + \mu X_\mu(x) = 0$ $X_\lambda''(x) + \lambda X_\lambda(x) = 0$, 两侧乘 $X_\lambda(x)$ 和 $X_\mu(x)$ 后在 $[0, L]$ 上积分

$$\int_0^L X_\mu''(x)X_\lambda(x) + \mu X_\mu(x)X_\lambda(x)dx = 0 \quad \int_0^L X_\lambda''(x)X_\mu(x) + \lambda X_\lambda(x)X_\mu(x)dx = 0$$

$$\text{即 } X_\lambda(x)X_\mu'(x)\Big|_0^L - \int_0^L X_\lambda'(x)X_\mu'(x)dx + \mu \int_0^L X_\mu(x)X_\lambda(x)dx = 0, \quad X_\mu(x)X_\lambda'(x)\Big|_0^L - \int_0^L X_\lambda'(x)X_\mu'(x)dx + \lambda \int_0^L X_\mu(x)X_\lambda(x)dx = 0$$

相减后有

$$(\mu - \lambda) \int_0^L X_\mu(x)X_\lambda(x)dx = X_\mu(x)X_\lambda'(x) - X_\lambda(x)X_\mu'(x)\Big|_0^L = \begin{vmatrix} X_\mu(x) & X_\mu'(x) \\ X_\lambda(x) & X_\lambda'(x) \end{vmatrix}\Big|_0^L = 0$$

$$\text{则有 } \int_0^L X_\mu X_\lambda(x)dx = 0$$

□

2. 求解 Sturm-Liouville 问题, 求出全部特征值和特征函数, 并求出相应的 $T(t)$ 的表达式
结合上面的定理和 ODE 的问题, 我们事实上得到了这样的结果:

$$X_n(x) = \sin \frac{n\pi}{l}x, \quad n = 1, 2, \dots$$

$$T_n(t) = A_n \sin \frac{an\pi}{l}t + B_n \cos \frac{an\pi}{l}t, \quad n = 1, 2, \dots$$

则 $u_n(x, t) = \left(A_n \sin \frac{an\pi}{l}t + B_n \cos \frac{an\pi}{l}t \right) \sin \frac{n\pi}{l}x, \quad \forall n \in \mathbb{N}_+, \quad u_n(x, t)$ 适合方程和边界条件。

3. 将所有变量分离形式的特解叠加起来, 并利用初值定出所有待定系数
此时我们取

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin \frac{an\pi}{l}t + B_n \cos \frac{an\pi}{l}t \right) \sin \frac{n\pi}{l}x$$

其中根据傅里叶分析, 我们有这样的结果:

$$\varphi(x) = u \Big|_{t=0} = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{l}x, \quad \psi(x) = u_t \Big|_{t=0} = \sum_{n=1}^{\infty} \frac{an\pi}{l} A_n \sin \frac{n\pi}{l}x$$

结合上面的定理, 我们有

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi}{l}x, \quad \psi(x) = \sum_{n=1}^{\infty} \psi_n \sin \frac{n\pi}{l}x$$

其中

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l}x dx, \quad \psi_n = \frac{2}{l} \int_0^l \psi(x) \sin \frac{n\pi}{l}x dx$$

进而 $B_n = \varphi_n, \quad A_n = \frac{l}{an\pi} \psi_n$.

例 2.6.1. 求以下特征值问题的特征函数

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$$

解答 结合特征方程 $r^2 + \lambda = 0$, 我们有

$$\begin{aligned} X(x) &= C_1 \sin \sqrt{\lambda}x + C_2 \cos \sqrt{\lambda}x \\ X'(x) &= C_1 \sqrt{\lambda} \cos \sqrt{\lambda}x - C_2 \sqrt{\lambda} \sin \sqrt{\lambda}x \end{aligned}$$

$$\begin{cases} X'(0) = C_1 \sqrt{\lambda} = 0 \\ X'(l) = C_1 \sqrt{\lambda} \cos \sqrt{\lambda}l - C_2 \sqrt{\lambda} \sin \sqrt{\lambda}l = 0 \end{cases} \quad \begin{cases} C_1 = 0 \\ \sqrt{\lambda} = \frac{n\pi}{l}, \quad n = 1, 2, \dots \end{cases}$$

则特征函数为 $X_n(x) = C \sin \frac{n\pi}{l}x \quad n = 1, 2, \dots$

□

例 2.6.2. 用分离变量法求解如下问题

(1)

$$\begin{cases} Lu = 0, & (x, t) \in Q \\ u|_{x=0} = u_x|_{x=l} = 0, & t \geq 0 \\ u|_{t=0} = x(x-2l), \quad u_t|_{t=0} = 0, & 0 \leq x \leq l \end{cases} \quad * (2) \quad \begin{cases} Lu + u = 0, & (x, t) \in Q \\ u_x|_{x=0} = u_x|_{x=l} = 0, & t \geq 0 \\ u|_{t=0} = \varphi(x), \quad u_t|_{t=0} = \psi(x), & 0 \leq x \leq l \end{cases}$$

(3)

$$\begin{cases} Lu = 0, & (x, t) \in Q \\ u_x|_{x=0} = (u_x + hu)|_{x=l} = 0, & t \geq 0 \\ u|_{t=0} = \varphi(x), \quad u_t|_{t=0} = \psi(x), & 0 \leq x \leq l \end{cases}$$

$$1) \varphi(x) = \psi(x) = 1 \quad 2) \varphi(x) = \cos \frac{\pi x}{L}, \quad \psi(x) = 0$$

解答 (1) 由 $X(x)$ 满足

$$\begin{aligned} X(x) &= C_1 \sin \sqrt{\lambda}x + C_2 \cos \sqrt{\lambda}x, \\ X'(x) &= C_1 \sqrt{\lambda} \cos \sqrt{\lambda}x - C_2 \sqrt{\lambda} \sin \sqrt{\lambda}x \end{aligned}$$

$$\begin{cases} X(0) = C_2 = 0 \\ X'(l) = C_1 \sqrt{\lambda} \cos \sqrt{\lambda}l - C_2 \sqrt{\lambda} \sin \sqrt{\lambda}l = 0 \end{cases} \quad \begin{cases} \sqrt{\lambda} = \frac{n\pi + \frac{\pi}{2}}{l}, \quad n = 1, 2, \dots \\ C_2 = 0. \end{cases}$$

则 $X_n(x) = C \sin \frac{n\pi + \frac{\pi}{2}}{l}x$, $n = 1, 2, \dots$

同理, 我们可以得到 $T_n(t) = A_n \sin \frac{a(n\pi + \frac{\pi}{2})}{l}t + B_n \cos \frac{a(n\pi + \frac{\pi}{2})}{l}t$, $n = 1, 2, \dots$ 于是

$$u(x, t) = \sum_{n=1}^{\infty} X_n(x) T_n(t) = \sum_{n=1}^{\infty} (A_n \sin \frac{a(n\pi + \frac{\pi}{2})}{l}t + B_n \cos \frac{a(n\pi + \frac{\pi}{2})}{l}t) \sin \frac{n\pi + \frac{\pi}{2}}{l}x$$

而 $u|_{t=0} = u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi + \frac{\pi}{2}}{l}x = \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi + \frac{\pi}{2}}{l}x$. 其中 $\varphi_n = \frac{2}{l} \int_0^l x(x-2l) \sin \frac{n\pi + \frac{\pi}{2}}{l}x dx = -\frac{32l^2}{(2n+1)^3\pi^3}$. 结合正弦系的系数, 我们有 $B_n = \varphi_n = -\frac{32l^2}{(2n+1)^3\pi^3}$ 而结合 $u_t|_{t=0} = 0$, 我们有 $A_n = 0$, 因此

$$u(x, t) = \sum_{n=0}^{\infty} -\frac{32l^2}{(2n+1)^3\pi^3} \cos \frac{a(n\pi + \frac{\pi}{2})}{l}t \sin \frac{n\pi + \frac{\pi}{2}}{l}x.$$

(2) 仍设 $u(x, t) = T(t)X(x)$, 代入 $Lu + u = 0$, 我们有

$$T''(t)X(x) - a^2T(t)X''(x) + T(t)X(x) = 0$$

即 $\frac{T''(t)}{T(t)} - a^2\frac{X''(x)}{X(x)} + 1 = 0$.

设 $\frac{X''(x)}{X(x)} = -\lambda$, 则有 $\frac{T''(t)}{T(t)} = -(a^2\lambda + 1)$, 经过计算

$$X(x) = C_1 \cos \sqrt{\lambda}x + C_2 \sin \sqrt{\lambda}x$$

$$X'(x) = -C_1 \sqrt{\lambda} \sin \sqrt{\lambda}x + C_2 \sqrt{\lambda} \cos \sqrt{\lambda}x$$

由 $X'(0) = X'(l) = 0$. 有 $X_n(x) = C \cos \frac{n\pi}{l}x$ $n = 1, 2, \dots$, 且 $T_n(t) = A_n \sin \omega_n t + B_n \cos \omega_n t$, 其中 $\omega_n = \sqrt{a^2(\frac{n\pi}{l})^2 + 1}$.

结合 $n = 0$ 时 $\omega_0 = 1$, 我们有 (事实上这里我们可以直接写 $\sum_{n=0}^{\infty}$)

$$u(x, t) = A_0 \sin t + B_0 \cos t + \sum_{n=1}^{\infty} (A_n \sin \omega_n t + B_n \cos \omega_n t) \cos \frac{n\pi}{l}x.$$

其中 $A_0 = \psi_0 = \frac{1}{l} \int_0^l \psi(x) dx$ $B_0 = \varphi_0 = \frac{1}{l} \int_0^l \varphi(x) dx$

$$A_n = \frac{\psi_n}{\omega_n} = \frac{2}{\omega_n l} \int_0^l \psi(x) \cos \frac{n\pi}{l} x \quad B_n = \varphi_n = \frac{2}{l} \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx$$

即 $u(x, t) = \sum_{n=0}^{\infty} \left(\left(\frac{2}{l\omega_n} \int_0^l \psi(x) \cos \frac{n\pi}{l} x dx \right) \sin \omega_n t + \left(\frac{2}{l} \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx \right) \cos \omega_n t \right) \cos \frac{n\pi}{l} x$

(3) 我们取 $X(x) = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x$, 则 $X'(x) = -C_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + C_2 \sqrt{\lambda} \cos \sqrt{\lambda} x$

$$\begin{cases} u_x|_{x=0} = 0 \\ (u_x + hu)|_{x=l} = 0 \end{cases} \quad \begin{cases} C_2 = 0 \\ hC_1 \cos \sqrt{\lambda} l - \sqrt{\lambda} C_1 \sin \sqrt{\lambda} l = 0 \end{cases} \quad \text{则有 } \tan \sqrt{\lambda_n} l = \frac{h}{\sqrt{\lambda_n}} \quad n = 1, 2, \dots$$

其中 $\sqrt{\lambda_n}$ 满足 $\sqrt{\lambda_n} \in \left(\frac{(n-1)\pi}{l}, \frac{(n-\frac{1}{2})\pi}{l} \right)$. $X_n(x) = \cos \sqrt{\lambda_n} x$. 故有

$$T_n(t) = A_n \cos a\sqrt{\lambda_n} t + B_n \sin a\sqrt{\lambda_n} t$$

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} (A_n \cos a\sqrt{\lambda_n} t + B_n \sin a\sqrt{\lambda_n} t) \cos \sqrt{\lambda_n} x$$

其中 $A_n = \varphi_n = \frac{1}{\|X_n\|^2} \int_0^l \varphi(x) \cos \sqrt{\lambda_n} x dx$ $B_n = \frac{\varphi_n}{a\sqrt{\lambda_n}} = \frac{1}{a\sqrt{\lambda_n} \|X_n\|^2} \int_0^l \psi(x) \cos \sqrt{\lambda_n} x dx$ $\|X_n\|^2 = \int_0^l \cos^2 \sqrt{\lambda_n} x dx$

1) $A_n = \frac{1}{\|X_n\|^2} \int_0^l \cos \sqrt{\lambda_n} x dx = \frac{1}{\|X_n\|^2} \frac{\sin \sqrt{\lambda_n} x}{\sqrt{\lambda_n}} \Big|_0^l = \frac{\sin \sqrt{\lambda_n} l}{\sqrt{\lambda_n}} \Big/ \left(\frac{l}{2} + \frac{\sin 2\sqrt{\lambda_n} l}{4\sqrt{\lambda_n}} \right).$

$$\|X_n\|^2 = \frac{l}{2} + \frac{\sin 2\sqrt{\lambda_n} l}{4\sqrt{\lambda_n}} \quad B_n = \frac{1}{\|X_n\|^2 a\sqrt{\lambda_n}} \int_0^l \cos \sqrt{\lambda_n} x dx = \frac{\sin \sqrt{\lambda_n} l}{a\lambda_n} \Big/ \left(\frac{l}{2} + \frac{\sin 2\sqrt{\lambda_n} l}{4\sqrt{\lambda_n}} \right)$$

则 $u(x, t) = \sum_{n=1}^{\infty} \left(\frac{2 \sin \sqrt{\lambda_n} l}{l\sqrt{\lambda_n}} \cos a\sqrt{\lambda_n} t + \frac{2 \sin \sqrt{\lambda_n} l}{a\lambda_n} \sin a\sqrt{\lambda_n} t \right) \cos \sqrt{\lambda_n} x \Big/ \left(\frac{l}{2} + \frac{\sin 2\sqrt{\lambda_n} l}{4\sqrt{\lambda_n}} \right).$

2)

$$\begin{aligned}
A_n &= \varphi_n = \frac{1}{||X_n||^2} \int_0^l \cos \frac{\pi x}{L} \cos \sqrt{\lambda_n} x dx = \frac{1}{2||X_n||^2} \int_0^l \cos \left(\frac{\pi}{L} - \sqrt{\lambda_n} \right) x + \cos \left(\frac{\pi}{L} + \sqrt{\lambda_n} \right) x dx \\
&= \frac{1}{2||X_n||^2} \left(\frac{1}{\frac{\pi}{L} - \sqrt{\lambda_n}} \sin \left(\frac{\pi}{L} - \sqrt{\lambda_n} \right) x + \frac{1}{\frac{\pi}{L} + \sqrt{\lambda_n}} \sin \left(\frac{\pi}{L} + \sqrt{\lambda_n} \right) x \right) \Big|_0^l \\
&= \frac{1}{\pi - \sqrt{\lambda_n} L} \sin \left(\pi - \sqrt{\lambda_n} \right) L + \frac{1}{\pi + \sqrt{\lambda_n} L} \sin \left(\pi + \sqrt{\lambda_n} L \right) \Big/ \left(\frac{1}{2} + \frac{\sin 2\sqrt{\lambda_n} L}{4\sqrt{\lambda_n} L} \right).
\end{aligned}$$

$$B_n = \psi_n = 0 \text{ 则 } u(x, t) = \sum_{n=1}^{\infty} \left(\frac{1}{\pi - \sqrt{\lambda_n} l} \sin \left(\pi - \sqrt{\lambda_n} \right) L + \frac{1}{\pi + \sqrt{\lambda_n} l} \sin \left(\pi + \sqrt{\lambda_n} L \right) \right) \cos a \sqrt{\lambda_n} t \cos \sqrt{\lambda_n} x \Big/ \left(\frac{1}{2} + \frac{\sin 2\sqrt{\lambda_n} l}{4\sqrt{\lambda_n} l} \right) \quad \square$$

针对大量的波动方程混合问题，我们会遇见边值条件非齐次化的情形，即如下问题：

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0, & Q = \{0 < x < l, t > 0\} \\ \alpha_1 \frac{\partial u}{\partial x} + \beta_1 u \Big|_{x=0} = \mu_1(t), & t \geq 0 \\ \alpha_2 \frac{\partial u}{\partial x} + \beta_2 u \Big|_{x=0} = \mu_2(t), & t \geq 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \\ u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

这里我们可以通过引入辅助函数使边值问题齐次化，我们通过构造函数

$$U(x, t) = A(x)\mu_1(t) + B(x)\mu_2(t)$$

并保证

$$\begin{cases} \alpha_1 A'(0) + \beta_1 A(0) = 1 \\ \alpha_1 B'(0) + \beta_1 B(0) = 0 \\ \alpha_2 A'(l) + \beta_1 A(l) = 0 \\ \alpha_2 B'(l) + \beta_2 B(l) = 1 \end{cases}$$

此时我们只要取 $u(x, t) = U(x, t) + v(x, t)$ 我们就可以验证 $\alpha_1 \frac{\partial v}{\partial x} + \beta_1 v \Big|_{x=0} = 0$, $\alpha_1 \frac{\partial v}{\partial x} + \beta_1 v \Big|_{x=0} = 0$ 于是就可以按照正常的计算方法求解了。

例 2.6.3. 设 $u(x, t)$ 适合定解问题

$$\begin{cases} Lu = f(x, t), & (x, t) \in Q \\ -\frac{\partial u}{\partial x} + \alpha u|_{x=0} = \mu_1(t), & t \geq 0 \\ \frac{\partial u}{\partial x} + \beta u|_{x=l} = \mu_2(t), & t \geq 0 \\ u|_{t=0} = \varphi(x), \quad u_t|_{t=0} = \psi(x), & 0 \leq x \leq l \end{cases}$$

试引进辅助函数, 把边条件齐次化, 设 (a) $\alpha > 0, \beta > 0$. (b) $\alpha = \beta = 0$

解答 我们构造辅助函数 $U(x, t) = A(x)\mu_1(t) + B(x)\mu_2(t)$, 并令其满足

$$\begin{cases} -A'(0) + \alpha A(0) = 1 \\ -B'(0) + \alpha B(0) = 0 \\ A'(l) + \beta A(l) = 0 \\ B'(l) + \beta B(l) = 1 \end{cases}$$

$$\begin{aligned} \text{此时若取 } u(x, t) = U(x, t) + v(x, t). \text{ 则 } -\frac{\partial u}{\partial x} + \alpha u \Big|_{x=0} &= -\frac{\partial U}{\partial x} - \frac{\partial v}{\partial x} + \alpha U + \alpha v \Big|_{x=0} = -(A'(0)\mu_1(t) + B'(0)\mu_2(t)) - \frac{\partial v}{\partial x} \Big|_{x=0} + \\ \alpha A(0)\mu_1(t) + \alpha B(0)\mu_2(t) + \alpha v(0, t) &= \mu_1(t) - \frac{\partial v}{\partial x} \Big|_{x=0} + \alpha v(0, t) = \mu_1(t) \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial x} + \beta u \Big|_{x=l} &= \frac{\partial U}{\partial x} + \frac{\partial v}{\partial x} + \beta U + \beta v \Big|_{x=l} = (A'(l)\mu_1(t) + B'(l)\mu_2(t)) + \frac{\partial v}{\partial x} \Big|_{x=l} + \beta A(l)\mu_1(t) + \beta B(l)\mu_2(t) \\ &= \mu_2(t) + \frac{\partial v}{\partial x} \Big|_{x=l} + \beta v(l, t) = \mu_2, \end{aligned}$$

此时, 我们有 v 满足

$$\begin{cases} -\frac{\partial v}{\partial x} + \alpha v \Big|_{x=0} = 0, & t \geq 0 \\ \frac{\partial v}{\partial x} + \beta v \Big|_{x=l} = 0, & t \geq 0. \end{cases}$$

(a) 当 $\alpha > 0, \beta > 0$ 时, 结合

$$\begin{cases} -A'(0) + \alpha A(0) = 1 \\ A'(l) + \beta A(l) = 0 \end{cases} \quad \begin{cases} -B'(0) + \alpha B(0) = 0 \\ B'(l) + \beta B(l) = 1 \end{cases}$$

我们取 $A(x) = a_1x + b_1, B(x) = a_2x + b_2$, 则有

$$\begin{cases} -a_1 + \alpha b_1 = 1 \\ a_1 + \beta a_1 l + b_1 \beta = 0 \end{cases} \quad \begin{cases} -a_2 + \alpha b_2 = 0 \\ a_2 + \beta a_2 l + b_2 \beta = 1 \end{cases} \Rightarrow \begin{cases} a_2 = \frac{\beta}{\alpha + \beta + \alpha \beta l} \\ b_2 = \frac{1 + \beta l}{\alpha + \beta + \alpha \beta l} \end{cases} \quad \begin{cases} a_1 = \frac{\alpha}{\alpha + \beta + \alpha \beta l} \\ b_1 = \frac{1}{\alpha + \beta + \alpha \beta l} \end{cases}$$

$$\text{则 } U(x, t) = \left(-\frac{\alpha}{\alpha + \beta + \alpha \beta l}x + \frac{1 + \beta l}{\alpha + \beta + \alpha \beta l}\right)\mu_1(t) + \left(\frac{\alpha}{\alpha + \beta + \alpha \beta l}x + \frac{1}{\alpha + \beta + \alpha \beta l}\right)\mu_2(t)$$

(b) 当 $\alpha = \beta = 0$ 时

$$\begin{cases} -A'(0) = 1 \\ A'(l) = 0 \end{cases} \quad \begin{cases} -B'(0) = 0 \\ B'(l) = 1 \end{cases} \quad \text{取} \quad \begin{cases} A(x) = -\frac{2l}{\pi} \sin \frac{\pi x}{2l} \\ B(x) = -\frac{2l}{\pi} \cos \frac{\pi x}{2l} \end{cases}$$

$$\text{则有 } U(x, t) = -\frac{2l}{\pi} \sin \frac{\pi x}{2l} \mu_1(t) - \frac{2l}{\pi} \cos \frac{\pi x}{2l} \mu_2(t)$$

□

另外, 我们也会遇到这样的非齐次问题:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t), & Q = \{0 < x < l, t > 0\} \\ u(0, t) = u(l, t) = 0, & t > 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \\ u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

我们只需要按照下面的求解方式求解就可以了:

1. 首先把变量分离形式的解带入相应的齐次方程和其次边界条件, 以得到相应的 Sturm-Liowville 问题, 并求出全部特征值

$$\left\{\left(\frac{n\pi}{l}\right)^2\right\} \text{ 和特征函数 } \left\{\sin \frac{n\pi}{l}x\right\}$$

2. 把解 $u(x, t)$ 、方程的非齐次项 $f(x, t)$ 和初值 $\varphi(x)$, $\psi(x)$ 都按照特征函数系展开:

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi}{l} x$$

$$f(x, t) = \sum_{n=1}^{\infty} f_n(t) \sin \frac{n\pi}{l} x$$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi}{l} x, \quad \psi(x) = \sum_{n=1}^{\infty} \psi_n \sin \frac{n\pi}{l} x$$

其中 $\varphi_n, \psi_n, f_n(t)$ 分别是 $\varphi(x), \psi(x), f(x, t)$ 的傅里叶系数, $T_n(t)$ 是 $u(x, t)$ 的傅里叶系数, 当然 T_n 暂时还是未知的。

将他们带入方程和初值条件, 我们结合上面的重要定理, 就可以得到关于 $T_n(t)$ 的 ODE 问题:

$$\begin{cases} T_n''(t) + \left(\frac{an\pi}{l}\right)^2 T_n(t) = f_n(t) \\ T_n(0) = \varphi_n \\ T_n'(0) = \psi_n \end{cases}$$

3. 求解上述初值问题, 最终得到

$$T_n(t) = \varphi_n \cos \frac{an\pi}{l} t + \frac{l}{an\pi} \psi_n \sin \frac{an\pi}{l} t + \frac{l}{an\pi} \int_0^t f_n(\tau) \sin \frac{an\pi}{l} (t - \tau) d\tau$$

进一步展开化简, 代入就可以得到方程的解。

这里我们给出注记, 所有的上述问题都是标准形式, 其他形式可能存在 u 的一阶偏导等项, 其本质上是变换了 ODE 的初值问题, 但整体的求解思路是大致相同的。

另外, 我们可能需要考虑扰动问题:

$$\begin{cases} u_{tt} - a^2 u_{xx} = A(x) \sin \omega t, & (x, t) \in Q \\ u(0, t) = u(l, t), & t \geq 0 \\ u(x, 0) = u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

与前面的混合问题相同，我们都可以解出这样的函数结果：

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi x}{l}\right)$$

同时我们对 $A(x)$ 进行傅里叶展开：

$$A(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right)$$

其中

$$A_n = \frac{2}{l} \int_0^l A(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

则如果我们将所有的结果代入 $\square u = A(x) \sin \omega t$ ，就有

$$\sum_{n=1}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] \sin\left(\frac{n\pi x}{l}\right) = \sum_{n=1}^{\infty} A_n \sin \omega t \sin\left(\frac{n\pi x}{l}\right)$$

结合正交性，我们可以得到

$$T_n''(t) + \omega_n^2 T_n(t) = A_n \sin \omega t$$

其中 $\omega_n = a\sqrt{\lambda_n}$ 。下面我们结合初始条件不难列出 ODE 问题：

$$\begin{cases} T_n'' + \omega_n^2 T_n = A_n \sin \omega t \\ T_n(0) = T_n'(0) = 0 \end{cases}$$

不难计算方程的齐次解为

$$T_n^{(h)}(t) = -\frac{A_n \omega}{(\omega_n^2 - \omega^2) \omega_n} \sin \omega_n t$$

下面我们需要分两种情况讨论：

1. $\forall n \in \mathbb{N}_+, \omega \neq \omega_n$

此时我们有 ODE 的特解格式为 $T_n^{(p)}(t) = P_n \sin \omega t$ ，将其带入方程就有 $T_n^{(p)}(t) = \frac{A_n}{\omega_n^2 - \omega^2} \sin \omega t$ 。因此我们就有

$$T_n(t) = \frac{A_n}{\omega_n^2 - \omega^2} \left[\sin \omega t - \frac{\omega}{\omega_n} \sin \omega_n t \right]$$

2. $\exists k \in \mathbb{N}_+$, s.t. $\omega = \omega_k$, 则当 $n \neq k$ 时 $T_n(t)$ 和上面的表达方式是一样的, 而当 $n = k$ 时, 其特解形式为

$$T_k^{(p)}(t) = -\frac{A_k}{2\omega_k} t \cos \omega_k t$$

总的来说, 方程的解就是

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$$

例 2.6.4. 用分离变量法求解下列定解问题 (1)

$$\begin{cases} Lu = -2b \frac{\partial u}{\partial t} + g, & (x, t) \in Q \\ u|_{x=0} = u|_{x=l} = 0, & t \geq 0 \\ u|_{t=0} = u_t|_{t=0} = 0, & 0 \leq x \leq l \end{cases} \quad (2) \begin{cases} \frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, & (x, t) \in Q \\ u_x|_{x=0} = A \sin \omega t, & u|_{x=l} = 0, t \geq 0 \\ u|_{t=0} = 1, u_t|_{t=0} = 0, & 0 \leq x \leq l \end{cases}$$

解答

补充知识 1. 特殊 $T_n(t)$ ODE 求解

对问题

$$\begin{cases} T_n''(t) + 2bT_n'(t) + cT_n(t) = f_n(t) \\ T_n(0) = \varphi_n \\ T_n'(0) = \psi_n \end{cases}$$

(1) 当 $c > b^2$ 时, $r_{1,2} = -b \pm i\sqrt{c-b^2}$, 则对应齐次解为取 $\alpha = \sqrt{c-b^2}$ $T_n^{(h)}(t) = e^{-bt}(A \cos \alpha t + B \sin \alpha t)$ 特解我们写作 $T_n^{(p)}(t) = \int_0^t \frac{e^{-b(t-\tau)}}{\alpha} \sin(\alpha(t-\tau)) f_n(\tau) d\tau$, 此时将 $T_n(0) = \varphi_n$, $T_n'(0) = \psi_n$ 代入, 有

$$T_n(t) = e^{-bt}(\varphi \cos \alpha t + \frac{b\varphi + \psi}{\alpha} \sin \alpha t) + \int_0^t \frac{e^{-b(t-\tau)}}{\alpha} \sin(\alpha(t-\tau)) f_n(\tau) d\tau$$

(2) 当 $c = b^2$ 时 $T_n^{(h)}(t) = (A + Bt)e^{-bt}$, 通解为

$$T(t) = e^{-bt}[\varphi(1 + bt) + \psi t] + \int_0^t (t - \tau) e^{-b(t-\tau)} f_n(\tau) d\tau$$

(3) 当 $c < b^2$ 时, 取 $\beta = \sqrt{b^2 - c}$ $r_{1,2} = -b \pm \sqrt{b^2 - c}$, 对应齐次解为 $T_n^{(h)}(t) = e^{-bt}(A \cosh \beta t + B \sinh \beta t)$ 特解我们写作 $T_n^{(p)}(t) = \int_0^t \frac{e^{-b(t-\tau)}}{\beta} \sinh(\beta(t-\tau)) f_n(\tau) d\tau$. 将初值代入, 有

$$T_n(t) = e^{-bt}(\varphi \cosh \beta t + \frac{\psi}{\beta} \sinh \beta t) + \psi \frac{\sinh \beta t}{\beta} + \int_0^t \frac{e^{-b(t-\tau)}}{\beta} \sinh(\beta(t-\tau)) f_n(\tau) d\tau$$

补充知识 2. 非齐次线性方程的求解 我们只需要研究

$$T_n''(t) + 2bT_n'(t) + cT_n(t) = A \sin \omega t$$

由特征方程 $r^2 + 2br + c = 0$, 我们解出特征值 λ_1, λ_2 , 得到对应通解为

$$T_n^{(h)}(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

我们分两种情况讨论. 1) λ_1, λ_2 均不等于 ω , 取 $T_n^{(p)}(t) = C_1 \sin \omega t + C_2 \cos \omega t$. 代入求解 C_1, C_2 , 并将初值条件代入. 2) 存在某个 λ_i 有 $\omega = \lambda_i$, 取 $T_n^{(p)}(t) = t(C_1 \sin \omega t + C_2 \cos \omega t)$ 代入求解 C_1, C_2 , 并将初值条件代入. 3) $\omega = \lambda_1 = \lambda_2$, 取 $T_n^{(p)}(t) = t^2(C_1 \sin \omega t + C_2 \cos \omega t)$ 代入求解 C_1, C_2 , 并将初值条件代入.

(1) 解仍设 $u(x, t) = X(x)T(t)$, 则结合 $Lu = -2b \frac{\partial u}{\partial t} + g$,

$$T''(t)X(x) + 2bT'(t)X(x) - a^2X''(x)T(t) = g(t)$$

设 $\frac{X''(x)}{X(x)} = -\lambda$, 则 $X_n(x) = \sin \frac{n\pi}{l}x$, 记 $\omega_n = \sqrt{\lambda_n} = \frac{n\pi}{l}$, $n = 1, 2, \dots$ 则

$$\begin{cases} T_n''(t) + 2bT_n'(t) + a^2\omega_n^2 T_n(t) = g_n(t) \\ T_n(0) = 0 \\ T_n'(0) = 0 \end{cases} \quad \text{若记 } u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi}{l}x \text{ .. 则来解决.}$$

(a) 当 $b^2 > a^2\omega_n^2$ 时. 记 $\alpha_n = \sqrt{b^2 - a^2\omega_n^2}$

$$T_n(t) = e^{-bt}(\varphi_n \cosh \alpha_n t + \frac{b\varphi_n + \psi_n}{\alpha_n} \sinh \alpha_n t) + \int_0^t \frac{e^{-b(t-\tau)}}{\alpha_n} \sinh(\alpha_n(t-\tau)) f_n(\tau) d\tau$$

(b) 当 $b^2 = a^2\omega_n^2$ 时 $T_n(t) = e^{-bt}[\varphi_n(1+bt) + \psi_nt] + \int_0^t (t-\tau)e^{-b(t-\tau)}f_n(\tau)d\tau.$

(c) 当 $b^2 < a^2\omega_n^2$ 时记 $\beta_n = \sqrt{a^2\omega_n^2 - b^2}$

$$T_n(t) = e^{-bt}(\varphi_n \cos \alpha_n t + \frac{b\varphi_n + \psi_n}{\alpha_n} \sin \alpha_n t) + \int_0^t \frac{e^{-b(t-\tau)}}{\alpha_n} \sin(\alpha_n(t-\tau))f_n(\tau)d\tau$$

(2) 取 $U(x, t) = A(x-l) \sin \omega t$, 并令 $v(x, t) = u(x, t) - U(x, t)$, 此时

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0 \quad \text{即} \quad \frac{\partial^2 v}{\partial t^2} - \frac{\partial^2 v}{\partial x^2} = A\omega^2(l-x) \sin \omega t. \text{我们将问题转化为}$$

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} - \frac{\partial^2 v}{\partial x^2} = A\omega^2(l-x) \sin \omega t, & x \in Q \\ v_x|_{x=0} = v_x|_{x=l} = 0, & t \geq 0 \\ v|_{t=0} = 1 & v_t|_{t=0} = A\omega(l-x), \quad 0 \leq x \leq l \end{cases} \quad \text{解得 } X_n(x) = \cos \omega_n x. \quad \omega_n = \frac{(n + \frac{1}{2})\pi}{l} \quad n = 1, 2, \dots \quad \sqrt{\lambda_n} = \omega_n$$

此时取 $\|X_n\|^2 = \int_0^l \cos^2 \omega_n x dx = \frac{l}{2} + \frac{\sin 2\omega_n l}{4\omega_n}$, 则有令

$$a_n = \frac{1}{\|X_n\|^2} \int_0^l (l-x) \cos \omega_n x dx \quad c_n = \frac{1}{\|X_n\|^2} \int_0^l \cos \omega_n x dx$$

$T_n(t)$ 应满足如下条件

$$\begin{cases} T_n''(t) + a^2\omega_n T_n(t) = f_n(t) = A\omega^2 a_n \sin \omega t. \\ T_n(0) = c_n \\ T_n'(0) = A\omega a_n. \end{cases}$$

方程的通解为 $T_n^{(h)}(t) = C_1 \sin a\sqrt{\omega_n}t + C_2 \cos a\sqrt{\omega_n}t.$

1) 当 $a\sqrt{\omega_n} = \omega$ 时, 取 $T_n^{(p)}(t) = t(M \sin \omega t + N \cos \omega t)$, 代入求解, 有 $T_n^{(p)}(t) = -\frac{A\omega a_n}{2}t \cos \omega t \quad T_n(t) = C_1 \cos \omega t + \frac{3A}{2}a_n \sin \omega t - \frac{A\omega a_n}{2}t \cos \omega t$

2) 当 $a\sqrt{\omega_n} \neq \omega$ 时, 取 $T_n^{(p)}(t) = M \sin \omega t + N \cos \omega t$, 代入求解, 有

$$T_n^{(p)}(t) = \frac{A\omega^2 a_n}{a^2\omega_n - \omega^2} \sin \omega t.$$

结合 $T_n(0) = c_n$, $T'_n(0) = A\omega a_n$, 有 $C_2 = c_n$, $C_1 = \frac{A\omega a_n}{a\omega_n} - \frac{a^2\omega_n - 2\omega^2}{a^2\omega_n - \omega^2}$ 则对有关 n 的情形 $u(x, t) = \sum_{n \neq n_1} T_n(t)X_n(t) + T_{n_1}(t)X_{n_1}(t)$.

代入即可

□

2.6.1 分离变量法的推广与应用

有了分离变量法的证明思路, 我们就可以解决其他的一些方程, 比如下面的例子: 取区域 $D: [0, 2] \times [0, 1] \subset \mathbb{R}^2$, 对方程

$$\begin{cases} \Delta u + \lambda u = 0 & , (x, y) \in D \\ u = 0 & , (x, y) \in \partial D \end{cases}$$

与混合问题相类似, 我们取

$$u(x, y) = X(x)Y(y)$$

结合条件, 我们不难推出

$$\begin{cases} \frac{X''}{X} + \frac{Y''}{Y} + \lambda = 0 \\ X(0) = X(2) = Y(0) = Y(1) \end{cases}$$

此时如果我们取 $\frac{X''}{X} = -\alpha$, $\frac{Y''}{Y} = -\beta$, 若 $u = X(x)Y(y)$ 是方程的解, 我们就有 $\alpha + \beta = \lambda$, 于是我们就得到了两个 ODE 问题:

$$\begin{cases} X'' + \alpha X = 0 \\ X(0) = X(2) = 0 \end{cases}, \quad \begin{cases} Y'' + \beta Y = 0 \\ Y(0) = Y(1) = 0 \end{cases}$$

通过简单的计算我们不难解得 $\mu_m = \frac{m\pi}{2}$, $m = 1, 2, \dots$, $X_m(x) = \cos \frac{m\pi}{2}x$, $Y_n(y) = \cos n\pi y$, $n = 1, 2, \dots$, 结合 $\alpha + \beta = \lambda$, 我们有

$$\left(\frac{m\pi}{2}\right)^2 + (n\pi)^2 = \lambda$$

则特征函数为 $u_{mn}(x, y) = \sin\left(\frac{m\pi x}{2}\right) \sin n\pi y$, 则方程的解为

$$u(x, y) = \sum_{(\frac{m\pi}{2})^2 + (n\pi)^2 = \lambda} A_{mn} \sin \frac{m\pi x}{2} \sin n\pi y$$

其中 A_{mn} 需要由其他初始条件确定。

例 2.6.5. 用分离变量法求解问题

$$\begin{cases} \Delta u = 0, [0, a] \times [0, b] \\ u(x, 0) = u(x, b) = 0 \\ u(0, y) = \varphi(y), u(a, y) = 0 \end{cases}$$

解答

我们将 $u(x, y) = X(x)Y(y)$ 代入, 有

$$X''(x)Y(y) + X(x)Y''(y) = 0,$$

若设 $\frac{Y''(y)}{Y(y)} = -\lambda$ 。结合 $u(x, 0) = u(x, b) = 0$, 我们有 $Y_n(y) = C \sin \frac{n\pi}{b}y$ 。此时我们有

$$u(x, y) = \sum_{n=1}^{\infty} X_n(x) \sin \frac{n\pi}{b}y$$

则有

$$\begin{cases} \sum_{n=1}^{\infty} (X_n''(x) - (\frac{n\pi}{b})^2 X_n(x)) \sin \frac{n\pi}{b}y = 0 \\ \sum_{n=1}^{\infty} X_n(0) \sin \frac{n\pi}{b}y = \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi}{b}y, \\ \sum_{n=1}^{\infty} X_n(a) \sin \frac{n\pi}{b}y = 0 \end{cases} \quad \begin{aligned} &\text{其中 } \varphi_n = \frac{2}{b} \int_0^b \varphi(x) \sin \frac{n\pi}{b}x dx. \\ &\text{则有 } X_n(x) = \varphi_n \frac{\sinh(\frac{n\pi}{b}(a-x))}{\sinh(\frac{n\pi}{b}a)} \end{aligned}$$

则

$$u(x, y) = \sum_{n=1}^{\infty} \frac{2}{b} \int_0^b \varphi(x) \sin \frac{n\pi}{b}x dx \frac{\sinh(\frac{n\pi}{b}(a-x))}{\sinh(\frac{n\pi}{b}a)} \sin \frac{n\pi}{b}y$$

□

按照同样的思路, 我们就可以计算热传导方程:

例 2.6.6. 热方程的分离变量法
$$\begin{cases} u_t - a^2 u_{xx} = f(x, t), & (0, l) \times (0, +\infty) \\ u(x, 0) = \varphi(x) \\ u_x(0, t) = u_x(l, t) = 0 \end{cases}$$

解答 题中傅里叶级数应当都是从 0 开始, 请各位注意 取 $u(x, t) = T(t)X(x)$. 则若取 $\frac{X'(x)}{X(x)} = -\lambda$. 结合 $u_x(0, t) = u_x(l, t) = 0$, 则有

$$X_n(x) = \cos \frac{n\pi}{l} x. \quad n = 1, 2, \dots$$

此时将 u 写作 $u(x, t) = \sum_{n=1}^{\infty} T_n(t) \cos \frac{n\pi}{l} x$, 代入方程, 有

$$\begin{cases} \sum_{n=0}^{\infty} (T'_n(t) + a^2 (\frac{n\pi}{l})^2 T_n(t)) \cos \frac{n\pi}{l} x = \sum_{n=0}^{\infty} f_n \cos \frac{n\pi}{l} x & \text{其中 } f_n(t) = \frac{2}{l} \int_0^l f(x, t) \cos \frac{n\pi}{l} x dx \\ \sum_{n=0}^{\infty} T_n(0) \cos \frac{n\pi}{l} x = \sum_{n=0}^{\infty} \varphi_n \cos \frac{n\pi}{l} x & \varphi_n = \frac{2}{l} \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx \end{cases}$$

则有

$$\begin{cases} T'_n(t) + (a \frac{n\pi}{l})^2 T_n(t) = f_n(t) \\ T_n(0) = \varphi_n \end{cases} \quad e^{(a \frac{n\pi}{l})^2 t} T'_n(t) + e^{(a \frac{n\pi}{l})^2 t} (a \frac{n\pi}{l})^2 T_n(t) = e^{(a \frac{n\pi}{l})^2 t} f_n(t)$$

$$(e^{(a \frac{n\pi}{l})^2 t} T_n(t))' = T_n(0) + \int_0^t (e^{(a \frac{n\pi}{l})^2 \tau} T_n(\tau))' d\tau = \varphi_n + \int_0^t e^{(a \frac{n\pi}{l})^2 \tau} f_n(\tau) d\tau.$$

$$\text{则 } T_n(t) = \exp\left(-\left(a \frac{n\pi}{l}\right)^2 t\right) \varphi_n + \exp\left(-\left(a \frac{n\pi}{l}\right)^2 t\right) \int_0^t \exp\left(\left(a \frac{n\pi}{l}\right)^2 \tau\right) f_n(\tau) d\tau$$

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) \cos \frac{n\pi}{l} x$$

□

2.6.2 混合问题形式解的适定性

通过分离变量法, 我们得到混合问题的形式解为:

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{L} t + B_n \sin \frac{n\pi a}{L} t \right) \sin \frac{n\pi x}{L} \quad (2)$$

其中, 系数 A_n 和 B_n 由初始条件决定:

$$A_n = \frac{2}{L} \int_0^L \varphi(x) \sin \frac{n\pi x}{L} dx, \quad n = 1, 2, 3, \dots \quad (3)$$

$$B_n = \frac{2}{n\pi a} \int_0^L \psi(x) \sin \frac{n\pi x}{L} dx, \quad n = 1, 2, 3, \dots \quad (4)$$

我们需要探究, 在何种条件下, 此形式解即为该定解问题的经典解, 我们需要确定, 在什么样的条件下, 形式解 $u(x, t)$ 是问题的经典解。即:

1. $u(x, t) \in C^2([0, L] \times (0, \infty))$, 即 u 在区域内部具有连续的二阶偏导数。

2. 在区域 $0 < x < L, t > 0$ 内满足波动方程: $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$ 。

3. 满足边界条件: 对任意 $t > 0$, 有 $u(0, t) = u(L, t) = 0$ 。

4. 满足初始条件: $\lim_{t \rightarrow 0^+} u(x, t) = \varphi(x)$ 且 $\lim_{t \rightarrow 0^+} \frac{\partial u}{\partial t}(x, t) = \psi(x)$ 。

证明的核心在于通过分部积分和兼容性条件, 推导出傅里叶系数 A_n, B_n 的衰减速率, 进而控制级数及其导数的收敛性。

引理 2.6.1. 如果初始函数 $\varphi(x)$ 和 $\psi(x)$ 满足以下充分条件:

1. $\varphi(x) \in C^3[0, L], \psi(x) \in C^2[0, L]$ 。(即 φ 三阶连续可导, ψ 二阶连续可导)

2. 满足兼容性条件:

$$\varphi(0) = \varphi(L) = 0, \quad \varphi''(0) = \varphi''(L) = 0 \quad (5)$$

$$\psi(0) = \psi(L) = 0 \quad (6)$$

则系数 A_n, B_n 满足以下衰减估计:

$$A_n = O\left(\frac{1}{n^2}\right) \quad (7)$$

$$B_n = O\left(\frac{1}{n^3}\right) \quad (8)$$

解答 令 $k = \frac{n\pi}{L}$, 我们按照这样的步骤证明:

1. 证明 $A_n = O(1/n^2)$

由 (3), $A_n = \frac{2}{L} \int_0^L \varphi(x) \sin(kx) dx$. 对积分进行两次分部积分:

$$\begin{aligned} \int_0^L \varphi(x) \sin(kx) dx &= \left[-\frac{\varphi(x) \cos(kx)}{k} \right]_0^L + \frac{1}{k} \int_0^L \varphi'(x) \cos(kx) dx \quad (\text{利用 } \varphi(0) = \varphi(L) = 0) \\ &= \frac{1}{k} \int_0^L \varphi'(x) \cos(kx) dx \\ &= \frac{1}{k} \left[\frac{\varphi'(x) \sin(kx)}{k} \right]_0^L - \frac{1}{k^2} \int_0^L \varphi''(x) \sin(kx) dx \quad (\text{第一项再次为 } 0) \\ &= -\frac{1}{k^2} \int_0^L \varphi''(x) \sin(kx) dx \end{aligned}$$

由于 $\varphi''(x)$ 连续, 最后的积分有界, 设其绝对值上界为 M_1 . 因此,

$$|A_n| \leq \frac{2}{L} \cdot \frac{1}{k^2} \cdot M_1 = \frac{2M_1}{L} \cdot \frac{L^2}{n^2\pi^2} = \frac{C_1}{n^2}$$

即 $A_n = O(1/n^2)$.

2. 证明 $B_n = O(1/n^3)$

由 (4), $B_n = \frac{2}{n\pi a} \int_0^L \psi(x) \sin(kx) dx$. 对积分进行一次分部积分 (利用 $\psi(0) = \psi(L) = 0$):

$$\int_0^L \psi(x) \sin(kx) dx = \frac{1}{k} \int_0^L \psi'(x) \cos(kx) dx$$

由于 $\psi'(x)$ 连续, 该积分有界, 设其绝对值上界为 M_2 。因此,

$$|B_n| \leq \frac{2}{n\pi a} \cdot \frac{1}{k} \cdot M_2 = \frac{2}{n\pi a} \cdot \frac{L}{n\pi} \cdot M_2 = \frac{C_2}{n^3}$$

即 $B_n = O(1/n^3)$ 。

□

定理 2.6.2. 如果初始函数 $\varphi(x)$ 和 $\psi(x)$ 满足以下充分条件:

1. $\varphi(x) \in C^3[0, L]$, $\psi(x) \in C^2[0, L]$ 。(即 φ 三阶连续可导, ψ 二阶连续可导)

2. 满足兼容性条件:

$$\varphi(0) = \varphi(L) = 0, \quad \varphi''(0) = \varphi''(L) = 0 \quad (9)$$

$$\psi(0) = \psi(L) = 0 \quad (10)$$

那么, 形式解 $u(x, t)$ 是该波动方程混合问题的经典解。

解答 结合推导

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos \frac{an\pi}{L}t + B_n \sin \frac{an\pi}{L}t) \sin \frac{n\pi}{L}x.$$

结合 $A_n = O(\frac{1}{n^4})$ $B_n = O(\frac{1}{n^3})$ $|\cos \frac{an\pi}{L}t| \leq 1$ $|\sin \frac{an\pi}{L}t| \leq 1$, $\sum \frac{1}{n^4} < +\infty$ $\sum \frac{1}{n^3} < \infty$

则 $\sum_{n=1}^{\infty} A_n \cos \frac{an\pi}{L}t + B_n \sin \frac{an\pi}{L}t$ 收敛, 由 $|\sin \frac{n\pi}{L}x| \leq 1$, 有级数一致收敛, $u(x, t)$ 连续.

则我们来验证方程

$$1) \frac{\partial^2 u}{\partial t^2} = - \sum_{n=1}^{\infty} (A_n (\frac{an\pi}{L})^2 \cos \frac{an\pi}{L}t + B_n (\frac{an\pi}{L})^2 \sin \frac{an\pi}{L}t) \sin \frac{n\pi}{L}x$$

$$\frac{\partial^2 u}{\partial x^2} = - \sum_{n=1}^{\infty} (A_n \cos \frac{an\pi}{L}t + B_n \sin \frac{an\pi}{L}t) (\frac{n\pi}{L})^2 \sin \frac{n\pi}{L}x. \quad , \text{故有 } u_{tt} = a^2 u_{xx}$$

$$2) u(0, t) = 0 = u(L, t) \quad (\sin 0 = \sin n\pi = 0)$$

$$3) u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}x = \varphi(x),$$

$$4) \frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} -A_n \frac{an\pi}{L} \sin \frac{an\pi}{L} t + B_n \frac{an\pi}{L} \cos \frac{an\pi}{L} t) \sin \frac{n\pi}{L} x$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} \frac{an\pi}{L} B_n \sin \frac{n\pi}{L} x = \sum_{n=1}^{\infty} \psi_n \sin \frac{n\pi}{L} x = \psi(x).$$

则有 $u(x, t)$ 即为原方程的解

□

针对混合问题我们同样有能量不等式的相关知识:

定理 2.6.3. 若 $u(x, t) \in C^1(\bar{Q}_\tau) \cap C^2(Q_\tau)$ 是波动方程混合问题的解, 其中 $Q_\tau = (0, l) \times (0, T)$ 则存在只依赖于 T 的常数 M , 使得

$$\int_0^l [u_t^2(x, \tau) + a^2 u_x^2(x, \tau)] dx \leq M \left[\int_0^l (\psi^2 + a^2 \varphi_x^2) dx + \iint_{Q_\tau} f^2 dx dt \right]$$

$$\iint_{Q_\tau} (u_t^2 + a^2 u_x^2) dx dt \leq M \left[\int_0^l (\psi^2 + a^2 \varphi_x^2) dx + \iint_{Q_\tau} f^2 dx dt \right]$$

其中 $0 \leq \tau \leq T$, 且在第一个不等式中, 当 $f = 0$ 时, $M = 1$, 不等号由等号代替。

2.6.3 混合问题的广义解

在上面问题讨论的基础之上, 我们希望进一步扩大解的范围, 即提出广义解的概念:

定义 2.6.1. $u(x, t) \in C(\bar{Q}_\tau)$ 称为混合问题的广义解, 如果对于任意的 $\zeta(x, t) \in \mathcal{D} = \{\zeta(x, t) \in C^2(\bar{Q}_\tau) | \zeta(x, T) = \zeta_t(x, T) = \zeta(0, t) = \zeta(l, t) = 0\}$, 积分等式

$$\iint_{Q_\tau} u(\zeta_{tt} - a^2 \zeta_{xx}) dx dt + \int_0^l \varphi \zeta_t(x, 0) dx - \int_0^l \psi \zeta(x, 0) dx = 0$$

总成立。

针对广义解我们要求其满足两个条件:

1. 古典解一定是广义解
2. 广义解是唯一的, 且按某种度量连续依赖于定解资料。

下面我们来讨论广义解的相容性。

定理 2.6.4. 混合问题的广义解必唯一，且一定是稳定的。

解答 假设 $u_1(x, t), u_2(x, t)$ 均为混合问题的解，且满足

$$\int_{Q_T} u_i(\zeta_{tt} - a^2 \zeta_{xx}) dx dt + \int_0^l \varphi \zeta_t(x, 0) dx - \int_0^l \psi \zeta(x, 0) dx = 0 \quad i = 1, 2$$

对两者作差，就有

$$\int_{Q_T} (u_1 - u_2) \square \zeta dx dt = 0$$

我们考虑这样的问题

$$\begin{cases} \square \zeta = g(x, t) \\ \zeta(l, t) = \zeta(0, t) = 0 \\ \zeta(x, T) = \zeta_t(x, T) = 0 \end{cases}, \quad 0 \leq t \leq T, 0 \leq x \leq l$$

此时取 $\tau = T - t$ $\bar{\zeta}(x, \tau) = \zeta(x, t)$ ，则我们有

$$\begin{cases} \square \bar{\zeta} = g(x, T - \tau) \\ \bar{\zeta}(0, \tau) = \bar{\zeta}(l, \tau) = 0 \\ \bar{\zeta}(x, 0) = \bar{\zeta}_\tau(x, 0) = 0, \quad 0 \leq x \leq l \end{cases}$$

满足混合条件. 结合 $g \in C_0^\infty(Q_T)$ ，共满足

$$(\square \bar{\zeta})|_{t=0} = g(x, T) = 0 \quad (\square \bar{\zeta})|_{x=0} = g(0, T) = 0 \quad (\square \bar{\zeta})|_{x=l} = g(l, T) = 0$$

则 $\exists \bar{\zeta}(x, \tau) \in C^2(\bar{Q}_T)$ st $\zeta(x, t) = \bar{\zeta}(x, T - t)$ ，满足 $\square \zeta = g(x, t)$.

因此结合 $g \in C_0^\infty(Q_T)$ 与第一章引理，由

$$\int_{Q_T} (u_1 - u_2) g dx dt = 0 \quad \text{我们有 } u_1 = u_2 \text{ 成立}$$

□

定理 2.6.5. 不考 若 $\varphi(x) \in C[0, l]$, $\varphi(0) = \varphi(l) = 0$, $\varphi'(x), \psi(x)$ 在 $[0, l]$ 上分别除了有限个点以外连续, 而在这些点两侧存在左右极限, 则定解问题存在唯一的广义解, 且他仍可表示成如下形式:

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin \frac{an\pi}{l} t + B_n \cos \frac{an\pi}{l} t \right) \sin \frac{n\pi}{l} x$$

解答 (教材的原证明是一坨狗屎, 请避雷!!!)

结合前面的讨论, 我们知道, 可以将 $\varphi(x), \psi(x)$ 写为如下形式

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi x}{L} \quad \psi(x) = \sum_{n=1}^{\infty} \psi_n \sin \frac{n\pi x}{L}$$

此时我们取 $S_N(x) = \sum_{n=1}^N \varphi_n \sin \frac{n\pi x}{L}$, $T_N(x) = \sum_{n=1}^N \psi_n \sin \frac{n\pi x}{L}$, 则结合 Parseval 定理, 我们有

$$\|\varphi(x) - S_N(x)\|_{L^2} \rightarrow 0 \quad \|\psi(x) - T_N(x)\|_{L^2} \rightarrow 0$$

此时, 我们来研究这样的混合问题

$$\begin{cases} u_{N,tt} - a^2 u_{N,xx} = 0, & x \in Q_t \quad (*) \\ u_N(0, t) = u_N(L, t) = 0, & 0 \leq t \leq T \\ u_N(x, 0) = S_N(x), & 0 \leq x \leq L \\ u_{N,t}(x, 0) = T_N(x), & 0 \leq x \leq L \end{cases}$$

其解为 $u_N(x, t) = \sum_{n=1}^N (A_n \cos \frac{n\pi}{L} t + B_n \sin \frac{n\pi}{L} t) \sin \frac{n\pi}{L} x$. (记住即可) 取 $w(x, t) = u(x, t) - u_N(x, t) = \sum_{n=N+1}^{\infty} (A_n \cos \frac{n\pi}{L} t + B_n \sin \frac{n\pi}{L} t) \sin \frac{n\pi}{L} x$

我们有 $|w_N| \leq \sum_{n=N+1}^{\infty} |A_n| + |B_n|$, 结合前面的引理 $A_n = O(\frac{1}{n^4})$ $B_n = O(\frac{1}{n^3})$ 有 $\lim_{N \rightarrow \infty} |w_N| = 0$, 则我们只要证明其为广义解即可

任取 $\zeta(x, t) \in D$, 其中 $D = \{\zeta(x, t) \in C^2(\bar{Q}_t) \mid \zeta(x, T) = \zeta_t(x, T) = \zeta(0, t) = \zeta(L, t) = 0\}$ 我们将 (*) 两侧乘 ζ , 并求积分, 有

$$\iint_{Q_t} \zeta(x, t) (u_{N,tt} - a^2 u_{N,xx}) dx dt = 0 \triangleq I_t + I_x$$

其中

$$\begin{aligned}
 I_t &= \iint u_{N,tt}\zeta(x,t)dxdt = \int u_{N,t}\zeta(x,t)dx \Big|_0^T - \iint u_{N,t}\zeta_t dxdt \\
 &= \int u_{N,t}\zeta(x,t)dx \Big|_0^T - \int u_N\zeta_t(x,t)dx \Big|_0^T + \iint u_N\zeta_{tt}dxdt \\
 &= \int u_{N,t}(x,T)\zeta(x,T)dx - \int u_{N,t}(x,0)\zeta(x,0)dx - \int u_N(x,T)\zeta_t(x,T)dx + \int u_N(x,0)\zeta_t(x,0)dx + \iint u_N\zeta_{tt}dxdt \\
 &= \iint u_N\zeta_{tt}dxdt - \int u_{N,t}(x,0)\zeta(x,0)dx + \int u_N(x,0)\zeta_t(x,0)dx
 \end{aligned}$$

$$\begin{aligned}
 I_x &= -a^2 \iint u_{N,xx}\zeta dxdt = \iint u_N\zeta_{tt}dxdt - \int T_N(x)\zeta(x,0)dx + \int S_N(x)\zeta_t(x,0)dx \\
 &= -a^2 \left(\int u_{N,x}\zeta dt \Big|_0^L - \iint u_{mx}\zeta_x dxdt \right) \\
 &= -a^2 \int u_{N,x}(L,t)\zeta(L,t)dt + a^2 \int u_{N,x}(0,t)\zeta(0,t)dt + a^2 \left(\int u_N(x,t)\zeta_x dt \Big|_0^L - \iint u_N\zeta_{xx}dxdt \right) \\
 &= -a^2 \iint u_N\zeta_{xx}dxdt
 \end{aligned}$$

则

$$\iint u_N(\zeta_{tt} - a^2\zeta_{xx})dxdt + \int_0^L S_N(x)\zeta_t(x,0)dx - \int_0^L T_N(x)\zeta(x,0)dx = 0$$

有原命题成立

□

3 热传导方程

3.1 初值问题

在讲解热传导问题之前, 我们需要回顾傅里叶分析中的一些知识, 首先是关于 Fourier 变换和 Fourier 反演。

定义 3.1.1. 若 $f(x) \in L(-\infty, \infty)$ 那么积分

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\lambda x} dx = \hat{f}(\lambda)$$

有意义, 它定义了一个从 $f(x) \rightarrow \hat{f}(\lambda)$ 的变换, 通常把上式中的积分所定义的变换称为 *Fourier* 变换, 记作 $\hat{f}$, $\hat{f}(\lambda)$ 也称为 $f(x)$ 的 *Fourier* 变式。

定理 3.1.1. 若 $f(x) \in L(-\infty, \infty) \cap C^1(-\infty, \infty)$, 那么我们有

$$\lim_{N \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-N}^N \hat{f}(\lambda) e^{i\lambda x} d\lambda = f(x)$$

上面的公式就称为反演公式。一般的我们将对 Fourier 变式 $\hat{f}(\lambda)$ 的反演记作 $(\hat{f}(\lambda))^{\vee}$, 且有 $(\hat{f}(\lambda))^{\vee} = f$ 。

例 3.1.1. 按定义求下列函数的 *Fourier* 变换. 1) $f(x) = \begin{cases} 0, & |x| > a \\ |x|, & |x| \leq a \end{cases}$ 2) $f(x) = \begin{cases} \sin \lambda_0 x, & |x| \leq a \\ 0, & |x| > a \end{cases}$ 3) $f(x) = \cos x e^{-a|x|}, \quad a > 0$

0

解答 (1)

$$\begin{aligned} \hat{f}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x) e^{-i\lambda x} dx = \frac{1}{\sqrt{2\pi}} \int_{-a}^a |x| (\cos \lambda x - i \sin \lambda x) dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^a x \cos \lambda x dx = \sqrt{\frac{2}{\pi}} \left(\frac{x}{\lambda} \sin \lambda x \Big|_0^a - \int_0^a \frac{1}{\lambda} \sin \lambda x dx \right) \\ &= \sqrt{\frac{2}{\pi}} \left(\frac{1}{\lambda} x \sin \lambda x \Big|_0^a + \frac{1}{\lambda^2} \cos \lambda x \Big|_0^a \right) = \sqrt{\frac{2}{\pi}} \left(\frac{a \sin \lambda a}{\lambda} + \frac{\cos a \lambda - 1}{\lambda^2} \right). \end{aligned}$$

(2)

$$\begin{aligned}
\hat{f}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x) e^{-i\lambda x} dx = \frac{1}{\sqrt{2\pi}} \int_{-a}^a \sin \lambda_0 x (\cos \lambda x - i \sin \lambda x) dx \\
&= -i \sqrt{\frac{2}{\pi}} \int_0^a \sin \lambda_0 x \sin \lambda x dx = -i \frac{1}{\sqrt{2\pi}} \int_0^a \cos(\lambda - \lambda_0)x - \cos(\lambda + \lambda_0)x dx \\
&= -\frac{i}{\sqrt{2\pi}} \left(\frac{\sin(\lambda - \lambda_0)x}{\lambda - \lambda_0} - \frac{\sin(\lambda + \lambda_0)x}{\lambda + \lambda_0} \right) \Big|_0^a = \frac{i}{\sqrt{2\pi}} \left(\frac{\sin(\lambda + \lambda_0)a}{\lambda + \lambda_0} - \frac{\sin(\lambda - \lambda_0)a}{\lambda - \lambda_0} \right)
\end{aligned}$$

(3)

$$\begin{aligned}
\hat{f}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x) e^{-i\lambda x} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \cos x e^{-a|x|} (\cos \lambda x - i \sin \lambda x) dx \\
&= \sqrt{\frac{2}{\pi}} \int_0^{+\infty} \cos x \cos \lambda x e^{-a|x|} dx = \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} (\cos(1 - \lambda)x + \cos(1 + \lambda)x) e^{-ax} dx \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{(1 - \lambda) \sin(1 - \lambda)x - a \cos(1 - \lambda)x}{a^2 + (1 - \lambda)^2} e^{-ax} + \frac{(1 + \lambda) \sin(1 + \lambda)x - a \cos(1 + \lambda)x}{a^2 + (1 + \lambda)^2} e^{-ax} \right) \Big|_0^{+\infty} \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{a}{a^2 + (1 - \lambda)^2} + \frac{a}{a^2 + (1 + \lambda)^2} \right)
\end{aligned}$$

□

关于 Fourier 变换, 我们有如下性质:

1. (线性性质) 若 $f_i(x) \in L(-\infty, \infty)$, $a_i \in \mathbb{C}$ ($i = 1, 2$), 则

$$(a_1 f_1 + a_2 f_2)^\wedge = a_1 \hat{f}_1 + a_2 \hat{f}_2$$

2. (微商性质) 若 $f(x), f'(x) \in C(-\infty, \infty) \cap L(-\infty, \infty)$, 则

$$\left(\frac{df}{dx} \right)^\wedge = i\lambda \hat{f}$$

3. (乘多项式) 若 $f(x), xf(x) \in L(-\infty, \infty)$, 则有

$$(xf(x))^\wedge = i \frac{d}{d\lambda} \hat{f}(\lambda)$$

4. (平移性质) 若 $f(x) \in L(-\infty, \infty)$, 则

$$(f(x-a))^\wedge(\lambda) = e^{-i\lambda a} \hat{f}(\lambda)$$

5. (伸缩性质) 若 $f(x) \in L(-\infty, \infty)$, 则

$$(f(kx))^\wedge(\lambda) = \frac{1}{|k|} \hat{f}\left(\frac{\lambda}{k}\right), \quad k \neq 0$$

6. (对称性质) 若 $f(x) \in L(-\infty, \infty)$, 则

$$(f(x))^\vee(\lambda) = \hat{f}(-\lambda) = [f(-x)]^\wedge(\lambda)$$

7. (卷积性质) 若 $f(x), g(x) \in L(-\infty, \infty)$, 则

$$f * g(x) = \int_{-\infty}^{\infty} f(x-t)g(t)dt \in L(-\infty, \infty)$$

且有

$$(f * g)^\wedge(\lambda) = \sqrt{2\pi} \hat{f} \hat{g}$$

例 3.1.2. 利用 *Fourier* 变换性质求下列函数的 *Fourier* 变换:

$$(1) f(x) = xe^{-a|x|}, \quad a > 0$$

$$(2) f(x) = \begin{cases} e^{\mu x}, & |x| < a \\ 0, & |x| \geq a \end{cases}$$

$$(3) f(x) = \begin{cases} e^{i\lambda_0 x}, & |x| < L \\ 0, & |x| \geq L \end{cases}$$

$$(4) f(x) = \frac{1}{a^2 + x^2}$$

解答 (1) 解设 $g(x) = e^{-a|x|}$, 则我们有

$$\begin{aligned} \hat{g}(\lambda) &= \frac{1}{\sqrt{2\pi}} \left(\int_0^{+\infty} e^{-(a+i\lambda)x} dx + \int_{-\infty}^0 e^{-(i\lambda-a)x} dx \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(-\frac{e^{-(a+i\lambda)x}}{a+i\lambda} \Big|_0^{+\infty} - \frac{e^{-(i\lambda-a)x}}{i\lambda-a} \Big|_{-\infty}^0 \right) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2 + \lambda^2} \end{aligned}$$

则 $\hat{f}(\lambda) = (xg(x))^\wedge(\lambda) = i \frac{d}{d\lambda} \hat{g}(\lambda) = -\sqrt{\frac{2}{\pi}} \frac{2ia\lambda}{(a^2 + \lambda^2)^2}$

(2) 由 $\hat{f}(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{\mu x - i\lambda x} dx$, 则 $\hat{f}(\lambda - i\mu) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{-i\lambda x} dx = \sqrt{\frac{2}{\pi}} \frac{\sin a\lambda}{\lambda}$ 故有 $\hat{f}(\lambda) = \sqrt{\frac{2}{\pi}} \frac{\sin a(\lambda + i\mu)}{\lambda + i\mu}$

(3) 利用 (2) 有 $\hat{f}(\lambda) = \sqrt{\frac{2}{\pi}} \frac{\sin L(\lambda - \lambda_0)}{\lambda - \lambda_0}$

(4) 结合 $\hat{g}(\lambda) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2 + \lambda^2}$, 则 $f(x) = \sqrt{\frac{\pi}{2}} \frac{1}{a} \hat{g}(x)$ 。结合 $\hat{f}(\lambda) = (f(x))^\vee(-\lambda) = \sqrt{\frac{\pi}{2}} \frac{1}{a} g(-\lambda)$ 我们有 $\hat{f}(\lambda) = \sqrt{\frac{\pi}{2}} \frac{1}{a} e^{-a|\lambda|}$ $\square$ $\square$

例 3.1.3. 求下列函数的 Fourier 反演

$$\hat{f}(\lambda) = \exp\{-a^2\lambda^2 + ib\lambda + ct\}$$

其中 a, b, c 为常数, $t > 0$ 为参数.

解答

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \hat{f}(\lambda) e^{i\lambda x} d\lambda = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\{-a^2\lambda^2 + i(bt+x)\lambda + ct\} d\lambda \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{-a^2t\left(\lambda^2 - \frac{i(x+bt)}{a^2t}\lambda - \frac{(x+bt)^2}{4a^4t^2}\right) - \frac{(x+bt)^2}{4a^2t} + ct\right\} d\lambda \\ &= \frac{1}{\sqrt{2\pi}} \exp\left\{ct - \frac{(x+bt)^2}{4a^2t}\right\} \int_{-\infty}^{+\infty} \exp\left\{-a^2t\left(\lambda - \frac{i(x+bt)}{2a^2t}\right)^2\right\} d\lambda \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{a\sqrt{t}} \exp\left\{ct - \frac{(x+bt)^2}{4a^2t}\right\} \int_{-\infty}^{+\infty} e^{-x^2} dx \\ &= \frac{1}{a\sqrt{2t}} \exp\left\{ct - \frac{(x+bt)^2}{4a^2t}\right\}. \end{aligned}$$

$\square$

当然我们将上面的参数修改为向量形式, 就得到了高维 Fourier 变换和高维 Fourier 反演

定义 3.1.2. 设 $f(\mathbf{x}) = f(x_1, x_2, \dots, x_n) \in L(\mathbb{R}^n)$, 那么积分

$$\frac{1}{(\sqrt{2\pi})^n} \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-i\lambda \cdot \mathbf{x}} d\mathbf{x} = \hat{f}(\boldsymbol{\lambda})$$

有意义, 称为 $f(\mathbf{x})$ 的 Fourier 变换, $\hat{f}(\boldsymbol{\lambda})$ 称为 $f(\mathbf{x})$ 的 Fourier 变式。

定理 3.1.2. 若 $f(\mathbf{x}) \in C^1(\mathbb{R}^n) \cap L(\mathbb{R}^n)$, 则有

$$(\hat{f}(\lambda))^\vee(\mathbf{x}) = \frac{1}{(\sqrt{2\pi})^n} \lim_{N \rightarrow \infty} \int_{|\lambda| \leq N} \hat{f}(\lambda) e^{i\lambda \cdot \mathbf{x}} d\lambda = f(\mathbf{x})$$

其性质除如下性质以外都与一维是一样的:

性质 3.1.1. 若 $f(\mathbf{x}) = f_1(x_1)f_2(x_2)\cdots f_n(x_n)$, 其中 $f_i(x_i) \in L(-\infty, \infty)$, 则有

$$\hat{f}\lambda = \prod_{i=1}^n \hat{f}_i(\lambda_i)$$

有了这样的基础, 我们来看这样的问题:

例 3.1.4. 设 $f_4(x) = e^{-x^2}$, 求 $\hat{f}_4(\lambda)$.

$$\begin{aligned} \text{解答} \quad \text{由 } \hat{f}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2} e^{-i\lambda x} dx = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\lambda} e^{-x^2} e^{-i\lambda x} \Big|_{-\infty}^{+\infty} + \frac{2i}{\lambda} \int_{-\infty}^{+\infty} x e^{-x^2} e^{-i\lambda x} dx \right) \\ &= \frac{2i}{\sqrt{2\pi}\lambda} \int_{-\infty}^{+\infty} x e^{-x^2} e^{-i\lambda x} dx = \frac{2i}{\lambda} [x e^{-x^2}]^\wedge = -\frac{2}{\lambda} \frac{d}{d\lambda} \hat{f}(\lambda) \end{aligned}$$

且 $\hat{f}(0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2} dx = \frac{1}{\sqrt{2}}$, 求解该 ODE, 有

$$\hat{f}(\lambda) = \frac{1}{\sqrt{2}} e^{-\frac{\lambda^2}{4}}$$

□

例 3.1.5. 设 $f_5(x) = e^{-Ax^2}$ ($A > 0$), 求 $\hat{f}_5(\lambda)$.

解答 结合 $\hat{g}(\lambda) = [g(x)]^\wedge = [f(\sqrt{A}x)]^\wedge = \frac{1}{\sqrt{A}} \hat{f}\left(\frac{\lambda}{\sqrt{A}}\right) = \frac{1}{\sqrt{2A}} e^{-\frac{\lambda^2}{4A}}$ 利用上面的结果, 若我们记 $\hat{g}(\lambda, t) = a\sqrt{2t}e^{-a^2\lambda^2 t}$, 则有 $g(x) = e^{-\frac{x^2}{4a^2t}}$ $\hat{h}(\lambda, t) = a\sqrt{2(t-\tau)}e^{-a^2\lambda^2(t-\tau)}$, 则有 $h(x) = e^{-\frac{x^2}{4a^2(t-\tau)}}$

□

在这样的知识基础之上, 我们来求解热传导方程的初值问题:

$$\begin{cases} \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t) & , -\infty < x < \infty, t > 0 \\ u(x, 0) = \varphi(x) & , -\infty < x < \infty \end{cases}$$

我们对方程做 Fourier 变换, 结合微商性质, 我们得到

$$\begin{cases} \frac{\partial \hat{u}}{\partial t} + a^2 \lambda^2 \hat{u} = \hat{f}(\lambda, t) \\ \hat{u}(\lambda, 0) = \hat{\varphi}(\lambda) \end{cases}$$

利用前面例题的性质, 我们取

$$g(x, t) = \frac{1}{a\sqrt{2t}} \exp\left\{-\frac{x^2}{4a^2t}\right\}, \quad h(x, t) = \frac{1}{a\sqrt{2(t-\tau)}} \exp\left\{-\frac{x^2}{4a^2(t-\tau)}\right\}$$

有 $\hat{g}(\lambda, t) = \exp\{-a^2\lambda^2t\}$, $\hat{h}(\lambda, t) = \exp\{-a^2\lambda^2(t-\tau)\}$. 则有 $\hat{u}(\lambda, t) = \hat{\varphi}\hat{g} + \int_0^t \hat{f}\hat{h}d\tau$, 对其做反演, 我们有

$$\begin{aligned} u(x, t) &= (\hat{\varphi}\hat{g})^\vee + \int_0^t (\hat{f}\hat{h})^\vee d\tau = \frac{1}{\sqrt{2\pi}}\varphi * g + \int_0^t \frac{1}{\sqrt{2\pi}}f * h d\tau \\ &= \frac{1}{2a\sqrt{\pi}} \left(\frac{1}{\sqrt{t}} \int_{-\infty}^{+\infty} \varphi(x-\xi) \exp\left\{-\frac{\xi^2}{4a^2t}\right\} d\xi + \int_0^t \frac{d\tau}{a\sqrt{t-\tau}} \int_{-\infty}^{+\infty} f(\xi, \tau) \exp\left\{-\frac{(x-\xi)^2}{4a^2(t-\tau)}\right\} d\xi \right) \end{aligned}$$

或者我们将式子写作如下形式:

$$u(x, t) = \int_{-\infty}^{+\infty} K(x-\xi, t) \varphi(\xi) d\xi + \int_0^t d\tau \int_{-\infty}^{+\infty} K(x-\xi, t-\tau) f(\xi, \tau) d\xi$$

其中

$$K(x, t) = \begin{cases} \frac{1}{2a\sqrt{\pi t}} \exp\left\{-\frac{x^2}{4a^2t}\right\} & , t > 0 \\ 0 & , t \leq 0 \end{cases}$$

其中, 我们称 $u(x, t)$ 为方程的解, $K(x-\xi, t-\tau)$ 为热传导方程的基本解, 关于基本解的概念我们将在后面进行讨论。

该问题的存在性定理表述如下:

定理 3.1.3. 若 $\varphi(x) \in C(-\infty, \infty)$ 且有界, $f(x, t) \equiv 0$, 则由 Poisson 公式确定的函数是热传导方程初值问题的解.

解答 欲证明该定理, 我们需要 Leibniz 法则

定理 (Leibniz 法则) 设 $F(x, t, \xi) = K(x - \xi, t)\varphi(\xi)$, 若满足 1. F 及其偏导 $\frac{\partial F}{\partial t}, \frac{\partial F}{\partial x}, \frac{\partial^2 F}{\partial x^2}$ 在 $\mathbb{R}^3$ 上连续 2. 积分 $\int_{-\infty}^{+\infty} F d\xi$ 及其导数积分一致收敛 (对 x, t 在紧集上) 则 F 可交换微分与积分次序.

证明: 我们只需要验证三个条件

(1) 验证 $u(x, t)$ 有界, 此时 $u(x, t) = \int_{-\infty}^{+\infty} K(x - \xi, t)\varphi(\xi) d\xi$. 结合

$$\int_{-\infty}^{+\infty} K(x - \xi, t) d\xi = \int_{-\infty}^{+\infty} \frac{1}{2a\sqrt{\pi t}} \exp\left\{-\frac{(x - \xi)^2}{4a^2 t}\right\} d\xi \stackrel{\eta = \frac{x - \xi}{\sqrt{4a^2 t}}}{=} \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-\eta^2} d\eta = 1$$

由 $\varphi(x)$ 有界, $\exists M > 0$ s.t. $\forall x \in (-\infty, \infty)$, 都有 $|\varphi(x)| \leq M$. 则 $|u(x, t)| \leq \int_{-\infty}^{+\infty} |\varphi(\xi)| K(x - \xi, t) d\xi \leq M \int_{-\infty}^{+\infty} K(x - \xi, t) d\xi = M$

(2) 验证热传导方程 $\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = 0$ (通过 $\frac{\partial^k K}{\partial x^k}$ 中表达式为 $\frac{1}{t}, \frac{x}{t}$ 构成的多项式与 $\exp\{-\frac{x^2}{8a^2 t}\}$ 的乘积, 我们可以保证 $\int_{-\infty}^{+\infty} F d\xi$ 一致收敛) 结合 Leibniz 法则, 结合

$$\begin{aligned} \frac{\partial K}{\partial t}(x - \xi, t) &= -\frac{1}{2\sqrt{4\pi a^2 t^3}} \exp\left\{-\frac{|x - \xi|^2}{4a^2 t}\right\} + \frac{1}{\sqrt{4\pi a^2 t}} \exp\left\{-\frac{|x - \xi|^2}{4a^2 t}\right\} \frac{|x - \xi|^2}{4a^2 t^2} \\ &= K(x - \xi, t) \left(-\frac{1}{2t} + \frac{(x - \xi)^2}{4a^2 t^2}\right) \frac{\partial K}{\partial x} = \frac{1}{\sqrt{4\pi a^2 t}} \left(-\frac{(x - \xi)}{2a^2 t}\right) \exp\left\{-\frac{(x - \xi)^2}{4a^2 t}\right\} \end{aligned}$$

$$\frac{\partial^2 K}{\partial x^2} = \frac{1}{\sqrt{4\pi a^2 t}} \left(-\frac{1}{2a^2 t} \exp\left\{-\frac{(x - \xi)^2}{4a^2 t}\right\} - \frac{|x - \xi|}{2a^2 t} \exp\left\{-\frac{(x - \xi)^2}{4a^2 t}\right\} \cdot -\frac{|x - \xi|}{2a^2 t}\right) = \frac{1}{a^2} K(x - \xi, t) \left(-\frac{1}{2t} + \frac{|x - \xi|^2}{4a^2 t^2}\right)$$

$$\text{则有 } \frac{\partial K}{\partial t} - a^2 \frac{\partial^2 K}{\partial x^2} = 0 \Rightarrow \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = \int_{-\infty}^{+\infty} \varphi(\xi) \left(\frac{\partial K}{\partial t} - a^2 \frac{\partial^2 K}{\partial x^2}\right) d\xi = 0$$

(3) 验证初值条件 $\lim_{t \rightarrow 0^+} u(x, t) = \varphi(x)$. 取 $\eta = \frac{\xi - x}{2a\sqrt{t}}$ $\xi = x + 2a\sqrt{t}\eta$ $d\xi = 2a\sqrt{t}d\eta$. 则有

$$u(x, t) = \frac{1}{\sqrt{4\pi a^2 t}} \int_{-\infty}^{+\infty} \varphi(x + 2a\sqrt{t}\eta) e^{-\eta^2} 2a\sqrt{t} d\eta = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-\eta^2} \varphi(x + 2a\sqrt{t}\eta) d\eta.$$

结合 $\int_{-\infty}^{+\infty} e^{-x^2} dx$ 的一致收敛性, $\forall \varepsilon > 0, \exists R > 0$ s.t $\int_{|\eta|>R} e^{-\eta^2} d\eta < \frac{\sqrt{\pi}\varepsilon}{2M}$. 且 $\forall \varepsilon > 0, \exists \delta > 0$ s.t 当 $t_0 \in (0, \frac{\delta^2}{4a^2R^2})$ 时, 有 $|\varphi(x + 2a\sqrt{t_0}\eta) - \varphi(x)| < \varepsilon$

$$\begin{aligned} |u(x, t_0) - \varphi(x)| &= \frac{1}{\sqrt{\pi}} \left| \int_{-\infty}^{+\infty} e^{-\eta^2} (\varphi(x + 2a\sqrt{t_0}\eta) - \varphi(x)) d\eta \right| \\ &\leq \frac{1}{\sqrt{\pi}} \int_{|\eta|>R} e^{-\eta^2} |\varphi(x + 2a\sqrt{t_0}\eta) - \varphi(x)| d\eta + \frac{1}{\sqrt{\pi}} \int_{|\eta|<R} e^{-\eta^2} |\varphi(x + 2a\sqrt{t_0}\eta) - \varphi(x)| d\eta \\ &\leq \frac{2M}{\sqrt{\pi}} \int_{|\eta|>R} e^{-\eta^2} d\eta + \frac{\varepsilon}{\sqrt{\pi}} \int_{|\eta|<R} e^{-\eta^2} d\eta < 2\varepsilon \end{aligned}$$

综上有原命题成立

□

结合解的存在性定理的证明, 我们可以得到如下 5 条性质:

1. 若 $\varphi(x)$ 有界, 则当 $t > 0$ 时, $u(x, t)$ 在 $\mathbb{R}_+^2$ 内必无穷次可微
2. 若热源杆, 即 $\varphi(x)$ 只在一段 $(x_0 - \delta, x_0 + \delta)$ 上不为零, 则在顷刻之间, 热量就传递到杆上的任意一点。
3. 若 φ 是奇 (偶, 周期) 函数, 则 $u(x, t)$ 也是关于 x 的奇 (偶, 周期) 函数。
4. 存在性定理中的 φ 有界的假设可以改进为

$$|\varphi(x)| \leq Me^{Ax^2}, \quad -\infty < x < \infty, \quad t > 0$$

此时 $u(x, t)$ 在区域 $\left\{ 0 \leq t < \frac{1}{4a^2A}, -\infty < x < \infty \right\}$ 仍是原问题的解。

5. 针对多维热传导方程的初值问题

$$\begin{cases} \frac{\partial u}{\partial t} - a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = f(x, y, t) & , \mathbb{R}^2 \times (0, \infty) \\ u(x, y, 0) = \varphi(x, y) & , \mathbb{R}^2 \end{cases}$$

我们有

$$u(x, y, t) = \iint_{\mathbb{R}^2} K(x - \xi, y - \eta, t) \varphi(\xi, \eta) d\xi d\eta + \int_0^t d\tau \iint_{\mathbb{R}^2} K(x - \xi, y - \eta, t - \tau) f(\xi, \eta, \tau) d\xi d\eta$$

其中

$$K(x, y, t) = \begin{cases} \frac{1}{4\pi a^2 t} \exp\left\{-\frac{x^2 + y^2}{4a^2 t}\right\} & , t > 0 \\ 0 & , t \leq 0 \end{cases}$$

例 3.1.6. 推导高维热传导方程。

解答 对初值问题作 Fourier 变换, 我们有

$$\begin{cases} \frac{\partial u}{\partial t} - a^2 \Delta u = f(\mathbf{x}, t) \\ u(\mathbf{x}, 0) = \varphi(\mathbf{x}) \end{cases} \Rightarrow \begin{cases} \frac{\partial \hat{u}}{\partial t} + a^2 |\boldsymbol{\lambda}|^2 \hat{u}(\boldsymbol{\lambda}, t) = \hat{f}(\boldsymbol{\lambda}, t) \\ \hat{u}(\boldsymbol{\lambda}, 0) = \hat{\varphi}(\boldsymbol{\lambda}) \end{cases}$$

求解该 ODE, 我们有

$$\hat{u}(\boldsymbol{\lambda}, t) = \hat{\varphi}(\boldsymbol{\lambda}) \exp\{-a^2 |\boldsymbol{\lambda}|^2 t\} + \int_0^t \hat{f}(\boldsymbol{\lambda}, \tau) \exp\{-a^2 |\boldsymbol{\lambda}|^2 (t - \tau)\} d\tau$$

对 $\psi(\mathbf{x}) = e^{-|\mathbf{x}|^2} = \prod_{i=1}^n e^{-x_i^2}$, 我们有

$$\hat{\psi}(\boldsymbol{\lambda}) = \prod_{i=1}^n (e^{-x_i^2})^\wedge = \left(\frac{1}{\sqrt{2}}\right)^n \prod_{i=1}^n e^{-\frac{\lambda_i^2}{4}} = \left(\frac{1}{\sqrt{2}}\right)^n e^{-\frac{|\boldsymbol{\lambda}|^2}{4}}$$

$\psi_1(\mathbf{x}) = e^{-A|\mathbf{x}|^2} = \prod_{i=1}^n e^{-Ax_i^2}$, 我们有

$$\hat{\psi}_1(\boldsymbol{\lambda}) = \prod_{i=1}^n (e^{-Ax_i^2})^\wedge = \frac{1}{(\sqrt{2A})^n} e^{-\frac{|\boldsymbol{\lambda}|^2}{4A}}$$

结合这里的讨论, 我们取

$$\hat{g}(\boldsymbol{\lambda}, t) = (\sqrt{2a^2 t})^n \exp\{-a^2 |\boldsymbol{\lambda}|^2 t\} \quad \hat{h}(\boldsymbol{\lambda}, t) = (\sqrt{2a^2(t - \tau)})^n \exp\{-a^2 |\boldsymbol{\lambda}|^2 (t - \tau)\}$$

则有 $g(\mathbf{x}, t) = \exp\left\{-\frac{|\mathbf{x}|^2}{4a^2 t}\right\}$ $h(\mathbf{x}, t) = \exp\left\{-\frac{|\mathbf{x}|^2}{4a^2(t - \tau)}\right\}$

对 $\hat{u}(\boldsymbol{\lambda}, t)$ 作 Fourier 反变换

$$\begin{aligned} u(\boldsymbol{x}, t) &= (\hat{\varphi}(\boldsymbol{\lambda}) \exp\{-a^2|\boldsymbol{\lambda}|^2 t\})^\vee + \int_0^t (\hat{f}(\boldsymbol{\lambda}, \tau) \exp\{-a^2|\boldsymbol{\lambda}|^2(t-\tau)\})^\vee d\tau \\ &= \frac{1}{(\sqrt{2a^2 t})^n} (\hat{\varphi}(\boldsymbol{\lambda}) \hat{g}(\boldsymbol{\lambda}, t))^\vee + \int_0^t \frac{1}{(\sqrt{2a^2(t-\tau)})^n} (\hat{f}(\boldsymbol{\lambda}, \tau) \hat{h}(\boldsymbol{\lambda}, t; \tau))^\vee d\tau \\ &= \frac{1}{(\sqrt{4\pi a^2 t})^n} \varphi * g + \int_0^t \frac{1}{(\sqrt{4\pi a^2(t-\tau)})^n} f * h d\tau. \end{aligned}$$

其中 $\varphi * g = \int_{\mathbb{R}^n} \varphi(\boldsymbol{y}) g(\boldsymbol{x} - \boldsymbol{y}, t) d\boldsymbol{y}$, $f * h = \int_{\mathbb{R}^n} f(\boldsymbol{y}, \tau) h(\boldsymbol{x} - \boldsymbol{y}, t; \tau) d\boldsymbol{y}$

则原问题的解为

$$u(\boldsymbol{x}, t) = \frac{1}{(2a\sqrt{\pi t})^n} \int_{\mathbb{R}^n} \varphi(\boldsymbol{y}) \exp\left\{-\frac{|\boldsymbol{x} - \boldsymbol{y}|^2}{4a^2 t}\right\} d\boldsymbol{y} + \int_0^t \frac{1}{(2a\sqrt{\pi(t-\tau)})^n} \int_{\mathbb{R}^n} f(\boldsymbol{y}, \tau) \exp\left\{-\frac{|\boldsymbol{x} - \boldsymbol{y}|^2}{4a^2(t-\tau)}\right\} d\boldsymbol{y} d\tau$$

□

例 3.1.7. 应用 Fourier 变换求解如下定解问题

$$(1) \begin{cases} \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} - b \frac{\partial u}{\partial x} - cu = f(x, t), & -\infty < x < \infty, t > 0 \\ u|_{t=0} = \varphi(x) & -\infty < x < \infty \end{cases} \quad (2) \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, & -\infty < x < \infty, y > 0 \\ u|_{y=0} = \varphi(x), & -\infty < x < \infty \end{cases}$$

其中 a, b, c 是常数, (2) 中设 $\varphi(x)$ 连续有界, 求问题的有界解

解答 对方程作 Fourier 变换, 我们有

$$\begin{cases} \frac{\partial \hat{u}}{\partial t} + (a^2 \lambda^2 - bi\lambda - c)\hat{u} = \hat{f}(\lambda, t) \\ \hat{u}(\lambda, 0) = \hat{\varphi}(\lambda) \end{cases}$$

解得

$$\hat{u}(\lambda, t) = \int_0^t \hat{f}(\lambda, \tau) \exp\{(-a^2 \lambda^2 + bi\lambda + c)(t - \tau)\} d\tau + \hat{\varphi}(\lambda) \exp\{(-a^2 \lambda^2 + bi\lambda + c)t\}$$

取 $\hat{g}(\lambda, t) = \exp\{-a^2\lambda^2 + bi\lambda + ct\}$, $\hat{h}(\lambda, t; \tau) = \exp\{(-a^2\lambda^2 + bi\lambda + c)(t - \tau)\}$ 则有

$$g(x, t) = \frac{1}{a\sqrt{2t}} \exp\left\{ct - \frac{(x + bt)^2}{4a^2t}\right\}, \quad h(x, t; \tau) = \frac{1}{a\sqrt{2(t - \tau)}} \exp\left\{c(t - \tau) - \frac{(x + b(t - \tau))^2}{4a^2(t - \tau)}\right\}$$

则对 $\hat{u}(\lambda, t)$ 作 Fourier 反演

$$\begin{aligned} u(x, t) &= \int_0^t (\hat{f}(\lambda, \tau) \hat{h}(\lambda, t; \tau))^\vee d\tau + (\hat{\varphi}(\lambda) \hat{g}(\lambda, t))^\vee \\ &= \frac{1}{\sqrt{2\pi}} \left(\int_0^t f(x, \tau) * h(x, t; \tau) d\tau + \varphi(x) * g(x, t) \right) \\ &= \frac{1}{\sqrt{2\pi}} \int_0^t d\tau \int_{-\infty}^{+\infty} \frac{1}{a\sqrt{2(t - \tau)}} f(x - \xi, \tau) \exp\left\{c(t - \tau) - \frac{(\xi + b(t - \tau))^2}{4a^2(t - \tau)}\right\} d\xi \\ &\quad + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \frac{1}{a\sqrt{2t}} \varphi(x - \xi) \exp\left\{ct - \frac{(\xi + bt)^2}{4a^2t}\right\} d\xi \end{aligned}$$

(2) 对 x 作 Fourier 变换, 我们得到如下 ODE 问题

$$\begin{cases} \frac{\partial^2 \hat{u}}{\partial y^2} - \lambda^2 \hat{u} = 0 \\ \hat{u}(\lambda, 0) = \hat{\varphi}(\lambda) \end{cases}$$

有 $\hat{u}(\lambda, y) = C_1(\lambda)e^{\lambda y} + C_2(\lambda)e^{-\lambda y}$ 为得到有界解, 我们需要保证 $\forall \lambda < 0, C_2(\lambda) = 0$; $\forall \lambda > 0, C_1(\lambda) = 0$ 则 $\hat{u}(\lambda, y) = \hat{\varphi}(\lambda)e^{-|\lambda|y}$ 取 $\hat{f}(\lambda, y) = e^{-|\lambda|y}$, 对其做 Fourier 反演

$$\begin{aligned} f(x, y) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-|\lambda|y} e^{i\lambda x} d\lambda = \frac{1}{\sqrt{2\pi}} \left(\int_0^{+\infty} e^{(ix-y)\lambda} d\lambda + \int_{-\infty}^0 e^{(ix+y)\lambda} d\lambda \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\left. \frac{e^{(ix-y)\lambda}}{ix - y} \right|_0^{+\infty} + \left. \frac{e^{(ix+y)\lambda}}{ix + y} \right|_{-\infty}^0 \right) = \frac{1}{\sqrt{2\pi}} \frac{2y}{x^2 + y^2} \end{aligned}$$

对 $\hat{u}(\lambda, y)$ 作 Fourier 反演

$$\begin{aligned} u(x, y) &= (\hat{\varphi}(\lambda)e^{-|\lambda|y})^\vee = (\hat{\varphi}(\lambda)\hat{f}(\lambda, y))^\vee = \frac{1}{\sqrt{2\pi}} (\varphi * f)(x, y) \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \varphi(\xi) \frac{2y}{(x - \xi)^2 + y^2} d\xi \\ &= \frac{y}{\pi} \int_{-\infty}^{+\infty} \frac{\varphi(\xi)}{(x - \xi)^2 + y^2} d\xi \end{aligned}$$

□

3.2 广义函数

广义函数本质上是一类泛函，它的一个总体可以看成是一种分布。首先我们做如下补充：以 $C_0^\infty(\mathbb{R})$ 表示所有当 $|x|$ 充分大时恒等于零的无穷次连续可微函数 $\varphi(x)$ 构成的集合。首先我们给出试验函数的概念：

定义 3.2.1. 如果 $\varphi(x) \in C_0^\infty(\mathbb{R})$, $\varphi_n(x) \in C_0^\infty(\mathbb{R})$, $n = 1, 2, \dots$, 且

(1) 存在 $M > 0$, 使得当 $|x| \geq M$ 时, $\varphi(x) \equiv 0$, $\varphi_n(x) \equiv 0$, $n = 1, 2, \dots$

(2) $\lim_{n \rightarrow \infty} \max_{[-M, M]} |\varphi_n(x) - \varphi(x)| = 0$, $\lim_{n \rightarrow \infty} \max_{[-M, M]} |\varphi_n^{(k)}(x) - \varphi^{(k)}(x)| = 0$, $k = 1, 2, \dots$.

则称序列 $\{\varphi_n\}$ 收敛于 φ , 规定了上述收敛性的线性空间 $C_0^\infty(\mathbb{R})$ 称为基本空间 $\mathcal{D}(\mathbb{R})$, $\varphi \in \mathcal{D}(\mathbb{R})$ 称为试验函数。

则广义函数就是定义在基本空间 $\mathcal{D}(\mathbb{R})$ 中的连续线性泛函：

定义 3.2.2. 如果 $f: \mathcal{D}(\mathbb{R}) \rightarrow \mathbb{R}$ 是连续线性泛函，则称 f 是一个广义函数，设 $\varphi \in \mathcal{D}(\mathbb{R})$ 是一个试验函数，用 $\langle f, \varphi \rangle$ 表示它所对应的数值，称为对偶积，所有广义函数构成的集合记作 $\mathcal{D}'(\mathbb{R})$ 。

例 3.2.1. 使任意试验函数 $\varphi(x)$ 与它在 $x = 0$ 处的函数值 $\varphi(0)$ 对应的广义函数称为 δ 函数，即 δ 函数满足

$$\langle \delta, \varphi \rangle = \varphi(0), \quad \forall \varphi \in \mathcal{D}(\mathbb{R})$$

此外，我们记 $L_{loc}(\mathbb{R})$ 为所有在 $\mathbb{R}$ 上局部绝对可积的函数组成的集合，其上对偶积定义为

$$\langle f, \varphi \rangle = \int_{-\infty}^{+\infty} f(x)\varphi(x)dx$$

所谓局部可积函数 f ，即对 $\mathbb{R}$ 中任意区间 $[a, b]$ ，我们都有 $\int_a^b |f(x)|dx$ 存在，则若我们将函数 f 与其与试验函数的成绩的积分确定的广义函数不加区分，则有

$$L_{loc}(\mathbb{R}) \subset \mathcal{D}'(\mathbb{R})$$

这里我们来解释一下为什么广义函数又被称为广义函数分布，在这样的例子中，如果 $f \in \mathcal{D}'(\mathbb{R})$ 那么我们就有其对偶积为 $\langle f, \varphi \rangle = \int_{-\infty}^{+\infty} f(x)\varphi(x)dx$ ，这是由 $L_{loc}(\mathbb{R})$ 确定的。此外，我们来补充泛函分析中强收敛的概念：

定义 3.2.3. 设 $f \in \mathcal{D}'(\mathbb{R})$, $\{f_n\} \subset \mathcal{D}'(\mathbb{R})$, 如果

$$\lim_{n \rightarrow \infty} \langle f_n, \varphi \rangle = \langle f, \varphi \rangle, \quad \forall \varphi \in \mathcal{D}(\mathbb{R})$$

则称广义函数 f 是广义函数序列 $\{f_n\}$ 当 $n \rightarrow \infty$ 时的极限, 记作

$$f_n \rightarrow f \quad (\text{在 } \mathcal{D}'(\mathbb{R}) \text{ 的意义下})$$

另外有广义函数的极限:

定义 3.2.4. 设 $f \in \mathcal{D}'(\mathbb{R})$, $f_\lambda \in \mathcal{D}'(\mathbb{R})$, 其中 λ 是参数, 如果

$$\lim_{\lambda \rightarrow \lambda_0} \langle f_\lambda, \varphi \rangle = \langle f, \varphi \rangle, \quad \forall \varphi \in \mathcal{D}(\mathbb{R})$$

则称广义函数 f 是广义函数 f_λ 当 $\lambda \rightarrow \lambda_0$ 时的极限, 记作

$$f_\lambda \rightarrow f \quad (\lambda \rightarrow \lambda_0, \text{在 } \mathcal{D}'(\mathbb{R}) \text{ 的意义下})$$

针对泛函的微商, 我们结合在 $L_{loc}(\mathbb{R})$ 中关于 $\int_{-\infty}^{+\infty} f(x)\varphi(x)dx$ 的性质, 定义如下弱导数:

定义 3.2.5. 广义函数 f 的弱导数 f' 也是广义函数, 用下式定义:

$$\langle f', \varphi \rangle = -\langle f, \varphi' \rangle, \quad \forall \varphi \in \mathcal{D}(\mathbb{R})$$

我们可以这样计算 δ 函数的弱导数:

$$\langle \delta^{(k)}, \varphi \rangle = (-1)^k \langle \delta, \varphi^{(k)} \rangle = (-1)^k \varphi^{(k)}(0), \quad \forall \varphi \in \mathcal{D}(\mathbb{R}), \quad k = 1, 2, \dots$$

再比如说, 针对 Heaviside 函数 $H(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$, 我们可以这样计算:

$$\langle H', \varphi \rangle = -\langle H, \varphi' \rangle = -\int_{-\infty}^{+\infty} H(x)\varphi'(x)dx = \varphi(0)\langle \delta, \varphi \rangle$$

也就是说 $H' = \delta$

例 3.2.2. 证明在 $D'(-\infty, +\infty)$ 意义下

$$1) \varphi(x)\delta'(x) = -\varphi'(0)\delta(x) + \varphi(0)\delta'(x)$$

$$2) x^m\delta^{(m)}(x) = (-1)^m m! \delta(x).$$

解答 (1) 证明由

$$\begin{aligned}\langle \varphi(x)\delta'(x), \psi(x) \rangle &= \langle \delta'(x), \varphi(x)\psi(x) \rangle = -\langle \delta(x), (\varphi(x)\psi(x))' \rangle \\ &= -(\varphi(0)\psi'(0) + \varphi'(0)\psi(0)) \\ &= -(\varphi(0)\psi'(0) + \varphi'(0)\psi(0))\end{aligned}$$

$$\langle -\varphi'(0)\delta(x) + \varphi(0)\delta'(x), \psi(x) \rangle = -\varphi'(0)\langle \delta(x), \psi(x) \rangle + \varphi(0)\langle \delta'(x), \psi(x) \rangle = -\varphi'(0)\psi(0) - \varphi(0)\psi'(0)$$

则 $\varphi(x)\delta'(x) = -\varphi'(0)\delta(x) + \varphi(0)\delta'(x)$ 成立.

(2)

$$\begin{aligned}\langle x^m\delta^{(m)}(x), \varphi(x) \rangle &= \langle \delta^{(m)}(x), \varphi(x)x^m \rangle = (-1)^m \langle \delta(x), (\varphi(x)x^m)^{(m)} \rangle \text{ (Leibniz)} \\ &= (-1)^m \sum_{k=0}^m \binom{m}{k} (x^m)^{(m-k)} (\varphi(x))^{(k)} \Big|_{x=0} \\ &= (-1)^m m! \varphi(0) = \langle (-1)^m m! \delta(x), \varphi(x) \rangle.\end{aligned}$$

则 $x^m\delta^{(m)}(x) = (-1)^m m! \delta(x)$.

□

同样的思路，我们也可以定义广义函数的平移：

定义 3.2.6. 广义函数 f 的平移也是广义函数，若右移 ξ ，则把平移所得的广义函数记为 $f(x - \xi)$ ，且用下式定义：

$$\langle f(x - \xi), \varphi \rangle = \langle f, \varphi(x + \xi) \rangle, \quad \forall \varphi \in \mathcal{D}(\mathbb{R})$$

而广义函数的乘积，我们做这样的定义：

$$\langle fg, \varphi \rangle = \langle f, g\varphi \rangle, \quad \forall \varphi \in \mathcal{D}(\mathbb{R})$$

在这样的定义下，我们可以做这样的操作：

$$\langle (fg)', \varphi \rangle = \langle f'g + fg', \varphi \rangle$$

具体的证明可以参考知乎。

最后我们可以定义二元广义函数 $f: \mathcal{D}(\mathbb{R}^2) \rightarrow \mathbb{R}$ ，其主要的应用在于二元 δ 函数：

$$\langle \delta(x - \xi, y - \eta), \varphi \rangle = \varphi(\xi, \eta), \quad \forall \varphi \in \mathcal{D}(\mathbb{R}^2)$$

例 3.2.3. 计算 1) $(|x|)^{(m)} \quad (m \geq 1)$

2) $(H(x)e^{ax})'' \quad H(x)$ 为 Heaviside 函数

3) $f'(x)$, 其中 ① $f(x) = \begin{cases} \sin x, & x \geq 0 \\ 0, & x < 0 \end{cases}$ ② $f(x) = \begin{cases} x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$

解答 (1) 解

$$(|x|)' = (xH(x) - xH(-x))' = H(x) - H(-x) + x(H(x) - H(-x))'$$

即

$$\langle H'(-x), f(x) \rangle = - \int_{-\infty}^{+\infty} H(-x) f'(x) dx = - \int_{-\infty}^0 f'(x) dx = -f(0)$$

$$\langle H'(x), f(x) \rangle = - \int_0^{+\infty} f'(x) dx = f(0) \quad (H(x) - H(-x))' = 2\delta(x)$$

则

$$\langle x(H(x) - H(-x))', f(x) \rangle = \langle (H(x) - H(-x))', xf(x) \rangle = \langle 2\delta(x), xf(x) \rangle = 0$$

$$(|x|)' = H(x) - H(-x) \quad (|x|)^{(m)} = (H(x) - H(-x))^{(m-1)} = 2\delta(x)^{(m-2)}$$

$$(2) (H(x)e^{ax})'' = (H'(x)e^{ax} + aH(x)e^{ax})' = (\delta(x)e^{ax} + aH(x)e^{ax})'$$

$$= (\delta(x) + aH(x)e^{ax})' = \delta'(x) + a\delta(x) + a^2H(x)e^{ax}.$$

$$(3) \text{ ① } f'(x) = (\sin x H(x))' = \cos x H(x) + \sin x \delta(x) = \cos x H(x)$$

②

$$f'(x) = (x^2(H(x+1) - H(x-1)))' = 2x(H(x+1) - H(x-1)) + x^2(\delta(x+1) - \delta(x-1)) = 2x(H(x+1) - H(x-1)) + \delta(x+1) - \delta(x-1).$$

□

3.3 Poisson 公式与基本解

在前面, 我们已经推导出了 Poisson 公式, 并指出 $K(x-\xi, t-\tau)$ 是热传导方程的基本解, 但是我们事实上并没有明确什么是基本解, 现在我们对它做出严格定义:

定义 3.3.1. 设 $Q = \{(x, t) | -\infty < x < \infty, t > 0\}$, 对任意 $(\xi, \tau) \in Q$, 如果函数 $u(x, t) \in L_{loc}(Q) \cap C(\bar{Q} \setminus (\xi, \tau))$ 且在广义函数意义下满足方程与初始条件:

$$\begin{cases} u_t - a^2 u_{xx} = \delta(x - \xi, t - \tau) & , (x, t) \in Q \\ u(x, 0) = 0 & , -\infty < x < \infty \end{cases}$$

则称它为热传导方程的基本解, 且记为 $\Gamma(x, t; \xi, \tau)$.

我们可以证明, $K(x-\xi, t-\tau)$ 是热传导方程的基本解:

解答 此时我们记 $u(x, t) = K(x-\xi, t-\tau)$. 我们将证明分为两步

(1) 验证 $u \in L_{loc}(Q) \cap C(Q \setminus \{(\xi, \tau)\})$

① 验证 $u \in L_{loc}(Q)$, 即任意紧集 $K \subset Q$, 有 $\iint_K |u| dx dt < \infty$

取 $s = t - \tau$ $y = x - \xi$, 则有

$$\iint_K |u(x, t)| dx dt = \iint_{K'} |K(y, s)| dy ds < \infty$$

因对 s , 我们有 $\int_{-\infty}^{+\infty} \frac{1}{2a\sqrt{\pi s}} \exp\{-\frac{y^2}{4a^2 s}\} ds \stackrel{x=\frac{y}{2a\sqrt{s}}}{=} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{\pi}} e^{-x^2} dx = 1$ 则 $\iint_{K'} |K(y, s)| dy ds \leq \int_{K'_s} ds$, 又结合 K 是 Q 中紧集, 取 T 为 K' 中关于 s 的上界, 又由于 $t < \tau$ 时 $K(x-\xi, t-\tau) = 0$, 我们仅需考虑 $t > \tau$ 的情形, 则 $\exists \delta > 0, s.t. t - \tau \geq \delta$. 有

$$\iint_{K'} |K(y, s)| dy ds \leq \int_{\delta}^T ds = T - \delta < \infty. \quad \text{则有 } u \in L_{loc}(Q)$$

② 验证 $u \in C(Q \setminus \{(\xi, \tau)\})$ 由 $t \leq \tau$ 时 $u(x, t) = 0$ 是连续的, $t > \tau$ 时 $u(x, t)$ 由有理数与指数函数构成, 则有其连续. 且对 $t = \tau, x = \xi$, 取 $t \rightarrow \tau^+$, 结合

$$\lim_{t \rightarrow \tau^+} \frac{1}{\sqrt{4a^2 \pi(t - \tau)}} \exp\left\{-\frac{y^2}{4a^2(t - \tau)}\right\} = \delta_0 = u(x, \tau).$$

我们有 $u \in C(\bar{Q} \setminus \{\xi, \tau\})$.

2) 验证 $u(x, 0) = 0$. 结合 $u(x, 0) = k(x - \xi, -\tau) = 0$. 我们有 $u(x, 0) = 0$ 成立.

3) 验证在广义函数意义下 $u_t - a^2 u_{xx} = \delta(x - \xi, t - \tau)$. 即证 $\langle u_t - a^2 u_{xx}, \varphi \rangle = \varphi(\xi, \tau)$. 结合

$$\langle u_t - a^2 u_{xx}, \varphi \rangle = \iint_{\mathbb{R}^2} u(x, t) \left(-\frac{\partial \varphi}{\partial t} - a^2 \frac{\partial^2 \varphi}{\partial x^2} \right) dx dt$$

我们取 $s = t - \tau$, $y = x - \xi$. 我们将原式转换为

$$\langle u_t - a^2 u_{xx}, \varphi \rangle = \iint_{\mathbb{R}^2} k(y, s) \left(-\frac{\partial \varphi}{\partial s}(s + \tau) - a^2 \frac{\partial^2 \varphi}{\partial y^2}(y + \xi) \right) ds dy$$

我们先研究 $\tau = \xi = 0$ 时情形, 取 $\varepsilon > 0$, 我们将域写为

$$\langle u_t - a^2 u_{xx}, \varphi \rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^{+\infty} ds \int_{-\infty}^{+\infty} k(y, s) \left(-\frac{\partial \varphi}{\partial s} - a^2 \frac{\partial^2 \varphi}{\partial y^2} \right) dy$$

对

$$\int_{\varepsilon}^{+\infty} ds \int_{-\infty}^{+\infty} k(y, s) \frac{\partial \varphi}{\partial s} dy = \int_{-\infty}^{+\infty} \left[k \varphi \Big|_{\varepsilon}^{+\infty} - \int_{\varepsilon}^{+\infty} \frac{\partial k}{\partial s} \varphi ds \right] dy = - \int_{-\infty}^{+\infty} [k(y, \varepsilon) \varphi(y, \varepsilon) + \int_{\varepsilon}^{+\infty} \frac{\partial k}{\partial s} \varphi ds] dy$$

由 $k(y, s) = \frac{1}{2a\sqrt{\pi s}} \exp\{-\frac{y^2}{4a^2 s}\} \rightarrow 0$ 当 $|y| \rightarrow \infty$ 时对

$$\begin{aligned} \int_{\varepsilon}^{+\infty} ds \int_{-\infty}^{+\infty} k(y, s) \frac{\partial^2 \varphi}{\partial y^2} dy &= \int_{\varepsilon}^{+\infty} ds [k \frac{\partial \varphi}{\partial y} \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{\partial k}{\partial y} \frac{\partial \varphi}{\partial y} dy] = - \int_{\varepsilon}^{+\infty} ds \int_{-\infty}^{+\infty} \frac{\partial k}{\partial y} \frac{\partial \varphi}{\partial y} dy \\ &= - \int_{\varepsilon}^{+\infty} ds [\frac{\partial k}{\partial y} \varphi \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{\partial^2 k}{\partial y^2} \varphi dy] = \int_{\varepsilon}^{+\infty} ds \int_{-\infty}^{+\infty} \frac{\partial^2 k}{\partial y^2} \varphi dy \end{aligned}$$

则有

$$\begin{aligned} \langle u_t - a^2 u_{xx}, \varphi \rangle &= \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^{+\infty} [k(y, \varepsilon) \varphi(y, \varepsilon) + \int_{\varepsilon}^{+\infty} \frac{\partial k}{\partial s} \varphi ds] dy - a^2 \int_{\varepsilon}^{+\infty} ds \int_{-\infty}^{+\infty} \frac{\partial^2 k}{\partial y^2} \varphi dy \\ &= \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^{+\infty} k(y, \varepsilon) \varphi(y, \varepsilon) dy + \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^{+\infty} ds \int_{-\infty}^{+\infty} \varphi \left(\frac{\partial k}{\partial s} - a^2 \frac{\partial^2 k}{\partial y^2} \right) dy \\ &= \lim_{\varepsilon \rightarrow 0^+} \int_{-\infty}^{+\infty} k(y, \varepsilon) \varphi(y, \varepsilon) dy \end{aligned}$$

结合 $k(y, \varepsilon) = \frac{1}{2a\sqrt{\pi\varepsilon}} \exp\{-\frac{y^2}{4a^2\varepsilon}\}$, 取 $x = \frac{y}{2a\sqrt{\varepsilon}}$, $dy = 2a\sqrt{\varepsilon}dx$, 则

$$\langle u_t - a^2 u_{xx}, \varphi \rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{\pi}} e^{-x^2} \varphi(2a\sqrt{\varepsilon}x, \varepsilon) dx = \varphi(0, 0).$$

结合平移不变性, 当 $\xi \neq 0, \tau \neq 0$ 时, 有

$$\langle u_t - a^2 u_{xx}, \varphi \rangle = \varphi(\xi, \tau) = \delta(x - \xi, t - \tau).$$

则有 $u(x, t) = k(x - \xi, t - \tau)$ 构成热传导方程的基本解.

□

另外, 我们还有基本解的另一个定义:

定义 3.3.2. 设 $Q = \{(x, t) | -\infty < x < \infty, t > 0\}$, 对任意 $\xi \in \mathbb{R}$, 如果函数 $u(x, t) \in L_{loc}(Q)$, 且在广义函数意义下满足

$$\begin{cases} u_t - a^2 u_{xx} = 0 & , (x, t) \in Q \\ u|_{t=0} = \delta(x - \xi) & , \end{cases}$$

则称它为热传导方程的基本解。

最后我们来讨论基本解的性质:

1. 当 $t > \tau$ 时, $\Gamma(x, t; \xi, \tau) > 0$.

2. $\Gamma(x, t; \xi, \tau) = \Gamma(\xi, t; x, \tau)$

3. 当 $t > \tau$, $-\infty < x < \infty$ 时

$$\int_{-\infty}^{+\infty} \Gamma(x, t; \xi, \tau) d\xi = 1$$

4. 当 $t > \tau, x, \xi \in (-\infty, \infty)$ 时,

$$\left(\frac{\partial}{\partial t} - a^2 \frac{\partial^2}{\partial x^2} \right) \Gamma(x, t; \xi, \tau) = 0, \quad \left(\frac{\partial}{\partial \tau} - a^2 \frac{\partial^2}{\partial \xi^2} \right) \Gamma(x, t; \xi, \tau) = 0$$

5. 当 $\varphi(x)$ 在 $(-\infty, \infty)$ 上有界连续时, 则

$$\lim_{t \rightarrow 0^+} \int_{-\infty}^{+\infty} \Gamma(x, t; \xi, 0) \varphi(\xi) d\xi = \varphi(x)$$

6. 当 $(x, t) \neq (\xi, \tau)$ 时, $\Gamma(x, t; \xi, \tau)$ 无穷次连续可微 (当 $t = \tau, x \neq \xi$ 时 $\Gamma(x, t; \xi, \tau)$ 定义为 0), 且有估计

$$|\Gamma(x, t; \xi, \tau)| \leq \frac{M}{\sqrt{t - \tau}}, \quad \text{当 } t > \tau \text{ 时}$$

其中 M 是常数。

3.4 半无界问题

与波动方程问题相同, 在这里我们来讨论热传导方程的半无界问题, 我们将问题分为三类

3.4.1 第一边值问题

我们在区域 $\bar{Q} = \{0 < x < \infty, 0 < t < \infty\}$ 中求解如下定解问题

$$\begin{cases} u_t - a^2 u_{xx} = f(x, t) & , (x, t) \in Q \\ u(0, t) = g(t) & , t > 0 \\ u(x, 0) = \varphi(x) & , 0 \leq x < \infty \end{cases}$$

先考虑 $g(t) \equiv 0$ 的情形, 我们仍对 f 和 φ 作奇延拓, 即

$$\bar{f}(x, t) = \begin{cases} f(x, t) & , x \geq 0 \\ -f(-x, t) & , x < 0 \end{cases} \quad \bar{\varphi}(x) = \begin{cases} \varphi(x) & , x \geq 0 \\ -\varphi(-x) & , x < 0 \end{cases}$$

则原问题又变为了

$$\begin{cases} \bar{u}_t - a^2 \bar{u}_{xx} = \bar{f}(x, t) \\ \bar{u}(x, 0) = \bar{\varphi}(x) \end{cases}$$

则方程的解为

$$\begin{aligned}\bar{u}(x, t) &= \int_{-\infty}^{+\infty} \Gamma(x - \xi, t) \bar{\varphi}(\xi) d\xi + \int_0^t d\tau \int_{-\infty}^{+\infty} \Gamma(x - \xi, t - \tau) \bar{f}(\xi, \tau) d\xi d\tau \\ &= \int_0^{+\infty} (\Gamma(x - \xi, t) - \Gamma(x + \xi, t)) \varphi(\xi) d\xi + \int_0^t d\tau \int_0^{+\infty} (\Gamma(x - \xi, t - \tau) - \Gamma(x + \xi, t - \tau)) f(\xi, \tau) d\xi\end{aligned}$$

此时我们只要将 $\bar{u}(x, t)$ 限制在 $x \geq 0$ 即可。当然，针对 $g(t) \neq 0$ 的问题，我们只要令 $u(x, t) = v(x, t) + g(t)$ ，经过变换即有对 $v(x, t)$ 有如原问题中 $g(t) = 0$ 的情形。

3.4.2 第二边值问题

我们在这里解决如下问题：

$$\begin{cases} u_t - a^2 u_{xx} = f(x, t) & , 0 < x < \infty, t > 0 \\ u_x(0, t) = g(t) & , t > 0 \\ u(x, 0) = \varphi(x) & , 0 \leq x < \infty \end{cases}$$

仍设 $g(t) = 0$ ，此时我们做偶延拓

$$\bar{\varphi}(x) = \begin{cases} \varphi(x) & , x \geq 0 \\ \varphi(-x) & , x < 0 \end{cases} \quad \bar{f}(x, t) = \begin{cases} f(x, t) & , x \geq 0 \\ f(-x, t) & , x < 0 \end{cases}$$

则原问题又变为了热传导方程形式，则有

$$\bar{u}(x, t) = \int_0^{+\infty} [\Gamma(x - \xi, t) + \Gamma(x + \xi, t)] \varphi(\xi) d\xi + \int_0^t d\tau \int_0^{+\infty} [\Gamma(x - \xi, t - \tau) + \Gamma(x + \xi, t - \tau)] f(\xi, \tau) d\xi$$

此时我们只要取 $u(x, t) = \bar{u}(x, t)|_{x \geq 0}$ 即可，同样地，若有 $g(t) \neq 0$ ，此时取 $u(x, t) = v(x, t) + xg(t)$ ，我们就可以将问题转化为 $g(t) = 0$ 的问题。

3.4.3 第三边值问题

此时我们将前两类问题结合在一起，求解如下问题：

$$\begin{cases} u_t - a^2 u_{xx} & , x > 0, t > 0 \\ u_x + \alpha u = 0 & , x = 0, t > 0 \\ u(x, 0) = \varphi(x) & , x \geq 0 \end{cases}$$

与前面的问题相似，我们希望利用延拓的思想，将问题转化为

$$\begin{cases} W_t - a^2 W_{xx} = 0 & , -\infty < x < \infty, t > 0 \\ W(x, 0) = \Phi(x) & , x \geq 0 \end{cases}$$

并且 $W(x, t)$ 应满足

$$u(x, t) = W(x, t)|_{x \geq 0}$$

其中 $\Phi(x)$ 是满足条件

$$\Phi(x) = \begin{cases} \varphi(x) & , x \geq 0 \\ \psi(x) & , x < 0 \end{cases}$$

的函数，同时我们希望 $W(x, t)$ 保证有条件

$$W_x(0, t) + \alpha W(0, t) = 0$$

成立，结合 Poisson 公式，我们有

$$W(x, t) = \int_{-\infty}^{+\infty} \Gamma(x - \xi, t) \Phi(\xi) d\xi$$

结合

$$\begin{aligned} W_x &= \int_{-\infty}^{+\infty} \Gamma_x(x - \xi, t) \Phi(\xi) d\xi \\ &= \Gamma(x, t)(\varphi(0) - \psi(0)) + \int_{-\infty}^{+\infty} \Gamma(x - \xi, t) \Phi'(\xi) d\xi \end{aligned}$$

此时如果我们假设 $\varphi(0) = \psi(0)$ ，结合条件：

$$W_x(0, t) + \alpha W(0, t) = 0$$

以及 $\Gamma(x, t)$ 是关于 x 的偶函数, 则有

$$\int_{-\infty}^{+\infty} \Gamma(-\xi, t) \Phi'(\xi) + \alpha \Gamma(\xi, t) \Phi(\xi) d\xi = 0$$

我们将 $(0, +\infty)$ 中的部分转化至 $(-\infty, 0)$ 部分, 有

$$\int_{-\infty}^0 \Gamma(\xi, t) (\psi'(\xi) + \alpha \psi(\xi) + \psi'(\xi) + \alpha \psi'(-\xi)) d\xi = 0$$

则问题转化为了如下 ODE 问题

$$\begin{cases} \psi'(\xi) + \alpha \psi(\xi) + \psi'(\xi) + \alpha \psi'(-\xi) & , \xi < 0 \\ \psi(0) = \varphi(0) \end{cases},$$

有 $\psi(\xi) = \varphi(-\xi) + 2\alpha e^{-\alpha\xi} \int_0^{-\xi} e^{-\alpha\eta} \varphi(\eta) d\eta$, $\xi < 0$, 只需代入 $W(x, t)$, 并将其限制在 $x \geq 0$ 即可。

$$W(x, t) = \int_0^{+\infty} [\Gamma(x + \xi, t) + \Gamma(x - \xi, t)] \varphi(\xi) d\xi + 2\alpha \int_0^{+\infty} e^{\alpha\xi} \Gamma(x + \xi, t) \int_0^{\xi} e^{-\alpha\eta} \varphi(\eta) d\eta d\xi$$

当然热传导方程同样有混合问题, 我们同样采用分离变量法求解问题, 这里我们就不再赘述了, 我们仅将结果陈述如下:

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi}{l} x$$

其中

$$T_n(t) = \varphi_n \exp \left\{ - \left(\frac{n\pi a}{l} \right)^2 t \right\} + \int_0^t f_n(\tau) \exp \left\{ - \left(\frac{n\pi a}{l} \right)^2 (t - \tau) \right\} d\tau, \quad n = 1, 2, \dots$$

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx, \quad f_n(t) = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx$$

例 3.4.1. 利用函数 $\Phi(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-y^2} dy$ 求解以下半无界问题 (1)

$$\begin{cases} u_t - a^2 u_{xx} = 0, & x > 0, t > 0 \\ u(x, 0) = 0, & x \geq 0 \\ u(0, t) = U_0, & t > 0 \end{cases} \quad (2) \begin{cases} u_t - a^2 u_{xx} = 0, & x > 0, t > 0 \\ u(x, 0) = \begin{cases} U_0, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases} \\ \frac{\partial u}{\partial x}(0, t) = 0, & t > 0 \end{cases}$$

例 3.4.2. (1) 解取 $u(x, t) = v(x, t) + U_0$, 则

$$\begin{cases} v_t - a^2 v_{xx} = 0, & x > 0, t > 0 \\ v(x, 0) = -U_0, & x \geq 0 \\ v(0, t) = 0, & t > 0 \end{cases}$$

取 $\bar{\varphi}(x) = \begin{cases} -U_0, & x \geq 0 \\ U_0, & x < 0 \end{cases}$, 则 $\bar{v}(x, t) = \frac{1}{2a\sqrt{\pi t}} \int_{-\infty}^{+\infty} \bar{\varphi}(\xi) \exp\{-\frac{(x-\xi)^2}{4a^2 t}\} d\xi$.

$$= \frac{1}{2a\sqrt{\pi t}} \left(\int_0^{+\infty} -U_0 \exp\{-\frac{(x-\xi)^2}{4a^2 t}\} d\xi + \int_{-\infty}^0 U_0 \exp\{-\frac{(x-\xi)^2}{4a^2 t}\} d\xi \right)$$

第一个积分中 $\eta = \frac{|x-\xi|}{2a\sqrt{t}}$ $d\xi = -2a\sqrt{t}d\eta$, 第二个积分中取 $\eta = \frac{\xi-x}{2a\sqrt{t}}$ $d\xi = 2a\sqrt{t}d\eta$.

$$\begin{aligned} \bar{v}(x, t) &= \frac{1}{\sqrt{\pi}} \left(\int_{\frac{x}{2a\sqrt{t}}}^{-\infty} + U_0 e^{-\eta^2} d\eta + \int_{-\infty}^{-\frac{x}{2a\sqrt{t}}} U_0 e^{-\eta^2} d\eta \right) \\ &= -\frac{U_0}{\sqrt{\pi}} \int_{-\frac{x}{2a\sqrt{t}}}^{\frac{x}{2a\sqrt{t}}} e^{-\eta^2} d\eta = -\frac{2U_0}{\sqrt{\pi}} \int_0^{\frac{x}{2a\sqrt{t}}} e^{-\eta^2} d\eta = -U_0 \Phi\left(\frac{x}{2a\sqrt{t}}\right) \end{aligned}$$

则有 $u(x, t) = -U_0 \Phi\left(\frac{x}{2a\sqrt{t}}\right) + U_0$

$$(2) \text{ 解取 } \varphi(x) = \begin{cases} U_0, & -1 \leq x \leq 1 \\ 0, & |x| > 1 \end{cases}. \text{ 则有}$$

$$u(x, t) = \frac{1}{2a\sqrt{\pi t}} \int_{-\infty}^{+\infty} \varphi(\xi) \exp\left\{-\frac{(x-\xi)^2}{4a^2 t}\right\} d\xi = \frac{1}{2a\sqrt{\pi t}} \int_{-1}^1 U_0 \exp\left\{-\frac{(x-\xi)^2}{4a^2 t}\right\} d\xi$$

$$\text{取 } \eta = \frac{|x-\xi|}{2a\sqrt{t}} \quad d\xi = -2a\sqrt{t}d\eta. \text{ 则 } u(x, t) = -\frac{1}{\sqrt{\pi}} \int_{\frac{x+1}{2a\sqrt{t}}}^{\frac{x-1}{2a\sqrt{t}}} U_0 e^{-\eta^2} d\eta = \frac{U_0}{2} \left(\Phi\left(\frac{x+1}{2a\sqrt{t}}\right) - \Phi\left(\frac{x-1}{2a\sqrt{t}}\right) \right) \quad \square$$

沿用第二章中的分离变量法, 我们这里也可以解决混合问题:

例 3.4.3. 用分离变量法求解下列混合问题 (1)

$$\begin{cases} u_t = a^2 u_{xx}, & 0 < x < l, t > 0 \\ u|_{t=0} = x^2(l-x), & 0 \leq x \leq l \\ u_x|_{x=0} = u|_{x=l} = 0, & t > 0 \end{cases}$$

(2)

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u|_{t=0} = \varphi(x), & 0 \leq x \leq l \\ \left[\frac{\partial u}{\partial x} + h(u - U_1)\right]_{x=0} = 0, & t > 0 \\ \left[\frac{\partial u}{\partial x} + h(u - U_2)\right]_{x=l} = 0, & t > 0 \end{cases}$$

其中在 (2) 中 $\varphi(x)$ 是有界连续函数, U_1, U_2, h 都是常数, 且 $h > 0$.

解答 (1) 解取 $u(x, t) = T(t)X(x)$, 则有 $\frac{T'(t)}{T(t)} - a^2 \frac{X''(x)}{X(x)} = 0$. 取 $\frac{X''(x)}{X(x)} = -\lambda$. 则得 $X_n(x) = C_n \cos \frac{n\pi + \frac{\pi}{2}}{l} x$, $n = 1, 2, \dots$

同时我们可以得到 ODE 问题

$$\begin{cases} T'_n(t) + \left(a \frac{n\pi + \frac{\pi}{2}}{l}\right)^2 T_n(t) = 0 \\ T_n(0) = \varphi_n = \frac{2}{l} \int_0^l x^2(l-x) \cos \frac{n\pi + \frac{\pi}{2}}{l} x dx \end{cases}$$

有 $T_n(t) = \varphi_n \exp\{-(a\frac{n\pi + \frac{\pi}{2}}{l})^2 t\}$ $u(x, t) = \sum_{n=1}^{\infty} \varphi_n \exp\{-(a\frac{n\pi + \frac{\pi}{2}}{l})^2 t\} \cos \frac{n\pi + \frac{\pi}{2}}{l} x$ 其中 $\varphi_n = \frac{64\pi l^3(2n+1)(-1)^n - 192l^3}{\pi^4(16n^4 + 32n^3 + 24n^2 + 8n + 1)}$ (反正我没算, 抄答案的)

(2) 想算的自己算吧, 我懒得算

□

3.5 极值原理

在本部分我们将会讨论在前面所讨论过的热传导问题的不同情形的极值原理, 所谓极值原理, 即物体的最高或最低温度总在物体的边界和初值时刻上达到。

首先我们明确 $Q = \{(x, t) | 0 < x < l, 0 < t \leq T\}$, Q 的侧边和底边为 $\{(x, t) | x = 0 \wedge x = l, 0 < t < T\} \cup \{(x, t) | t = 0, 0 < x < l\}$ 记作 Γ 或者 ∂Q , $C^{2,1}$ 表示在 Q 内对 x 二次连续可微, 对 t 一次连续可微的函数的集合。

首先我们介绍弱极值原理, 这是后面内容的基础:

定理 3.5.1 (弱极值原理). 设 $u \in C^{2,1}(Q) \cap C(\bar{Q})$ 且满足 $Lu = f \leq 0$, 则 u 在 $\bar{Q}$ 上的最大值必在 Q 的抛物边界 Γ 上达到, 即

$$\max_{\bar{Q}} u(x, t) = \max_{\Gamma} u(x, t)$$

解答 我们先证明 $f < 0$ 时的情形, 反设 $u(x, t)$ 的最大值在 Q 内部点 (x_0, t_0) 取到, 则有

$$\left. \frac{\partial u}{\partial t} \right|_{(x_0, t_0)} = 0 \quad \left. \frac{\partial^2 u}{\partial x^2} \right|_{(x_0, t_0)} \leq 0, \quad \begin{cases} \frac{\partial u}{\partial t} = 0, & \text{当 } t_0 < T \text{ 时 (在内部导数为零)} \\ \frac{\partial u}{\partial t} \geq 0, & \text{当 } t_0 = T \text{ 时 (可能在边界处仍在上升)} \end{cases}$$

则 $f(x, t) = \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} \geq 0$ 与 $f < 0$ 的假设矛盾。

下面我们考虑 $f \leq 0$ 的情形, 设 $v(x, t) = u(x, t) - \varepsilon t$, 其中 ε 为任意大于零的数, 则 v 应满足

$$\frac{\partial v}{\partial t} - a^2 \frac{\partial^2 v}{\partial x^2} = f - \varepsilon < 0.$$

则 $\max_{\bar{Q}} u(x, t) = \max_{\bar{Q}} (v(x, t) + \varepsilon t) \leq \max_{\bar{Q}} v(x, t) + \varepsilon T = \max_{\Gamma} v(x, t) + \varepsilon T \leq \max_{\Gamma} u(x, t) + \varepsilon T$. 结合 $\max_{\bar{Q}} u(x, t) \geq \max_{\Gamma} u(x, t)$ 与 ε 的任意性, 我们有

$$\max_{\bar{Q}} u(x, t) = \max_{\Gamma} u(x, t) \quad \text{成立} \quad \square$$

□

他有这样两条推论:

推论 3.5.1. 设 $u \in C^{2,1}(Q) \cap C(\bar{Q})$ 且满足 $Lu = f \geq 0$, 则 u 在 $\bar{Q}$ 上的最小值必在 Γ 上达到, 即

$$\min_{\bar{Q}} u = \min_{\Gamma} u$$

如果 $Lu = 0$ 则 u 在 $\bar{Q}$ 上的最大值与最小值都必在抛物边界 Γ 上达到。

推论 3.5.2 (比较原理). 设 $u, v \in C^{2,1}(Q) \cap C(\bar{Q})$, 且有 $Lu \leq Lv$, $u|_{\Gamma} \leq v|_{\Gamma}$, 则在 $\bar{Q}$ 上 $u(x, t) \leq v(x, t)$.

例 3.5.1. 记 $Q_l = \{0 < x < l, 0 < t \leq T\}$, 设 $u \in C^{2,1}(Q_l) \cap C(\bar{Q})$ 是定解问题

$$\begin{cases} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0 & (x, t) \in Q_l \\ u|_{t=0} = 0 & 0 \leq x \leq l \\ u(0, t) = g(t), u(l, t) = 0 & 0 \leq t \leq T \end{cases}$$

的解. 其中 $g(t) \geq 0$, 证明 $u(x, t)$ 关于 l 是递增的, 即对于 $l_1 < l_2$

$$u_{l_1}(x, t) \leq u_{l_2}(x, t) \quad (x, t) \in \bar{Q}_{l_1}$$

解答 设 $w(x, t) = u_{l_2}(x, t) - u_{l_1}(x, t)$, 则 $w(x, t)$ 满足关系式:

$$\begin{cases} \frac{\partial w}{\partial t} - \frac{\partial^2 w}{\partial x^2} = 0, & (x, t) \in Q_{l_1} \\ w|_{t=0} = 0 & 0 \leq x \leq l_1 \\ w(0, t) = 0, \quad w(l_1, t) = u_{l_2}(l_1, t), & 0 \leq t \leq T. \end{cases}$$

结合弱极值原理, 由 $w(x, 0) = 0$, $w(0, t) = 0$, 我们只要保证 $u_{l_2}(l_1, t) \geq 0$ 即可. 再结合弱极值原理, 由 $u_{l_2}(x, 0) = 0$, $u_{l_2}(l_2, t) = 0$, $u_{l_2}(0, t) = g(t) \geq 0$ 则 $u_{l_2}(l_1, t) \geq 0$ 成立. 则原命题成立. □

例 3.5.2. 设 $u \in C(\bar{Q}) \cap C^{2,1}(Q)$ 且满足

$$Lu \equiv u_t - a^2 u_{xx} + c(x, t)u \leq 0, \quad (x, t) \in Q$$

其中 $c(x, t)$ 有界, 且 $c(x, t) \geq 0$, 试证明: 如果 u 在 $\bar{Q}$ 上存在非负最大值, 则 u 必在抛物边界 Γ 上达到其在 $\bar{Q}$ 上的非负最大值.

解答 先研究 $Lu < 0$ 的情形, 若 u 在 $\bar{Q}$ 上的非负最大值不在 Γ 上取到, 即 $\exists (x_0, t_0) \in Q \setminus \Gamma$ s.t.

$$u(x_0, t_0) = \max_{\bar{Q}} u(x, t).$$

则有 $\frac{\partial u}{\partial t}(x_0, t_0) = 0$ (当 $t_0 < T$ 时), $\frac{\partial u}{\partial t}|_{(x_0, t_0)} \geq 0$ (当 $t_0 < T$ 时), $u(x_0, t_0) \geq 0$, $\frac{\partial u}{\partial x}|_{(x_0, t_0)} = 0$, $\frac{\partial^2 u}{\partial x^2}|_{(x_0, t_0)} \leq 0$, 则 $Lu = u_t - a^2 u_{xx} + c(x, t)u \geq 0$, 矛盾.

再考虑 $Lu \leq 0$ 的情形, 引入辅助函数 $v(x, t) = u(x, t) - \varepsilon t$, 则其有非负最大值时一定在边界上, 记其最大值为负则有 $v < 0$, 则取 $v^+ = \max\{0, v\}$, 则 $\max_{\bar{Q}} v \leq \max_{\Gamma} \{0, v\} = \max_{\Gamma} v^+$. 则有

$$\begin{aligned} \max_{\bar{Q}} u(x, t) &= \max_{\bar{Q}} (v(x, t) + \varepsilon t) \leq \max_{\bar{Q}} v(x, t) + \varepsilon T \leq \max_{\Gamma} v^+ + \varepsilon T = \max_{\Gamma} (u - \varepsilon t)^+ + \varepsilon T \\ &= \max_{\Gamma} \max\{0, u - \varepsilon t\} + \varepsilon T \leq \max_{\Gamma} \max\{\varepsilon t, u\} + 2\varepsilon T. \end{aligned}$$

取 $\varepsilon \rightarrow 0$, 有 $\max_{\bar{Q}} u(x, t) \leq \max_{\Gamma} \{0, u\}$, 则若 u 在 $\bar{Q}$ 上存在非负最大值.

$$\max_{\bar{Q}} u \leq \max_{\Gamma} u. \quad \text{而} \quad \max_{\bar{Q}} u \geq \max_{\Gamma} u.$$

综上 $\max_{\bar{Q}} u = \max_{\Gamma} u$. □

解答 证明半无界问题

$$\begin{cases} \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t), & 0 < x < \infty, t > 0 \\ u|_{t=0} = \varphi(x), & 0 \leq x < \infty \\ u|_{x=0} = \mu(t), & t \geq 0 \end{cases}$$

的有界解是唯一的. □

解答 设 $u_1(x, t), u_2(x, t)$ 为方程的解, 则令 $w(x, t) = u_1(x, t) - u_2(x, t)$. 则 $w(x, t)$ 满足

$$\begin{cases} \frac{\partial w}{\partial t} - a^2 \frac{\partial^2 w}{\partial x^2} = 0, & 0 < x < T, \quad 0 \leq t \leq T \\ w|_{t=0} = 0, & 0 \leq x < T \\ w|_{x=0} = 0, & 0 \leq t \leq T \end{cases}$$

我们将 w 进行奇延拓, 表达式与上式相同. 我们只要证 $w(x, t) \equiv 0$ 即可. 我们取 $v(x, t) = \frac{k}{T^2}(x^2 + 2a^2t) \pm w(x, t)$ (令 $k = \sup |w(x, t)|$) 则

$$\begin{cases} \frac{\partial v}{\partial t} - a^2 \frac{\partial^2 v}{\partial x^2} = \frac{\partial w}{\partial t} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ v(x, 0) = \frac{k}{T^2}x^2 \\ v(0, t) = \frac{k}{T^2}2a^2t \\ v(T, t) = k + 2a^2\frac{k}{T^2}t \pm w(T, t). \end{cases}$$

在 $\bar{Q}_T$, 则结合弱极值原理, v 的最小值在边界上取到当 $t = 0$ 时 $v(x, 0) = \frac{k}{T^2}x^2 \geq 0$ $x = 0$ 时 $v(0, t) = \frac{2a^2k}{T^2}t \geq 0$ $x = T$ 时 $v(T, t) = k + 2a^2\frac{k}{T^2}t \pm w(T, t) \geq 0$ 则 $|w(x, t)| \leq \frac{k}{T^2}x^2 + 2a^2\frac{k}{T^2}t$ □

下面讨论第一边值混合问题, 即

$$\begin{cases} Lu = u_t - a^2 u_{xx} = f & , (x, t) \in Q \\ u \Big|_{t=0} = \varphi(x) & , 0 \leq x \leq l \\ u \Big|_{x=0} = g_1(t), u \Big|_{x=l} = g_2(t) & , 0 \leq t \leq T \end{cases}$$

为了研究这个方程的性质, 我们需要如下引理

引理 3.5.1. 设 $u(x, t) \in C^{2,1}(Q) \cap C^{1,0}(\bar{Q})$ 满足

$$\begin{aligned} & Lu = u_t - a^2 u_{xx} \geq 0, \quad (x, t) \in Q \quad u|_{t=0} \geq 0, \quad 0 \leq x \leq l \\ & \left[-\frac{\partial u}{\partial x} + \alpha(t)u \right] \Big|_{x=0} \geq 0, \quad 0 \leq t \leq T \quad \left[\frac{\partial u}{\partial x} + \beta(t)u \right] \Big|_{x=l} \geq 0, \quad 0 \leq t \leq T. \end{aligned}$$

则在 $\bar{Q}$ 上 $u(x, t) \geq 0$

解答 先证 $-\frac{\partial u}{\partial x} + \alpha(t)u \Big|_{x=0} > 0, \frac{\partial u}{\partial x} + \beta(t)u \Big|_{x=l} > 0$ 的情形, 反设 $u(x, t)$ 在 $\bar{Q}$ 上取到负极大值, 结合 $Lu \geq 0$, 则 Q 的最小值在 Γ 上取到, 设 (x_0, t_0) 为极小值点. 若 $x_0 = 0$, 则有 $-\frac{\partial u}{\partial x} \Big|_{P_0} \leq 0$, 且 $\alpha(t)u \Big|_{P_0} \leq 0$. 则 $-\frac{\partial u}{\partial x} + \alpha(t)u \Big|_{x=0} \leq 0$ 与假设矛盾. 若 $x_0 = l$ 与 $x_0 = 0$ 相同. 若 $t = 0$, 结合 $u|_{t=0} \geq 0$, 我们有假设不成立, 则 $u(x, t) \geq 0$. 下面讨论一般情形, 设

$$v(x, t) = u(x, t) + \varepsilon[2a^2t + (x - \frac{l}{2})^2]$$

$$\text{则有 } Lv = v_t - a^2v_{xx} = u_t - a^2u_{xx} + 2a^2\varepsilon - 2\varepsilon a^2 = Lu \geq 0 \quad v|_{t=0} = u|_{t=0} + \varepsilon(x - \frac{l}{2})^2 \geq 0$$

$$\begin{aligned} -\frac{\partial v}{\partial x} + \alpha(t)v \Big|_{x=0} &= -\frac{\partial u}{\partial x} + \alpha(t)u \Big|_{x=0} - 2\varepsilon(x - \frac{l}{2}) + \varepsilon\alpha(t)[2a^2t + (x - \frac{l}{2})^2] \Big|_{x=0} \\ &= -\frac{\partial u}{\partial x} + \alpha(t)u \Big|_{x=0} + \varepsilon(l + \alpha(t)(2a^2t + \frac{l^2}{4})) > 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial v}{\partial x} + \beta(t)v \Big|_{x=l} &= \frac{\partial u}{\partial x} + \beta(t)u \Big|_{x=l} + 2\varepsilon(x - \frac{l}{2}) + \varepsilon\beta(t)[2a^2t + (x - \frac{l}{2})^2] \Big|_{x=l} \\ &= \frac{\partial u}{\partial x} + \beta(t)u \Big|_{x=l} + \varepsilon(l + \beta(t)(2a^2t + \frac{l^2}{4})) > 0 \end{aligned}$$

则有 $v(x, t) \geq 0$ 在 $\bar{Q}$ 上成立. 结合 $u(x, t) = v(x, t) - \varepsilon[2a^2t + (x - \frac{l}{2})^2]$ 取 $\varepsilon \rightarrow 0$, 有 $u(x, t) \geq 0$ 成立. □ □

定理 3.5.2. 设 $u(x, t) \in C^{2,1}(Q) \cap C(\bar{Q})$ 是上述问题的解, 则

$$\max_{\bar{Q}} |u| \leq FT + B$$

其中

$$F = \sup_Q |f|, \quad B = \max \left\{ \sup_{[0,l]} |\varphi|, \sup_{[0,T]} |g_1|, \sup_{[0,T]} |g_2| \right\}$$

解答 我们引入辅助函数 $w(x, t) = Ft + Bz(x, t) \pm u(x, t)$, 其中 $z(x, t) = 1 + \frac{1}{l}[2a^2t + (x - \frac{l}{2})^2]$, 则

$$Lz = z_t - a^2z_{xx} = \frac{1}{l}(2a^2 - 2a^2) = 0 \quad z(x, 0) = 1 + \frac{1}{l}(x - \frac{l}{2})^2 \geq 1$$

$$-\frac{\partial z}{\partial x} + \alpha(t)z \Big|_{x=0} = -\frac{2}{l}(x - \frac{l}{2}) + \alpha(t)(1 + \frac{1}{l}[2a^2t + (x - \frac{l}{2})^2]) \Big|_{x=0} = 1 + \alpha(t)(1 + \frac{1}{l}(2a^2t + \frac{l^2}{4})) \geq 1$$

$$\frac{\partial z}{\partial x} + \beta(t)z \Big|_{x=l} = \frac{2}{l}(x - \frac{l}{2}) + \beta(t)(1 + \frac{1}{l}[2a^2t + (x - \frac{l}{2})^2]) \Big|_{x=l} = 1 + \beta(t)(1 + \frac{1}{l}(2a^2t + \frac{l^2}{4})) \geq 1$$

则有

$$Lw = w_t - a^2w_{xx} = F \pm Lu \geq 0 \quad w(x, 0) = Bz(x, 0) \pm u(x, 0) \geq 0$$

$$-\frac{\partial w}{\partial x} + \alpha(t)w \Big|_{x=0} = B(-\frac{\partial z}{\partial x} + \alpha(t)z) \pm (-\frac{\partial u}{\partial x} + \alpha(t)u) \Big|_{x=0} \geq B \pm (u_x(0, t) + \alpha(t)u(0, t)) \geq 0$$

$$\frac{\partial w}{\partial x} + \beta(t)w \Big|_{x=l} = B(\frac{\partial z}{\partial x} + \beta(t)z) \pm (\frac{\partial u}{\partial x} + \beta(t)u) \Big|_{x=l} \geq B \pm g_2(t) \geq 0$$

则 $w(x, t) \geq 0$, 则有

$$|u(x, t)| \leq Ft + B(1 + \frac{1}{l}[2a^2t + (x - \frac{l}{2})^2]) \leq FT + B(1 + \frac{1}{l}[2a^2T + (x - \frac{l}{2})^2])$$

取 $C = \max\{T, 1 + \frac{1}{l}[2a^2T + \frac{l^2}{4}]\}$, 有 $|u(x, t)| \leq C(F + B)$

□

□ 它同样具有两条推论:

推论 3.5.3. 上述问题在 $C^{2,1}(Q) \cap C(\bar{Q})$ 中的解是唯一的。

推论 3.5.4. 上述问题的在 $C^{2,1}(Q) \cap C(\bar{Q})$ 中的解连续依赖于非齐次项 f , 初值 φ 与边值 g_1, g_2 。

例 3.5.3. 设 $u \in C^{2,1}(\bar{Q}), u_t \in C^{2,1}(Q)$ 且满足以下定解问题

$$\begin{cases} u_t - u_{xx} = f(x, t), & (x, t) \in Q \\ u|_{t=0} = \varphi(x), & 0 \leq x \leq l \\ u|_{x=0} = u|_{x=l} = 0, & 0 \leq t \leq T \end{cases}$$

则有以下估计

$$\max_{\bar{Q}} |u_t(x, t)| \leq C[\|f\|_{C^1(\bar{Q})} + \|\varphi''\|_{C[0, l]}],$$

其中 C 依赖于 T .

解答 取 $u_t = v$, 则 v 满足 $v - u_{xx} = f(x, t) \implies v_t - u_{xx} = f_t(x, t)$. 则

$$\begin{cases} v_t - v_{xx} = f_t(x, t) & (x, t) \in Q \\ v|_{t=0} = u_t|_{t=0} = u_{xx}|_{t=0} + f(x, 0) = \varphi''(x) + f(x, 0), & 0 \leq x \leq l \\ v|_{x=0} = v|_{x=l} = u_t|_{x=0} = u_t|_{x=l} = 0 & 0 \leq t \leq T. \end{cases}$$

则结合第一边值问题的最大模估计

$$\begin{aligned} \max_{\bar{Q}} |v(x, t)| &= \max_{\bar{Q}} |u_t(x, t)| \leq \sup_{\bar{Q}} |f_t| T + \sup_{x \in [0, l]} |f(x, 0) + \varphi''(x)| \\ &\leq \max\{T, 1\} (\sup_{\bar{Q}} |f_t| + \sup_{x \in [0, l]} |f(x, 0)| + \sup_{x \in [0, l]} |\varphi''(x)|) \\ &\leq \max\{T, 1\} [\|f\|_{C^1(\bar{Q})} + \|\varphi''\|_{C[0, l]}]. \end{aligned}$$

注:

$$\|u\|_{C^k(\Omega)} = \sup_{x \in \Omega} |u(x)| + \sum_{n=1}^k \sum_{\alpha_1 + \dots + \alpha_t = n} \sup_{x \in \Omega} \left| \frac{\partial^n u}{\partial x_1^{\alpha_1} \dots \partial x_t^{\alpha_t}} \right|$$

□

例 3.5.4. 设 $u \in C^{1,0}(\bar{Q}) \cap C^{2,1}(Q)$ 且满足定解问题

$$\begin{cases} u_t - u_{xx} = 0, & (x, t) \in Q \\ u|_{t=0} = \varphi(x), & 0 \leq x \leq l \\ u|_{x=0} = u|_{x=l} = 0, & 0 \leq t \leq T \end{cases}$$

(1) 试证明 $\max_{[0, T]} \left| \frac{\partial u(0, t)}{\partial x} \right|, \max_{[0, T]} \left| \frac{\partial u(l, t)}{\partial x} \right| \leq C$. 其中 C 仅依赖于 $\|\varphi\|_{C^1[0, l]}$

(2) 又设 $u_x \in C^{2,1}(Q)$, 则 $\max_{\bar{Q}} \left| \frac{\partial u(x, t)}{\partial x} \right| \leq \tilde{C}$, 其中 $\tilde{C}$ 也仅依赖于 $\|\varphi\|_{C^1[0, l]}$

解答 (1) 证明: 首先我们来说明 $\forall t \in [0, T]$, 都有 $\left| \frac{\partial u(0, t)}{\partial x} \right| \leq C(\|\varphi\|_{C^1[0, l]})$. 由 $u_x(0, t) = \lim_{x \rightarrow 0} \frac{u(x, t) - u(0, t)}{x} = \lim_{x \rightarrow 0} \frac{u(x, t)}{x}$ 则若记 $C = \|\varphi'\|$, 若我们可以证明 $\frac{u(x, t)}{x} \leq C \quad \forall (x, t) \in [0, l] \times [0, T]$ 成立, 命题就成立了. 取 $w(x, t) = Cx \pm u(x, t)$, 则 $w_t - w_{xx} = 0$,

则结合弱极值原理, 其最小值在边界上取到, 则对 $x = 0, l$ 有 $w(0, t) = \pm u(0, t) = 0$ $w(l, t) = lC \pm u(l, t) = lC \geq 0$ 对 $t = 0$, 有 $w(x, 0) = Cx \pm \varphi(x)$, 结合 $u(0, 0) = \varphi(0) = 0$ $\exists \xi \in (0, x)$ s.t. 则 $|\varphi(x)| = |\varphi(x) - \varphi(0)| = |\varphi'(\xi)|x \leq Cx$, 则 $w \geq 0$ 成立. 则 $|u(x, t)| \leq Cx$. 同理可证明 $\max_{[0, T]} \left| \frac{\partial u(l, t)}{\partial x} \right| \leq C$ 成立.

(2) 解取 $v = u_x$, 则有

$$\begin{cases} v_t - v_{xx} = 0 \\ v|_{t=0} = \varphi'(x) \\ v|_{x=0} = g_1(t), v|_{x=l} = g_2(t) \end{cases} \quad \text{其中 } |g_1(t)| \leq C, \quad |g_2(t)| \leq C$$

则结合第一边值条件, 有

$$\max_{\bar{Q}} |v(x, t)| \leq \max \left\{ \sup_{[0, l]} |\varphi'|, \sup_{(0, T)} |g_1(t)|, \sup_{(0, T)} |g_2(t)| \right\} \leq \max \left\{ \sup_{[0, l]} |\varphi'|, C \right\} \triangleq \tilde{C}$$

□

而对第二、第三边值条件, 我们表述如下:

$$\begin{cases} Lu = u_t - a^2 u_{xx} = f & , (x, t) \in Q \\ u|_{t=0} = \varphi(x) & , 0 \leq x \leq l \\ \left[-\frac{\partial u}{\partial x} + \alpha(t)u \right]_{x=0} = g_1(t) & , 0 \leq t \leq T \\ \left[\frac{\partial u}{\partial x} + \beta(t)u \right]_{x=l} = g_2(t) & , 0 \leq t \leq T \end{cases}$$

其中 $\alpha(t) \geq 0, \beta(t) \geq 0$, 本部分的估计式相对较为复杂:

定理 3.5.3. 设 $u(x, t) \in C^{2,1}(Q) \cap C^{1,0}(\bar{Q})$ 是上述问题的解, 则

$$|u(x, t)| \leq C(F + B)$$

其中常数 C 只依赖于 a, l 和 T , 而

$$F = \sup_Q |f|, \quad B = \max \left\{ \max_{[0, T]} |g_1|, \max_{[0, T]} |g_2|, \max_{[0, l]} |\varphi| \right\}$$

下面讨论初值问题解的最大模估计, 其帮助我们证明初值问题解的唯一性:

定理 3.5.4. 设 $u(x, t) \in C^{2,1}(Q) \cap C(\bar{Q})$ 是热传导方程初值问题的有界解, 则有估计

$$\sup_Q |u(x, t)| \leq T \sup_Q |f(x, t)| + \sup_{(-\infty, \infty)} |\varphi(x)|$$

最后我们来看边值问题解的能量模估计, 记 $Q_T = \{(x, t) | 0 < x < l, 0 < t \leq T\}$, 在 Q_T 上考虑混合问题:

$$\begin{cases} \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f & , (x, t) \in Q_T \\ u(x, 0) = \varphi(x) & , 0 \leq x \leq l \\ u(0, t) = u(l, t) = 0 & , 0 \leq t \leq T \end{cases}$$

我们有

定理 3.5.5. 设 $u \in C^{1,0}(\bar{Q}_T) \cap C^{2,1}(Q_T)$ 是上述问题的解, 则我们有估计

$$\sup_{0 \leq t \leq T} \int_0^l u^2(x, t) dx + 2a^2 \int_0^T \int_0^l \left(\frac{\partial u}{\partial x} \right)^2 dx dt \leq M \left(\int_0^l \varphi^2(x) dx + \int_0^T \int_0^l f^2(x, t) dx dt \right)$$

其中 M 只与 T 有关。

解答 对 $\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f$ 同乘 u , 并在 Q_t 上求积分

$$\begin{aligned} & \int_0^t d\tau \int_0^l u(x, \tau) \frac{\partial u}{\partial t}(x, \tau) dx - a^2 \int_0^t d\tau \int_0^l u(x, \tau) \frac{\partial^2 u}{\partial x^2}(x, \tau) dx \\ &= \frac{1}{2} \int_0^t d\tau \int_0^l \frac{\partial u^2(x, \tau)}{\partial t} dx - a^2 \left(\int_0^t u(x, \tau) \frac{\partial u}{\partial x} \Big|_0^l - \int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial x} \right)^2 dx \right) \\ &= \frac{1}{2} \int_0^t u^2(x, t) dx - \frac{1}{2} \int_0^l \varphi^2(x) dx + a^2 \int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial x} \right)^2 dx \quad (u(0, t) = u(l, t) = 0) \\ &= \int_0^t d\tau \int_0^l u f dx \leq \frac{1}{2} \int_0^t d\tau \int_0^l u^2(x, \tau) dx + \frac{1}{2} \int_0^t d\tau \int_0^l f^2(x, \tau) dx \end{aligned}$$

取 $G(t) = \int_0^t d\tau \int_0^l u^2(x, \tau) dx$, 则 $\frac{d}{dt}G(t) = \int_0^l u^2(x, t) dx$, $G(0) = 0$ 取 $F(t) = \int_0^t d\tau \int_0^l f^2(x, \tau) dx + \int_0^l \varphi^2(x) dx$, 则有

$$\frac{d}{dt}G(t) \leq F(t) + G(t).$$

结合 Gronwall 不等式, 我们有

$$\begin{aligned} \int_0^l u^2(x, t) dx + 2a^2 \int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial x}\right)^2 dx &\leq (1 + e^t) \left(\int_0^l \varphi^2(x) dx + \int_0^t d\tau \int_0^l f^2(x, \tau) dx \right) \\ &\leq (1 + e^T) \left(\int_0^l \varphi^2(x) dx + \int_0^T dt \int_0^l f^2(x, t) dx \right) \end{aligned}$$

则有 $\sup_{0 \leq t \leq T} \int_0^l u^2(x, t) dx + 2a^2 \int_0^T dt \int_0^l \left(\frac{\partial u}{\partial x}\right)^2 dx \leq M \left(\int_0^l \varphi^2(x) dx + \int_0^T dt \int_0^l f^2(x, t) dx \right)$ □ 我们使用同样的思路, 可以得到另一种模估计:

$$a^2 \sup_{0 \leq t \leq T} \int_0^l \left(\frac{\partial u}{\partial x}\right)^2 dx + 2 \int_0^T \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx dt \leq M_1 \left[\int_0^l (\varphi'(x))^2 dx + \int_0^T \int_0^l f^2(x, t) dx dt \right]$$

例 3.5.5. 设 $u(x, t) \in C^{2,1}(\bar{Q})$ 是问题

$$\begin{cases} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f, & (x, t) \in Q \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \\ u(0, t) = u(l, t) = 0, & 0 \leq t \leq T \end{cases}$$

的解, 满足以下估计

$$\sup_{0 \leq t \leq T} \int_0^l \left(\frac{\partial u}{\partial x}\right)^2 dx + \int_0^T \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx dt \leq M \left[\int_0^l (\varphi'(x))^2 dx + \int_0^T \int_0^l f^2(x, t) dx dt \right]$$

解答 对方程两侧同乘 $\frac{\partial u}{\partial t}$, 并在 Q_t 上求积分

$$\begin{aligned}
\int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 - \frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial x^2} dx &= \int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx - \int_0^t d\tau \left(\frac{\partial u}{\partial t} \frac{\partial u}{\partial x}\right) \Big|_0^l - \int_0^l \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial x \partial t} dx \\
&= \int_0^t d\tau \left(\int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx + \int_0^l \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial x \partial t} dx \right) \\
&= \int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx + \int_0^l dx \left(\frac{1}{2} \left(\frac{\partial u}{\partial x}\right)^2 \right) \Big|_0^t - \int_0^t \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial x \partial t} dt \\
&= \int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx + \frac{1}{2} \int_0^l u_x^2(x, t) - u_x^2(x, 0) dx = \int_0^t d\tau \int_0^l f \frac{\partial u}{\partial t} dx
\end{aligned}$$

则

$$\int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx + \frac{1}{2} \int_0^l \left(\frac{\partial u}{\partial x}\right)^2 dx \leq \frac{1}{2} \int_0^l (\varphi'(x))^2 dx + \frac{1}{2} \int_0^t d\tau \int_0^l f^2 dx + \frac{1}{2} \int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx$$

即

$$\int_0^t d\tau \int_0^l \left(\frac{\partial u}{\partial t}\right)^2 dx + \int_0^l \left(\frac{\partial u}{\partial x}\right)^2 dx \leq \int_0^l (\varphi'(x))^2 dx + \int_0^t d\tau \int_0^l f^2 dx$$

□

4 位势方程

本章我们主要来研究位势方程

$$-\Delta u = f$$

一般的, 我们称上述方程为泊松方程, 若 $f = 0$, 则我们称之为 Laplace 方程。

4.1 基本解

首先我们来讨论位势方程的基本解:

定义 4.1.1. 如果 $U \in L_{loc}(\mathbb{R}^n)$ 在广义函数意义下满足

$$-\Delta U = \delta(x - \xi), \quad x, \xi \in \mathbb{R}^n$$

即对于任意 $\varphi \in \mathcal{D}(\mathbb{R}^n)$, 有

$$\int \cdots \int_{\mathbb{R}^n} U(-\Delta \varphi) dx = \varphi(\xi)$$

则称 U 为 n 维 Laplace 方程的一个基本解, 记作 $\Gamma(x; \xi)$ 。

在推导基本解的过程中, 我们需要用到 Green 第二恒等式:

定理 4.1.1. 设 $\partial\Omega$ 分段光滑, $u, v \in C^2(\Omega) \cap C^1(\bar{\Omega})$, 则我们有以下 Green 公式:

$$\iint_{\Omega} (u \Delta v - v \Delta u) dx dy = \int_{\partial\Omega} \left(u \frac{\partial v}{\partial \mathbf{n}} - v \frac{\partial u}{\partial \mathbf{n}} \right) dl$$

其中 $\mathbf{n}$ 表示 $\partial\Omega$ 的单位外法向量。

下面我们来看 Laplace 方程的基本解具有什么样的形式:

对方程的解 $\Phi(x)$, 则有 $\Delta \Phi = \delta_0$. 结合 Laplace 算子的旋转对称性, 我们可以将 $\Phi(x)$ 写作 $\Phi(x) = \phi(r)$ 的形式, 其中 $r = \sqrt{\sum_{i=1}^n x_i^2}$.

则有

$$\frac{\partial \Phi}{\partial x_i} = \frac{\partial}{\partial x_i}(\phi(r)) = \phi'(r) \frac{\partial r}{\partial x_i} = \frac{x_i}{r} \phi'(r)$$

$$\frac{\partial^2 \Phi}{\partial x_i^2} = \frac{\partial}{\partial x_i} \left(\frac{x_i}{r} \phi'(r) \right) = \frac{1}{r} \phi'(r) - \frac{x_i}{r^2} \cdot \frac{x_i}{r} \phi'(r) + \left(\frac{x_i}{r} \right)^2 \phi''(r) = \frac{1}{r} \phi'(r) - \frac{x_i^2}{r^3} \phi'(r) + \frac{x_i^2}{r^2} \phi''(r)$$

$$\text{则 } \Delta \Phi = \sum_{i=1}^n \frac{\partial^2 \Phi}{\partial x_i^2} = \frac{n-1}{r} \phi'(r) + \phi''(r) = 0$$

$$\textcircled{1} \ n=1 \text{ 时 } \phi(r) = Ar + B \triangleq Ar \quad (C \text{ 一般取 } B=0)$$

$$\textcircled{2} \ n=2 \text{ 时取 } \psi(r) = \phi'(r), \text{ 即 } \frac{1}{r} \psi(r) + \psi'(r) = 0 \implies \psi(r) + r\psi'(r) = 0$$

$$\psi(r) = \frac{C}{r} \implies \phi(r) = A \log r$$

$$\textcircled{3} \ n \geq 3 \text{ 时 } \psi'(r) + \frac{n-1}{r} \psi(r) = 0, \text{ 即 } r^{n-1} \psi'(r) + (n-1)r^{n-2} \psi(r) = 0$$

$$\text{则 } \psi(r) = Cr^{1-n} \quad \text{则 } \phi(r) = Ar^{2-n}$$

为保证 $\forall \varphi \in C_0^\infty(\mathbb{R}^n)$, 都有

$$\langle -\Delta \Phi, \varphi \rangle = \delta(0) = \langle \delta, \varphi \rangle$$

则结合

$$\begin{aligned} \int_{\mathbb{R}^n} \Phi(x) \Delta \varphi(x) dx &= \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n \setminus B_\varepsilon} \Phi(x) \Delta \varphi(x) dx = \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n \setminus B_\varepsilon} (\Phi(x) \Delta \varphi(x) - \Delta \Phi(x) \varphi(x)) dx \\ &= \lim_{\varepsilon \rightarrow 0} \int_{\partial \Omega_\varepsilon} \left(\Phi \frac{\partial \varphi}{\partial n} - \varphi \frac{\partial \Phi}{\partial n} \right) dS \quad \text{其中 } \Omega_\varepsilon = \{(x, y) \mid \varepsilon^2 < |x|^2 + y^2 < \frac{1}{\varepsilon}\} \end{aligned}$$

$$\text{且有 } \int_{\partial B_\varepsilon} dS = \omega_n \varepsilon^{n-1}, \quad \omega_n = \frac{(2\pi)^{\frac{n}{2}}}{\Gamma(\frac{n}{2})} \text{ 结合 } \frac{\partial \Phi}{\partial n} = -\phi'(r), \quad \varphi \in C_c^\infty(\mathbb{R}^n).$$

我们有 $\exists M > 0$ s.t. $|\varphi(x)| < M$, 且 $|\frac{\partial \varphi}{\partial n}| < M \quad \forall x \in \partial \Omega_\varepsilon$ 且 $\varphi|_{\partial \Omega_\varepsilon} = \varphi(0) + O(\varepsilon)$. 则对

$$\textcircled{1} \ n=1 \text{ 时, 结合 } \frac{\partial^2 \Phi}{\partial x^2} = \delta_0. \text{ 由 } \frac{d}{dx} |x| = \text{sign}(x) \quad \frac{d^2}{dx^2} = \frac{d}{dx} \text{sign}(x) = 2\delta. \text{ 则 } \frac{d^2}{dx^2} \frac{1}{2} |x| = \delta_0, \text{ 则有 } \Phi(x) = \frac{1}{2} |x|.$$

$$\textcircled{2} \ n=2 \text{ 时 } \phi(r) = A \log r \quad \phi'(r) = \frac{A}{r}, \text{ 代入上式, 应有}$$

$$\int_{\mathbb{R}^2} \Phi(x) \Delta \varphi(x) dx = \lim_{\varepsilon \rightarrow 0} \int_{\partial \Omega_\varepsilon} A \log r \frac{\partial \varphi}{\partial n} + \varphi \phi'(r) dS$$

则对 $\int_{\partial\Omega_\varepsilon} A \log r \frac{\partial\varphi}{\partial n} dS \leq |A \log \varepsilon| \left| \int_{\partial\Omega_\varepsilon} \frac{\partial\varphi}{\partial n} dS \right| \leq |A \log \varepsilon| M \cdot 2\pi\varepsilon \rightarrow 0$

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0^+} \int_{\partial\Omega_\varepsilon} \varphi \phi'(r) dS &= \lim_{\varepsilon \rightarrow 0^+} \int_{\partial\Omega_\varepsilon} \varphi \frac{A}{\varepsilon} dS = \lim_{\varepsilon \rightarrow 0^+} \frac{A}{\varepsilon} \int_{\partial\Omega_\varepsilon} \varphi dS = \lim_{\varepsilon \rightarrow 0^+} \frac{A}{\varepsilon} (\varphi(0)2\pi\varepsilon + O(\varepsilon^2)) \quad (\text{积分中值定理}) \\ &= 2\pi A\varphi(0) = \varphi(0) = \int_{\mathbb{R}^n} \Phi(x) \delta\varphi(x) dx \end{aligned}$$

则 $A = \frac{1}{2\pi}$, 方程的解为 $\phi(r) = \frac{1}{2\pi} \log r$.

③ $n \geq 3$ 时 $\phi(r) = Ar^{2-n}$ $\phi'(r) = A(2-n)r^{1-n}$. 代入, 有

$$\int_{\mathbb{R}^n} \Phi(x) \Delta\varphi(x) dx = \lim_{\varepsilon \rightarrow 0^+} \int_{\partial\Omega_\varepsilon} \left(\Phi \frac{\partial\varphi}{\partial n} - \varphi \frac{\partial\Phi}{\partial n} \right) dS = \lim_{\varepsilon \rightarrow 0^+} \int_{\partial\Omega_\varepsilon} \left(Ar^{2-n} \frac{\partial\varphi}{\partial n} + \varphi A(2-n)r^{1-n} \right) dS$$

对

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\partial\Omega_\varepsilon} Ar^{2-n} \frac{\partial\varphi}{\partial n} dS = \lim_{\varepsilon \rightarrow 0^+} A\varepsilon^{2-n} \int_{\partial\Omega_\varepsilon} \frac{\partial\varphi}{\partial n} dS \leq AM\varepsilon^{2-n} \cdot \omega_n \varepsilon^{n-1} = \frac{AM\omega_n}{\varepsilon} \rightarrow 0$$

对

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0^+} \int_{\partial\Omega_\varepsilon} \varphi A(2-n)r^{1-n} dS &= \lim_{\varepsilon \rightarrow 0^+} A(2-n)\varepsilon^{1-n} \int_{\partial\Omega_\varepsilon} \varphi dS = \lim_{\varepsilon \rightarrow 0^+} A(2-n)\varepsilon^{1-n} (\varphi(0)\omega_n \varepsilon^{n-1} + O(\varepsilon^n)) \\ &= \lim_{\varepsilon \rightarrow 0^+} (A(2-n)\varphi(0)\omega_n + O(\varepsilon)) = A\varphi(0)\omega_n(2-n) \end{aligned}$$

则有 $\int_{\mathbb{R}^n} \Phi(x) \Delta\varphi(x) dx = A\varphi(0)\omega_n(2-n) = \varphi(0)$ 则 $A = \frac{1}{\omega_n(2-n)}$ 故有 $\phi(r) = -\frac{r^{2-n}}{\omega_n(n-2)}$ 综上, $\Phi(x)$ 的统一表达式为

$$\Phi(x) = \begin{cases} -\frac{1}{2}|x|, & n=1 \\ -\frac{1}{2\pi} \log \|x\|, & n=2 \\ -\frac{1}{\omega_n(n-2)} \|x\|^{2-n}, & n \geq 3 \end{cases}$$

下面这条定理保证了基本解的适定性:

定理 4.1.2. 由上面两条公式所定义的 $\Gamma(x; \xi)$ 是 *Laplace* 方程的基本解。

在证明上面的定理中，我们参考证明思路，得到这样的公式：

定理 4.1.3. 设 $\partial\Omega$ 分段光滑， $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ ，则

$$u(\xi, \eta) = - \iint_{\Omega} \Gamma(x, y; \xi, \eta) \Delta u(x, y) dx dy + \int_{\partial\Omega} \left[\Gamma(x, y; \xi, \eta) \frac{\partial u(x, y)}{\partial \mathbf{n}} - \frac{\partial \Gamma(x, y; \xi, \eta)}{\partial \mathbf{n}} u(x, y) \right] dl$$

针对位势方程，我们还存在特征值问题：

定义 4.1.2. 如果 $\lambda \in \mathbb{R}$ 使得

$$\begin{cases} -\Delta u + \lambda u = 0 & x \in \Omega \\ \frac{\partial u}{\partial n} \Big|_{\partial\Omega} = 0 \end{cases}.$$

有非零解，则称 λ 为上述问题的特征值，相应的非零解称为特征函数，记为 u_λ 。

其解的存在性定理表述如下：

定理 4.1.4. 问题

$$\begin{cases} -\Delta u + \lambda u = f & x \in \Omega \\ \frac{\partial u}{\partial n} \Big|_{\partial\Omega} = \varphi \end{cases}.$$

有属于 $C^1(\bar{\Omega}) \cap C^2(\Omega)$ 的解的充要条件是对于相应的齐次问题的任意特征函数 $u_\lambda(x, y)$ 都有

$$\iint_{\Omega} f(x, y) u_\lambda(x, y) dx dy + \int_{\partial\Omega} \varphi(x, y) u_\lambda(x, y) dl = 0$$

4.2 Green 函数

在前面的定理中，我们已经知道了位势方程的解应具有这样的条件：

$$u(\xi, \eta) = - \iint_{\Omega} \Gamma(x, y; \xi, \eta) \Delta u(x, y) dx dy + \int_{\partial\Omega} \left[\Gamma(x, y; \xi, \eta) \frac{\partial u(x, y)}{\partial \mathbf{n}} - \frac{\partial \Gamma(x, y; \xi, \eta)}{\partial \mathbf{n}} u(x, y) \right] dl$$

但我们不难发现等式右侧仍然存在 $\frac{\partial u}{\partial \mathbf{n}}$, 这部被认为是解出了 u , 若我们能够通过调整 Γ 使得最新得到的 G 满足 $G(x, y; \xi, \eta) \Big|_{\partial\Omega} = 0$, 我们也就解出了 u , 则我们便引出 Green 函数的概念:

定义 4.2.1. 定义于 $\bar{\Omega} \times \bar{\Omega} \setminus \{(x, y) = (\xi, \eta)\}$ 的连续函数

$$G(x, y; \xi, \eta) = \Gamma(x, y; \xi, \eta) + g(x, y; \xi, \eta)$$

称为 Green 函数, 如果其中的 $g(x, y; \xi, \eta)$ 在 $\Omega \times \Omega$ 上二次连续可微, 而且 $G(x, y; \xi, \eta)$ 满足以下问题 (在广义函数意义下):

$$\begin{cases} -\Delta_{(x,y)} G(x, y; \xi, \eta) = \delta(x - \xi, y - \eta) \\ G(x, y; \xi, \eta) \Big|_{\partial\Omega} = 0 \end{cases}$$

则位势方程的解应当满足

定理 4.2.1. 设 u 是问题

$$\begin{cases} -\Delta u = f(x, y) & (x, y) \in \Omega \\ u|_{\partial\Omega} = \varphi(x, y) \end{cases}$$

则 u 可以表示成这样的形式:

$$u(\xi, \eta) = \iint_{\Omega} G(x, y; \xi, \eta) f(x, y) dx dy - \int_{\partial\Omega} \frac{\partial G(x, y; \xi, \eta)}{\partial n} \varphi(x, y) dl$$

例 4.2.1. 求边值问题 $\begin{cases} -\Delta u = f(x, y), & (x, y) \in \Omega \\ u|_{\partial\Omega} = \varphi \end{cases}$ 的 Green 函数, 其中

(1) Ω 是上半平面 (2) Ω 是第一象限 (本题我们使用对称镜面法, 应保证函数在边界处为零)

解答 (1) 解对在 (ξ, η) 点处的正电荷, 我们在 $(\xi, -\eta)$ 处引入电荷量相同的负电荷, 则 Green 函数为

$$\begin{aligned} G(x, y; \xi, \eta) &= \Gamma(x, y; \xi, \eta) - \Gamma(x, y; \xi, -\eta) \\ &= \frac{1}{2\pi} \log \frac{((x - \xi)^2 + (y + \eta)^2)^{\frac{1}{2}}}{((x - \xi)^2 + (y - \eta)^2)^{\frac{1}{2}}} \end{aligned}$$

(2) 解对在第一象限的电荷 (ξ, η) , 我们在第二象限 $(-\xi, \eta)$ 引入负电荷, 第四象限 $(\xi, -\eta)$ 引入负电荷, 在第三象限 $(-\xi, -\eta)$ 引入正电荷

$$\begin{aligned} G(x, y; \xi, \eta) &= \Gamma(x, y; \xi, \eta) + \Gamma(x, y; -\xi, -\eta) - \Gamma(x, y; \xi, -\eta) - \Gamma(x, y; -\xi, \eta) \\ &= \log \frac{((x - \xi)^2 + (y + \eta)^2)^{\frac{1}{2}} ((x + \xi)^2 + (y - \eta)^2)^{\frac{1}{2}}}{((x - \xi)^2 + (y - \eta)^2)^{\frac{1}{2}} ((x + \xi)^2 + (y + \eta)^2)^{\frac{1}{2}}} \end{aligned}$$

□

例 4.2.2. 记 $B^+(R) = \{(x, y) \mid x^2 + y^2 < R^2, y > 0\}$, 求定解问题

$$\begin{cases} -\Delta u = f(x, y), & (x, y) \in B^+(R) \\ u|_{\partial B^+(R) \cap \{y > 0\}} = \varphi(x, y) \\ \frac{\partial u}{\partial y} \Big|_{y=0} = \psi(x, 0), & -R \leq x \leq R \end{cases}$$

的 Green 函数, 如果 $u \in C^1(\overline{B^+(R)}) \cap C^2(B^+(R))$ 是上述问题的解, 试给出解的表达式.

解答 我们可以将 u 写作如下形式

$$u(\xi, \eta) = \iint_{\Omega} G(x, y; \xi, \eta) f(x, y) dx dy + \int_{\partial \Omega} [G(x, y; \xi, \eta) \frac{\partial u}{\partial n} - \frac{\partial G}{\partial n}(x, y; \xi, \eta) u] dl$$

取 $\Gamma_1 = \partial B^+(R) \cap \{y > 0\}$ $\Gamma_2 = \{y = 0\} \cap \{-R \leq x \leq R\}$, 则有

$$u(\xi, \eta) = \iint_{\Omega} G(x, y; \xi, \eta) f(x, y) dx dy + \int_{\Gamma_1} \frac{\partial u}{\partial n} G(x, y; \xi, \eta) - \varphi \frac{\partial G}{\partial n} dl + \int_{\Gamma_2} G \frac{\partial u}{\partial n} - \frac{\partial G}{\partial n} \psi dl$$

则我们应保证 $G|_{\Gamma_1} = 0$, $\frac{\partial G}{\partial n} \Big|_{\Gamma_2} = \frac{\partial G}{\partial y} \Big|_{\Gamma_2} = 0$. (使 $y = 0$ 电势平滑) 则为使 $y = 0$ 处电势平滑, 若 (ξ, η) 有正电荷, 则 $(\xi, -\eta)$ 有

正电荷. 为使球面电荷为零, 则 (ξ^*, η^*) 有负电荷, $(\xi^*, -\eta^*)$ 有负电荷, 其中 $(\xi^*, \eta^*) = \frac{R^2}{\xi^2 + \eta^2}(\xi, \eta)$ 则

$$\begin{aligned} G(x, y; \xi, \eta) &= \Gamma(x, y; \xi, \eta) + \Gamma(x, y; \xi, -\eta) - \Gamma(x, y; \xi^*, \eta^*) - \Gamma(x, y; \xi^*, -\eta^*) \\ &= \frac{1}{2\pi} \log \frac{((x - \xi^*)^2 + (y - \eta^*)^2)^{\frac{1}{2}}}{((x - \xi)^2 + (y - \eta)^2)^{\frac{1}{2}}} + \frac{1}{2\pi} \log \frac{((x - \xi^*)^2 + (y + \eta^*)^2)^{\frac{1}{2}}}{((x - \xi)^2 + (y + \eta)^2)^{\frac{1}{2}}} \end{aligned}$$

$$u(\xi, \eta) = \iint_{\Omega} G f dx dy + \int_{\Gamma_1} \frac{\partial \varphi}{\partial n} G dl + \int_{\Gamma_2} G \frac{\partial \psi}{\partial n} dl$$

□

格林函数具有如下四条性质:

1. 当 $(x, y) \in \Omega \setminus \{(\xi, \eta)\}$ 时,

$$\Delta_{(x,y)} G(x, y; \xi, \eta) = 0$$

当 $(x, y) \rightarrow (\xi, \eta)$ 时 Green 函数 $G(x, y; \xi, \eta)$ 与 $\Gamma(x, y; \xi, \eta) = \frac{1}{2\pi} \log((x - \xi)^2 + (y - \eta)^2)^{-1}$ 等价。

2. 当 $(x, y) \in \Omega \setminus \{(\xi, \eta)\}$ 时

$$0 < G(x, y; \xi, \eta) \leq \frac{1}{2\pi} \log \frac{d}{\sqrt{(x - \xi)^2 + (y - \eta)^2}}$$

其中 d 是 Ω 的直径。

3. Green 函数具有对称性, 即

$$G(x, y; \xi, \eta) = G(\xi, \eta; x, y)$$

$$4. \int_{\partial\Omega} \frac{\partial G(x, y; \xi, \eta)}{\partial n} dl = -1.$$

基于 Green 函数, 我们可以得到球面上的 Poisson 公式: 针对二维问题

$$\begin{cases} -\Delta u = f(x, y) & (x, y) \in B_a \\ u|_{\partial B_a} = \varphi \end{cases}$$

他的解可以表示成

$$u(\rho, \theta) = \frac{1}{4\pi} \int_0^a r dr \int_0^{2\pi} \log \frac{\rho^2 r^2 + a^4 - 2r\rho a^2 \cos(\theta - \alpha)}{a^2(r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha))} f(r, \alpha) d\alpha + \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} \varphi(\alpha) d\alpha$$

特别地, 当 $f \equiv 0$ 时,

$$u(\rho, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} \varphi(\alpha) d\alpha$$

称为圆上的 Poisson 公式。

解答 我们构造 Green 函数

$$\begin{cases} -\Delta u = 0, & (x, y) \in B_a \\ u|_{\partial B_a} = \varphi \end{cases}$$

$$G(x, y; \xi, \eta) = \frac{1}{2\pi} \log \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2}} - \frac{1}{2\pi} \log \frac{C}{\sqrt{(x-\xi^*)^2 + (y-\eta^*)^2}} \triangleq \frac{1}{2\pi} \log \frac{r^*}{Cr}$$

其中 $r \triangleq \sqrt{(x-\xi)^2 + (y-\eta)^2}$ $r^* \triangleq \sqrt{(x-\xi^*)^2 + (y-\eta^*)^2}$, 对 (ξ, η) 与 (ξ^*, η^*) , 取 $\rho = \sqrt{\xi^2 + \eta^2}$, $\rho^* = \sqrt{\xi^{*2} + \eta^{*2}}$ 其满足 $\rho\rho^* = a^2$, 其中 a 为圆的半径, 则 $\xi^* = \frac{a^2}{\rho^2}\xi$, $\eta^* = \frac{a^2}{\rho^2}\eta$

现在我们首先确定常数 C , 当 $x^2 + y^2 = a^2$ 时, 我们有 $G = 0$, 即 $\frac{r^*}{Cr} = 1$, 则 $C = \frac{r^*}{r}$ 此时 $r^2 = (x-\xi)^2 + (y-\eta)^2 = x^2 + y^2 + \xi^2 + \eta^2 - 2(x\xi + y\eta) = a^2 + \rho^2 - 2(x\xi + y\eta)$

$$\begin{aligned} r^{*2} &= (x-\xi^*)^2 + (y-\eta^*)^2 = x^2 + y^2 + \xi^{*2} + \eta^{*2} - 2(x\xi^* + y\eta^*) = a^2 + \rho^{*2} - 2(x\xi^* + y\eta^*) \\ &= a^2 + \frac{a^4}{\rho^2} - 2(x\frac{a^2}{\rho^2}\xi + y\frac{a^2}{\rho^2}\eta) = \frac{a^2}{\rho^2}(\rho^2 + a^2 - 2(x\xi + y\eta)) = \frac{a^2}{\rho^2}r^2 \end{aligned}$$

则 $C = \frac{r^*}{r} \frac{a}{\rho} = \frac{a}{\rho}$. $G(x, y; \xi, \eta) = \frac{1}{2\pi} \log \frac{ar^*}{\rho r}$ 结合定理我们有 u 可以表示为如下形式

$$u(\xi, \eta) = - \int_{\partial\Omega} \varphi \frac{\partial G}{\partial n} dl$$

因此我们需计算 $\frac{\partial G}{\partial n} = \frac{\partial G}{\partial r}$. 取 $(x, y) = (s \cos \theta, s \sin \theta)$, $(\xi, \eta) = (\rho \cos \varphi, \rho \sin \varphi)$ 则

$$r^2 = s^2 + \rho^2 - 2\rho s \cos(\theta - \varphi) \quad r^{*2} = s^2 + \rho^{*2} - 2(x\xi^* + y\eta^*) = s^2 + \frac{a^4}{\rho^2} - 2\frac{a^2}{\rho^2}\rho s \cos(\theta - \varphi) = s^2 + \frac{a^4}{\rho^2} - \frac{2a^2}{\rho} s \cos(\theta - \varphi)$$

则

$$\frac{\partial r}{\partial s} = \frac{1}{r}(s - \rho \cos(\theta - \varphi)) \quad \frac{\partial r^*}{\partial s} = \frac{1}{r^*}(s - \frac{a^2}{\rho} \cos(\theta - \varphi))$$

则

$$\begin{aligned}
 \frac{\partial G}{\partial n} &= \frac{\partial G}{\partial s} \Big|_{s=a} = \frac{\partial}{\partial s} \left[\frac{1}{2\pi} (\log \frac{a}{\rho} + \log r^* - \log r) \right] \Big|_{s=a} \\
 &= \frac{1}{2\pi} \frac{1}{r^*} \frac{\partial r^*}{\partial s} - \frac{1}{2\pi} \frac{1}{r} \frac{\partial r}{\partial s} \Big|_{s=a} \\
 &= \frac{1}{2\pi} \left(\frac{1}{r^{*2}} (s - \frac{a^2}{\rho} \cos(\theta - \varphi)) - \frac{1}{r^2} (s - \rho \cos(\theta - \varphi)) \right) \Big|_{s=a} \\
 &= \frac{1}{2\pi} \left(\frac{\rho^2}{a^2 r^2} (a - \frac{a^2}{\rho} \cos(\theta - \varphi)) - \frac{1}{r^2} (a - \rho \cos(\theta - \varphi)) \right) = \frac{a^2 - \rho^2}{2\pi a r^2}
 \end{aligned}$$

我们将其代入, 有

$$u(\rho, \theta) = - \int_{\partial\Omega} \varphi \frac{\partial G}{\partial n} dl = \frac{a^2 - \rho^2}{2\pi a} \int_0^{2\pi} \frac{\varphi(\varphi)}{a^2 + \rho^2 - 2a\rho \cos(\theta - \varphi)} a d\varphi$$

若 $\Delta u = f$

$$u(\rho, \theta) = \frac{1}{4\pi} \int_0^a r dr \int_0^{2\pi} \log \frac{\rho^2 r^2 + a^4 - 2r\rho a^2 \cos(\theta - \alpha)}{a^2[r^2 + \rho^2 - 2r\rho \cos(\theta - \alpha)]} f(r, \alpha) d\alpha + \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} \varphi(\alpha) d\alpha.$$

□

其存在性定理表述如下:

定理 4.2.2. 对方程

$$\begin{cases} -\Delta u = 0, & (x, y) \in B_a, \\ u|_{\partial B_a} = \varphi \end{cases}, \text{ 的解可表示为 } u(\rho, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} \varphi(\alpha) d\alpha$$

解答 不难发现(书上写的也很简单) u 是一个解析函数(参考《复变函数》)结合前面的推导, 我们只需要证明 $\lim_{\rho \rightarrow a^-} u(\rho, \theta_0) = \varphi(\theta_0)$ 结合 Green 函数的性质, 我们有

$$1 = - \int_{\partial B_a} \frac{\partial G}{\partial n} dl = \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} d\alpha$$

则

$$\begin{aligned} |u(\rho, \theta_0) - \varphi(\theta_0)| &= \left| \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} (\varphi(\alpha) - \varphi(\theta_0)) d\alpha \right| \\ &= \frac{1}{2\pi} \left| \left(\int_{\theta_0-\pi}^{\theta_0-\delta} + \int_{\theta_0-\delta}^{\theta_0+\delta} + \int_{\theta_0+\delta}^{\theta_0+\pi} \right) \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} (\varphi(\alpha) - \varphi(\theta_0)) d\alpha \right| = \int_1 + \int_2 + \int_3 \end{aligned}$$

对

$$\int_2 = \frac{1}{2\pi} \left| \int_{\theta_0-\delta}^{\theta_0+\delta} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} (\varphi(\alpha) - \varphi(\theta_0)) d\alpha \right| \leq \frac{1}{2\pi} \int_{\theta_0-\delta}^{\theta_0+\delta} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} |\varphi(\alpha) - \varphi(\theta_0)| d\alpha$$

结合 $\varphi \in C^\infty(\mathbb{R})$, 则 $\forall \varepsilon > 0, \exists \delta > 0$, s.t. $|\varphi(\alpha) - \varphi(\theta_0)| < \varepsilon$. 取上面的 δ 为此处 δ , 就有 $\int_2 \rightarrow 0$. 对

$$\int_1 = \frac{1}{2\pi} \left| \int_{\theta_0-\pi}^{\theta_0-\delta} \frac{a^2 - \rho^2}{a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha)} (\varphi(\alpha) - \varphi(\theta_0)) d\alpha \right|$$

当 $|\alpha - \theta_0| > \delta$ 时

$$a^2 + \rho^2 - 2a\rho \cos(\theta - \alpha) \geq a^2 + \rho^2 - 2a\rho \cos \delta = (a - \rho)^2 + 2a\rho(1 - \cos \delta).$$

则

$$\int_1 \leq \frac{1}{2\pi} \int_{\theta_0-\pi}^{\theta_0-\delta} \frac{a^2 - \rho^2}{(a - \rho)^2 + 2a\rho(1 - \cos \delta)} |\varphi(\alpha) - \varphi(\theta_0)| d\alpha \leq \frac{\|\varphi\|}{\pi} \int_{\theta_0-\pi}^{\theta_0-\delta} \frac{a^2 - \rho^2}{(a - \rho)^2 + 2a\rho(1 - \cos \delta)} d\alpha \leq \frac{\|\varphi\|}{2\pi a\rho(1 - \cos \delta)} (a + \rho)(a - \rho) \rightarrow 0,$$

同理 $\int_3 \rightarrow 0$, 原命题成立. □

例 4.2.3. 证明第二边值问题 $\begin{cases} \Delta u = 0, & r < a \\ \frac{\partial u}{\partial r} \Big|_{r=a} = \varphi(\theta) \end{cases}$ 的解当 $\int_0^{2\pi} \varphi(\theta) d\theta = 0$ 时可表成

$$u(r, \theta) = -\frac{a}{2\pi} \int_0^{2\pi} \varphi(\omega) \log(a^2 + r^2 - 2ar \cos(\omega - \theta)) d\omega + C$$

其中 C 为任意常数

解答 取 $x = r \cos \theta$, $y = r \sin \theta$, 则 $r = \sqrt{x^2 + y^2}$, $\theta = \arctan \frac{y}{x}$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial \theta}{\partial x} = -\frac{y}{r^2}, \quad \frac{\partial \theta}{\partial y} = \frac{x}{r^2}$$

则

$$\begin{aligned} u_x &= u_r r_x + u_\theta \theta_x = \frac{x}{r} u_r - \frac{y}{r^2} u_\theta & u_y &= u_r r_y + u_\theta \theta_y = \frac{y}{r} u_r + \frac{x}{r^2} u_\theta \\ u_{xx} &= \frac{1}{r} u_r - \frac{x^2}{r^2} u_{rr} + \frac{x}{r} u_{rr} r_x + \frac{2xy}{r^4} u_\theta r_x - \frac{y}{r^2} u_{\theta\theta} \theta_x = \frac{1}{r} u_r - \frac{x^2}{r^3} u_{rr} + \frac{x^2}{r^2} u_{rr} + \frac{2xy}{r^4} u_\theta + \frac{y^2}{r^4} u_{\theta\theta} \\ u_{yy} &= \frac{1}{r} u_r - \frac{y^2}{r^3} u_{rr} + \frac{y}{r} u_{rr} r_y - \frac{2xy}{r^3} u_\theta r_y + \frac{x}{r^2} u_{\theta\theta} \theta_y = \frac{1}{r} u_r - \frac{y^2}{r^3} u_{rr} + \frac{y^2}{r^2} u_{rr} - \frac{2xy}{r^4} u_\theta + \frac{x^2}{r^4} u_{\theta\theta} \end{aligned}$$

结合 $\Delta u = 0$, 有 $\frac{2}{r} u_r - \frac{1}{r} u_{rr} + u_{rr} + \frac{1}{r^2} u_{\theta\theta} = 0$. 则

$$\frac{1}{r} \frac{\partial}{\partial r} (r u_r) = -\frac{1}{r^2} u_{\theta\theta}$$

取 $w = r \frac{\partial u}{\partial r}$, 则

$$\begin{aligned} \Delta w &= \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = \frac{1}{r} \frac{\partial w}{\partial r} + \frac{\partial^2 w}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 w}{\partial \theta^2} = \frac{1}{r} \frac{\partial}{\partial r} (r \frac{\partial u}{\partial r}) + \frac{\partial}{\partial r} (\frac{\partial u}{\partial r} + r \frac{\partial^2 u}{\partial r^2}) + \frac{1}{r} \frac{\partial}{\partial r} (\frac{\partial^2 u}{\partial \theta^2}) \\ &= \frac{1}{r} (\frac{\partial u}{\partial r} + r \frac{\partial^2 u}{\partial r^2}) + \frac{\partial^2 u}{\partial r^2} + \frac{\partial^2 u}{\partial r^2} + r \frac{\partial^3 u}{\partial r^3} - \frac{1}{r} \frac{\partial}{\partial r} (r \frac{\partial}{\partial r} (r \frac{\partial u}{\partial r})) \\ &= \frac{1}{r} \frac{\partial u}{\partial r} + 3 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial^3 u}{\partial r^3} - \frac{1}{r} \frac{\partial}{\partial r} (r \frac{\partial u}{\partial r} + r^2 \frac{\partial^2 u}{\partial r^2}) \\ &= \frac{1}{r} \frac{\partial u}{\partial r} + 3 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial^3 u}{\partial r^3} - \frac{1}{r} (\frac{\partial u}{\partial r} + r \frac{\partial^2 u}{\partial r^2} + 2r \frac{\partial^2 u}{\partial r^2} + r^2 \frac{\partial^3 u}{\partial r^3}) = 0 \end{aligned}$$

则 w 构成一个调和函数, 其满足 Dirichlet 问题

$$\begin{cases} \Delta w = 0, & w \in B_a(0) \\ w|_{\partial B_a(0)} = w(a, \theta) \end{cases}$$

则

$$\begin{aligned} w(r, \theta) &= \frac{1}{2\pi} \int_0^{2\pi} a\varphi(\alpha) \frac{a^2 - r^2}{a^2 + r^2 - 2ar \cos(\theta - \alpha)} d\alpha = r \frac{\partial u}{\partial r} \\ &= \frac{1}{2\pi} \left(\int_0^{2\pi} a\varphi(\alpha) d\alpha + \int_0^{2\pi} \frac{2ar \cos(\theta - \alpha) - 2r^2}{a^2 + r^2 - 2ar \cos(\theta - \alpha)} a\varphi(\alpha) d\alpha \right) \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{2ar \cos(\theta - \alpha) - 2r^2}{a^2 + r^2 - 2ar \cos(\theta - \alpha)} a\varphi(\alpha) d\alpha \end{aligned}$$

则

$$\frac{\partial u}{\partial r} = -\frac{a}{2\pi} \int_0^{2\pi} \frac{2r - 2a \cos(\theta - \alpha)}{a^2 + r^2 - 2ar \cos(\theta - \alpha)} \varphi(\alpha) d\alpha = -\frac{a}{2\pi} \int_0^{2\pi} \varphi(\alpha) \frac{\partial}{\partial r} \log(a^2 + r^2 - 2ar \cos(\theta - \alpha)) d\alpha$$

故有

$$u = -\frac{a}{2\pi} \int_0^{2\pi} \varphi(\alpha) \log(a^2 + r^2 - 2ar \cos(\theta - \alpha)) d\alpha$$

□

针对更高的 n 维, 我们可以这样表述结果

$$u(x) = \frac{R^2 - |x|^2}{\omega_n R} \int_{\partial B_R(0)} \frac{\varphi(y)}{(y - x)^n} dy$$

解答 在该前提下, 我们有基本解为 $\Phi(x) = \frac{|x|^{2-n}}{(n-2)\omega_n}$, 其中 $\omega_n = \frac{2\pi^{\frac{n}{2}}}{\Gamma(n/2)}$ (注意: 网上大多数资料会给出 $\Phi(x) = \frac{|x|^{2-n}}{n(n-2)\omega_n}$,

这是由 ω_n 定义不同导致的, 此时 $\omega_n = \frac{2\pi^{\frac{n}{2}}}{n\Gamma(n/2)}$)

则其具有性质 $\Delta\Phi = \delta_0$, 我们 $\forall x \in B_R(0) \setminus \{0\}$, 取 $x^* = \frac{R^2}{|x|^2}x$. 则有 $\forall y \in \partial B_R(0)$, $|y - x^*| = \frac{R}{|x|}|y - x|$

$$(|y - x^*|^2 = |y|^2 + |x^*|^2 - 2y \cdot x^* = R^2 + \frac{R^4}{|x|^4}|x|^2 - 2y \cdot \frac{R^2}{|x|^2}x = \frac{R^2}{|x|^2}(|x|^2 + |y|^2 - 2y \cdot x) = \frac{R^2}{|x|^2}|y - x|^2$$

定义 Green 函数 (镜面法) 对固定 $x \in B_R(0) \setminus \{0\}$

$$G(x, y) = \Phi(y - x) - \Phi\left(\frac{|x|}{R}(y - x^*)\right)$$

则当 $y \in \partial B_R(0)$ 时

$$\frac{|x|}{R}|y - x^*| = \frac{|x|}{R} \frac{R}{|x|}|y - x| = |y - x|. \quad \text{结合 } \Phi(x) \text{ 只必有 } \Phi(y - x) = \Phi\left(\frac{|x|}{R}(y - x^*)\right)$$

则有当 $y \in \partial B_R(0)$ 时 $G(x, y) = 0$, 另外, 结合 Green 函数定义

$$\Delta_y G(x, y) = -\delta(x).$$

即 $\forall \varphi \in C_0^\infty(\mathbb{R}^n)$, 有 $\int_{B_R(0)} \varphi(y) \Delta_y G(x, y) dy = \varphi(x)$, 则结合 Green 第二恒等式:

$$\int_{B_R(0)} u(y) \Delta_y G(x, y) - \Delta u(y) G(x, y) dy = \int_{\partial B_R(0)} \left(u(y) \frac{\partial G}{\partial n_y} - G \frac{\partial u}{\partial n_y} \right) dS(y)$$

结合 $\Delta u = 0$, $G|_{\partial\Omega} = 0 \quad \forall y \in \partial B_R(0)$ 成立, 则有

$$-u(\mathbf{x}) = \int_{\partial B_R(0)} u(\mathbf{y}) \frac{\partial G}{\partial n_y} dS(\mathbf{y})$$

$$\text{结合 } \frac{\partial G}{\partial n_y} = \frac{\partial}{\partial n_y} \Phi(\mathbf{y} - \mathbf{x}) - \frac{\partial}{\partial n_y} \Phi\left(\frac{|x|}{R}(\mathbf{y} - \mathbf{x}^*)\right) = A - B$$

$$A = \frac{\partial}{\partial n_y} \Phi(\mathbf{y} - \mathbf{x}) = \nabla_y \Phi(\mathbf{y} - \mathbf{x}) \cdot \mathbf{n}_y = -\frac{\mathbf{y} - \mathbf{x}}{\omega_n |\mathbf{y} - \mathbf{x}|^n} \cdot \frac{\mathbf{y}}{R}$$

(注: ① $\frac{\partial}{\partial n_y} \Phi(\mathbf{y} - \mathbf{x})$ 可以理解是 $\Phi(\mathbf{y} - \mathbf{x})$ 关于 $\mathbf{y}$ 的梯度与 $\mathbf{n}_y$ 方向的投影, 故有

$$\frac{\partial}{\partial n_y} = \nabla_y \Phi(\mathbf{y} - \mathbf{x}) \cdot \mathbf{n}_y$$

② $\nabla |\mathbf{z}|^{2-n}$ 的计算: 设 $r = \sqrt{z_1^2 + z_2^2 + \cdots + z_n^2} = |\mathbf{z}|$

则

$$\frac{\partial r}{\partial z_i} = \frac{2z_i}{2\sqrt{z_1^2 + \cdots + z_n^2}} = \frac{z_i}{r} \quad \frac{\partial |\mathbf{z}|^{2-n}}{\partial z_i} = \frac{\partial}{\partial r} r^{2-n} \frac{\partial r}{\partial z_i} = (2-n)r^{1-n} \frac{z_i}{r} = (2-n) \frac{z_i}{r^n}$$

$$\text{故有 } \nabla |\mathbf{z}|^{2-n} = \frac{(2-n)\mathbf{z}}{r^n} = \frac{(2-n)\mathbf{z}}{|\mathbf{z}|^n}$$

③ $\mathbf{n}_y$ 表示 $\partial B_R(0)$ 在 $\mathbf{y}$ 处的单位外法向量, 则 $\mathbf{n}_y = \frac{\mathbf{y}}{R}$ (在 $\partial B_R(0)$ 上)。

$$B = \frac{\partial}{\partial n_y} \Phi \left(\frac{|\mathbf{x}|}{R} (\mathbf{y} - \mathbf{x}^*) \right)$$

取 $\mathbf{z} = \frac{|\mathbf{x}|}{R} (\mathbf{y} - \mathbf{x}^*)$, 则 $\nabla_y \mathbf{z} = \frac{|\mathbf{x}|}{R}$

$$\frac{\partial}{\partial n_y} \Phi \left(\frac{|\mathbf{x}|}{R} (\mathbf{y} - \mathbf{x}^*) \right) = \nabla_y \Phi(\mathbf{z}) \cdot \mathbf{n}_y = \nabla_z \Phi(\mathbf{z}) \cdot \nabla_y \mathbf{z} \cdot \mathbf{n}_y = -\frac{\mathbf{z}}{\omega_n |\mathbf{z}|^n} \cdot \frac{|\mathbf{x}|}{R} \cdot \frac{\mathbf{y}}{R} = -\frac{\frac{|\mathbf{x}|}{R} (\mathbf{y} - \mathbf{x}^*) \cdot \mathbf{y}}{\omega_n R \left| \frac{|\mathbf{x}|}{R} (\mathbf{y} - \mathbf{x}^*) \right|^n} = -\frac{|\mathbf{x}|^2 \mathbf{y} - \mathbf{x}^* \cdot \mathbf{y} \cdot |\mathbf{x}|^2}{R^2 R \omega_n \frac{|\mathbf{x}|^n}{R^n} |\mathbf{y} - \mathbf{x}^*|^n}$$

则

$$\begin{aligned} u(\mathbf{x}) &= \int_{\partial B_R(0)} u(\mathbf{y}) \left(\frac{|\mathbf{x}|^2 |\mathbf{y}|^2 - \frac{R^2}{|\mathbf{x}|^2} \mathbf{x} \cdot \mathbf{y}}{R^3 \omega_n |\mathbf{y} - \mathbf{x}|^n} - \frac{(\mathbf{y} - \mathbf{x}) \cdot \mathbf{y}}{\omega_n R |\mathbf{y} - \mathbf{x}|^n} \right) dS(\mathbf{y}) \\ &= \int_{\partial B_R(0)} u(y) \frac{1}{\omega_n R |y - x|^n} (|x|^2 \frac{y^2}{R^2} - x \cdot y - y^2 + x \cdot y) dy \\ &= -\frac{1}{\omega_n R} \int_{\partial B_R(0)} \varphi(y) \frac{R^2(R^2 - |x|^2)}{R^2 |y - x|^n} dy \quad (y^2 = R^2) \end{aligned}$$

我们把负号去掉, 就有

$$u(x) = \frac{1}{\omega_n R} \int_{\partial B_R(0)} \varphi(y) \frac{R^2 - |x|^2}{|y - x|^n} dy = \frac{R^2 - |x|^2}{\omega_n R} \int_{\partial B_R(0)} \varphi(y) \frac{1}{|y - x|^n} dy$$

此即 n 维球面 Poisson 公式.

另外 $u(0) = \frac{1}{\omega_n R^{n-1}} \int_{\partial B_R(0)} \varphi(y) dy$ 称为平均值公式. 结合积分中值定理, 我们还可以得到

$$u(0) = \frac{1}{|B_R(0)|} \int_{\partial B_R(0)} u(x) dx. \quad \text{也是平均值公式.}$$

□

例 4.2.4. 设 $u(r, \theta)$ 是 $B(R)$ 外的有界调和函数, 令 $v(r, \theta) = u(R^2/r, \theta)$, 试证 $v(r, \theta)$ 是圆 $B(R)$ 内的调和函数, 由得解第一类外部边值问题

$$\begin{cases} \Delta u = 0, & (x, y) \in \mathbb{R}^2 \setminus \overline{B(R)} \\ v|_{\partial B(R)} = \varphi(x, y) \\ u \text{ 有界} \end{cases}$$

解答 取 $x = r \cos \theta$, $y = r \sin \theta$, $r = \sqrt{x^2 + y^2}$, $\theta = \arctan \frac{y}{x}$

$$r_x = \frac{x}{r}, \quad r_y = \frac{y}{r}, \quad \theta_x = -\frac{y}{r^2}, \quad \theta_y = \frac{x}{r^2}$$

结合 $\Delta u = 0$, 我们有

$$\frac{1}{r} \frac{\partial}{\partial r}(ru_r) = -\frac{1}{r^2} u_{\theta\theta} \quad (r > R) \quad (*)$$

则

$$\begin{aligned} \Delta v &= \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial v}{\partial \theta} \frac{\partial \theta}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial v}{\partial \theta} \frac{\partial \theta}{\partial y} \right) \\ &\stackrel{\text{令 } \rho = R^2/r}{=} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial \rho} \frac{\partial \rho}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial \rho} \frac{\partial \rho}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial \rho} \left(-\frac{R^2}{r^2} \right) \frac{x}{r} + \frac{\partial u}{\partial \theta} \left(-\frac{y}{r^2} \right) \right) + \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial \rho} \left(-\frac{R^2}{r^2} \right) \frac{y}{r} + \frac{\partial u}{\partial \theta} \frac{x}{r^2} \right) \\ &= -\frac{R^2}{r^3} \frac{\partial u}{\partial \rho} + \frac{3xR^2}{r^4} \frac{\partial u}{\partial \rho} \frac{\partial r}{\partial x} - \frac{xR^2}{r^3} \frac{\partial^2 u}{\partial \rho^2} \frac{\partial \rho}{\partial x} - \frac{xR^2}{r^3} \frac{\partial^2 u}{\partial \rho \partial \theta} \frac{\partial \theta}{\partial x} - \frac{y}{r^2} \frac{\partial^2 u}{\partial \theta \partial \rho} \frac{\partial \rho}{\partial x} - \frac{y}{r^2} \frac{\partial^2 u}{\partial \theta^2} \frac{\partial \theta}{\partial x} \\ &\quad - \frac{R^2}{r^3} \frac{\partial u}{\partial \rho} - \frac{3yR^2}{r^4} \frac{\partial u}{\partial \rho} \frac{\partial r}{\partial y} + \frac{yR^2}{r^3} \frac{\partial^2 u}{\partial \rho^2} \frac{\partial \rho}{\partial y} - \frac{2x}{r^3} \frac{\partial u}{\partial \theta} \frac{\partial r}{\partial y} + \frac{x}{r^2} \frac{\partial^2 u}{\partial \theta^2} \frac{\partial \theta}{\partial y} \\ &= \frac{R^2}{r^3} \frac{\partial u}{\partial \rho} + \frac{R^4}{r^4} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = \frac{1}{\rho} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = 0 \end{aligned}$$

这里的 ρ 就是 (*) 中的 r .

则 v 构成 B_R 中心调和函数, 则有

$$v(r, \theta) = \int_0^{2\pi} \frac{a^2 - r^2}{a^2 + r^2 - 2ar \cos(\theta - \alpha)} \varphi(\alpha) d\alpha.$$

结合 $u(r, \theta) = v(\frac{R^2}{r}, \theta)$, 则有

$$u(r, \theta) = \int_0^{2\pi} \frac{a^2 - (\frac{R^2}{r})^2}{a^2 + (\frac{R^2}{r})^2 - 2a\frac{R^2}{r} \cos(\theta - \alpha)} \varphi(\alpha) d\alpha = \int_0^{2\pi} \frac{a^2 r^2 - R^4}{a^2 r^2 + R^4 - 2arR^2 \cos(\theta - \alpha)} \varphi(\alpha) d\alpha. \quad \square$$

□

4.3 极值原理

我们考虑 $\mathbb{R}^n$ 中最一般的方程:

$$-\Delta u + c(x)u = f(x)$$

这里我们假定 $c(x) \geq 0$, 当 $x \in \Omega$ 时, 首先我们来介绍最基础的引理:

引理 4.3.1. 设 $c(x) \geq 0$, 函数 $u \in C^2(\Omega) \cap C(\bar{\Omega})$ 且在 Ω 内满足 $Lu = f < 0$, 则 u 不能在 Ω 内达到它在 $\bar{\Omega}$ 上的非负最大值。

由此我们可以引出弱极值原理:

定理 4.3.1 (弱极值原理). 设 $c(x) \geq 0$ 且在 Ω 上有界, $u \in C^2(\Omega) \cap C(\bar{\Omega})$ 内满足 $Lu = f \leq 0$, 则 u 在 Ω 上的非负最大值必在 $\partial\Omega$ 上达到, 即

$$\sup_{x \in \Omega} u(x) \leq \sup_{x \in \partial\Omega} u^+(x)$$

其中 $u^+(x) = \max\{u(x), 0\}$.

解答 我们取 $w(x) = u(x) + \varepsilon e^{\alpha x_1}$, 则有

$$\begin{aligned} Lw &= -\Delta(u + \varepsilon e^{\alpha x_1}) + c(x)(u + \varepsilon e^{\alpha x_1}) \\ &= -\Delta u + c(x)u - \alpha^2 \varepsilon e^{\alpha x_1} + c(x)\varepsilon e^{\alpha x_1} \\ &= f(x) + \varepsilon e^{\alpha x_1}(c(x) - \alpha^2) \end{aligned}$$

结合 $c(x) \geq 0$ 在 Ω 上有界, 我们取 α 足够大 s.t. $c(x) - \alpha^2 < 0$. 结合 $f \leq 0$, 则 $Lw < 0$. 结合引理 $w(x)$ 不能在 Ω 内达到常在 $\bar{\Omega}$ 上的非负最大值, 即

$$\sup_{x \in \Omega} w(x) \leq \sup_{x \in \partial\Omega} w^+(x).$$

则

$$\sup_{x \in \Omega} u(x) \leq \sup_{x \in \Omega} w(x) \leq \sup_{x \in \partial\Omega} w^+(x) \leq \sup_{x \in \partial\Omega} u^+(x) + \varepsilon \sup_{x \in \partial\Omega} e^{\alpha x_1}$$

此时取 $\varepsilon \rightarrow 0$, 立即有 $\sup_{x \in \Omega} u(x) \leq \sup_{x \in \partial\Omega} u^+(x)$ □

在这样的定理基础之上, 我们不难发现, 如果在 Ω 内 $Lu = f \geq 0$, 则 u 在 Ω 上的非正最小值必在 $\partial\Omega$ 上得到, 如果 $Lu \equiv 0$, 则 u 在 $\bar{\Omega}$ 上的非负最大值与非正最小值都必在边界达到。

例 4.3.1. 设 $u(x)$ 是定解问题 $\begin{cases} -\Delta u + c(x)u = f(x), & x \in \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$ 的一个解

(1) 如果 $c(x) \geq c_0 > 0$, 则有估计

$$\max_{\Omega} |u(x)| \leq c_0^{-1} \sup_{\Omega} |f(x)|$$

(2) 如果 $c(x) \geq 0$ 且有界, 则

$$\max_{\Omega} |u(x)| \leq M \sup_{\Omega} |f(x)|$$

其中 M 依赖于 $c(x)$ 的界与 Ω 的直径

(3) 如果 $c(x) < 0$, 试举反例说明上述最大模估计一般不成立

解答 (1) 证明取 $v(x) = c_0^{-1} \sup_{\Omega} |f(x)| \pm u(x)$. 则 $v|_{\partial\Omega} = c_0^{-1} \sup_{\Omega} |f(x)| \pm u|_{\partial\Omega} = c_0^{-1} \sup_{\Omega} |f(x)| \geq 0$ 且

$$-\Delta v + c(x)v = -(\pm \Delta u) + c(x)(c_0^{-1} \sup_{\Omega} |f(x)| \pm u(x)) = \pm f(x) + c(x)c_0^{-1} \sup_{\Omega} |f(x)| \geq \sup_{\Omega} |f(x)| \pm f(x) \geq 0.$$

结合弱极值原理, 由 $Lv \geq 0$, 则 v 的非负最小值必在边界上取到, 即 $v(x) \geq \min_{\partial\Omega} v \geq 0, \forall x \in \bar{\Omega}$. 此即 $\max_{\Omega} |u(x)| \leq c_0^{-1} \sup_{\Omega} |f(x)|$.

(2) 设 $z(x) = \frac{\sup_{\Omega} |f(x)|}{2n} (d^2 - |x|^2)$ 并令 $w(x) = z(x) \pm u(x)$. 其中 $d = \sup_{x, y \in \Omega} |x - y|$

$$w|_{\partial\Omega} = z|_{\partial\Omega} \pm u|_{\partial\Omega} = z|_{\partial\Omega} \geq 0$$

$$-\Delta w + c(x)w = (-\Delta z + c(x)z) \pm (-\Delta u + c(x)u) = \sup_{\Omega} |f(x)| + c(x) \frac{\sup_{\Omega} |f(x)|}{2n} (d^2 - |x|^2) \pm f(x) \geq 0$$

结合弱极值原理 $\max_{\Omega} |u(x)| \leq \frac{1}{2n} (d^2 - |x|^2) \sup_{\Omega} |f(x)| \triangleq M \sup_{\Omega} |f(x)|$

(3) 略

□

相对应地, 我们也有强极值原理, 但为了证明强极值原理, 我们需要这样的引理:

引理 4.3.2 (边界点引理). 设 S 是 $\mathbb{R}^n$ 的一个球, 在 S 上 $c(x) \geq 0$ 且有界, 如果 $u \in C^1(\bar{S}) \cap C^2(S)$ 且满足条件

$$(1) \quad Lu \leq 0$$

$$(2) \quad x^0 \in \partial S, u(x^0) \geq 0 \text{ 且当 } x \in S \text{ 时 } u(x) < u(x^0),$$

则

$$\frac{\partial u}{\partial \nu} \Big|_{x=x^0} > 0$$

其中 ν 与 ∂S 在点 x^0 的单位外法向量 n 的夹角小于 $\frac{\pi}{2}$.

解答 设 S 是 $\mathbb{R}^n$ 的一个球, 在 S 上 $c(x) \geq 0$ 且有界, 如果 $u \in C^1(\bar{S}) \cap C^2(S)$ 且满足条件 1) $Lu \leq 0$ 2) $x^0 \in \partial S, u(x^0) \geq 0$ 且当 $x \in S$ 时 $u(x) < u(x^0)$, 则

$$\frac{\partial u}{\partial \nu} \Big|_{x=x^0} > 0$$

其中 ν 与 ∂S 在点 x^0 的单位外法向量 n 的夹角小于 $\frac{\pi}{2}$.

证明结合 $\mathbb{R}^n$ 中心球可以平移至原点处, 我们考虑 $S^* = \{x \mid \frac{r_1}{2} < |x| < r_1\}$, 并令

$$w(x) = u(x) - u(x^0) + \varepsilon v(x).$$

若我们可以保证 $w(x)$ 在 x^0 处仍能取到非负最大值, 此时有

$$\frac{\partial w}{\partial r} \Big|_{x=x^0} = \left(\frac{\partial u}{\partial r} + \varepsilon \frac{\partial v}{\partial r} \right) \Big|_{x=x_0} \geq 0 \quad (\text{为啥???)}$$

此时若我们可以保证 $-\frac{\partial v}{\partial r}\Big|_{x=x^0} > 0$, 且 $Lv \leq 0$, 则有原命题成立. 取 $v(x) = e^{-\alpha|x|^2} - e^{-\alpha r_1^2}$, 则

$$\frac{\partial v}{\partial x_i} = -2\alpha x_i e^{-\alpha|x|^2}, \quad \frac{\partial^2 v}{\partial x_i^2} = -2\alpha e^{-\alpha|x|^2} - 2\alpha x_i(-2\alpha x_i)e^{-\alpha|x|^2} = -2\alpha e^{-\alpha|x|^2} + 4\alpha^2 x_i^2 e^{-\alpha|x|^2}$$

则

$$\Delta v = \sum_{i=1}^n \frac{\partial^2 v}{\partial x_i^2} = -2n\alpha e^{-\alpha|x|^2} + 4\alpha^2|x|^2 e^{-\alpha|x|^2}$$

$$\begin{aligned} Lv = -\Delta v + c(x)v(x) &= 2n\alpha e^{-\alpha|x|^2} - 4\alpha^2|x|^2 e^{-\alpha|x|^2} + c(x)(e^{-\alpha|x|^2} - e^{-\alpha r_1^2}) \\ &= (2n\alpha - 4\alpha^2|x|^2 + c(x))e^{-\alpha|x|^2} - c(x)e^{-\alpha r_1^2} \\ &\leq (2n\alpha - 4\alpha^2|x|^2 + c(x))e^{-\alpha|x|^2} \end{aligned}$$

我们取 α 足够大, 就有 $Lv < 0$, 则 $Lw = Lu + \varepsilon Lv < 0$

结合引理, $w(x)$ 不可能在 S^* 内部取到非负最大值, 结合

$$\min_{|x|=\frac{r_1}{2}} (u(x^0) - u(x)) = \beta > 0$$

我们有 $w|_{|x|=\frac{r_1}{2}} \leq -\beta + \varepsilon v < 0$. 又对 $|x| = r_1$, 我们有 $w(x) = u(x) - u(x^0) < 0$. 则 $w(x)$ 在 $x = x^0$ 处取到非负最大值, 则结合

$$-\frac{\partial v}{\partial \boldsymbol{\nu}}\Big|_{x^0} = -\frac{\partial v}{\partial n} \cos(\boldsymbol{\nu}, n)\Big|_{x^0} = 2\alpha r e^{-\alpha r_1^2} \cos(\boldsymbol{\nu}, n)\Big|_{x^0} > 0.$$

有 $\frac{\partial u}{\partial \boldsymbol{\nu}}\Big|_{x^0} > 0$ 成立.

□

例 4.3.2. 用辅助函数 $W(\boldsymbol{x}) = |\boldsymbol{x}|^{-a} - r^{-a}$ 证明边界点引理 (其中常数 $a > 0$ 待定, r 是 S 的半径)。

解答 由圆可以平移至原点处, 我们考虑球壳 $S^* = \{\boldsymbol{x} \in \mathbb{R}^n : \frac{r}{2} \leq |\boldsymbol{x}| \leq r\}$, 并构造函数

$$v(\boldsymbol{x}) = u(\boldsymbol{x}) - u(\boldsymbol{x}^0) + \varepsilon W(\boldsymbol{x})$$

若 $v(\mathbf{x})$ 的最大值仍在 $\mathbf{x}^0$ 处取到, 结合 $\frac{\partial v}{\partial \nu}\Big|_{\mathbf{x}=\mathbf{x}^0} = \frac{\partial u}{\partial \nu}\Big|_{\mathbf{x}=\mathbf{x}^0} + \varepsilon \frac{\partial W}{\partial \nu}\Big|_{\mathbf{x}=\mathbf{x}^0} \geq 0$. 此时若我们保证 $-\varepsilon \frac{\partial W}{\partial \nu}\Big|_{\mathbf{x}=\mathbf{x}^0} > 0$, $Lu \leq 0$, 则此时我们就有 $\frac{\partial u}{\partial \nu}\Big|_{\mathbf{x}=\mathbf{x}^0} > 0$. 取 $W(\mathbf{x}) = |\mathbf{x}|^{-a} - r^{-a}$. 其中 r 是 S 的半径. 则

$$\begin{aligned}\frac{\partial W}{\partial x_i} &= \frac{\partial}{\partial x_i} \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}} = -\frac{a}{2} \cdot 2x_i \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}-1} = -ax_i \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a+2}{2}} \\ \frac{\partial^2 W}{\partial x_i^2} &= -a \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}-1} + a(a+2)x_i^2 \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}-2}\end{aligned}$$

$$\text{则 } \Delta W = \sum_{i=1}^n \frac{\partial^2 W}{\partial x_i^2} = a(a+2) \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}-1} - na \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}-1} = \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}-1} (a^2 + 2a - na)$$

$$\begin{aligned}-\Delta W + c(\mathbf{x})W &= (-a^2 + na - 2a) \left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}-1} + c(\mathbf{x}) \left(\left(\sum_{i=1}^n x_i^2 \right)^{-\frac{a}{2}} - r^{-a} \right) \\ &\leq (-a^2 + na - 2a) |\mathbf{x}|^{-a-2}\end{aligned}$$

取 a 足够大, 我们就有 $-\Delta W + c(\mathbf{x})W < 0$. 则 $Lv = Lu + \varepsilon LW < 0$. 结合引理 $v(\mathbf{x})$ (此处手稿写的是 $w(x)$, 应为 $v(x)$) 不可能在 S^* 内部取到非负最大值

$$\min_{|\mathbf{x}|=\frac{r}{2}} (u(\mathbf{x}^0) - u(\mathbf{x})) = \beta > 0$$

则我们有 $v\Big|_{|\mathbf{x}|=\frac{r}{2}} = -\beta + \varepsilon v(\mathbf{x})\Big|_{|\mathbf{x}|=\frac{r}{2}} < 0$.

且当 $|\mathbf{x}| = r$ 时 $w(\mathbf{x}) = 0$. 则 $v(\mathbf{x})$ 在 $\mathbf{x}^0$ 处取到非负最大值. 结合 $-\frac{\partial v}{\partial r}\Big|_{\mathbf{x}=\mathbf{x}^0} = -\frac{\partial v}{\partial n} \cos(\nu, n)\Big|_{\mathbf{x}=\mathbf{x}^0} = 2\alpha |\mathbf{x}| e^{-\alpha|\mathbf{x}|^2} \cos(\nu, n)\Big|_{\mathbf{x}=\mathbf{x}^0} > 0$. 有 $\frac{\partial u}{\partial r}\Big|_{\mathbf{x}=\mathbf{x}^0} > 0$ 成立. □

定理 4.3.2 (强极值原理). 设 Ω 是有界连通开区域, 在 Ω 上 $c(x) \geq 0$ 且有界, $u \in C(\bar{\Omega}) \cap C^2(\Omega)$ 并满足 $Lu \leq 0$, 如果 u 在 Ω 内达到非负最大值, 则 u 恒为常数.

然后是最大模估计:

定理 4.3.3. 设 $u \in C^2(\Omega) \cap C(\bar{\Omega})$ 是 *Dirichlet* 问题

$$\begin{cases} -\Delta u = f(x), & x \in \Omega \\ u|_{\partial\Omega} = \varphi(x) \end{cases}$$

的解, 则

$$\max_{\Omega} |u(x)| \leq \Phi + CF$$

其中 $\Phi = \max_{\partial\Omega} |\varphi(x)|$, $F = \sup_{\Omega} |f(x)|$, C 是仅依赖于维数 n 与 Ω 的直径的常数。

解答 我们构造辅助函数

$$w(x) = z(x) + \frac{\Phi}{\alpha_0} \pm u(x).$$

其中 $z(x) = \frac{F}{2n}(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2)$, 设 $\vec{n} = (\beta_1(x), \beta_2(x), \dots, \beta_n(x))$, 则有 $\sum_{i=1}^n \beta_i^2(x) = 1$. 对 $x \in \partial\Omega$ 成立.

$$-\Delta z + c(x)z = -\sum_{i=1}^n \frac{\partial}{\partial x_i} \left(\frac{\partial z}{\partial x_i} \right) + c(x)z = -\sum_{i=1}^n \frac{\partial}{\partial x_i} (-2x_i) \cdot \frac{F}{2n} + c(x)z \geq F$$

$$\begin{aligned} \frac{\partial z}{\partial n} + \alpha(x)z \Big|_{\partial\Omega} &= \sum_{i=1}^n \frac{\partial z}{\partial x_i} \frac{\partial x_i}{\partial n} + \alpha(x) \frac{F}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2 \right) \Big|_{\partial\Omega} \\ &= \sum_{i=1}^n \frac{F}{2n} (-2x_i) \beta_i(x) + \alpha(x) \frac{F}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2 \right) \Big|_{\partial\Omega} \\ &\geq \frac{F}{2n} \left[-\sum_{i=1}^n x_i^2 - \sum_{i=1}^n \beta_i^2(x) + \alpha(x) \left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2 \right) \right] \Big|_{\partial\Omega} \\ &\geq \frac{F}{2n} [-|x|^2 + d^2] \geq 0 \end{aligned}$$

则有 $-\Delta w + c(x)w = -\Delta z + c(x)z + c(x)\frac{\Phi}{\alpha_0} \pm (-\Delta u + c(x)u) \geq F + c(x)\frac{\Phi}{\alpha_0} \pm f(x) \geq 0$

$$\begin{aligned} \frac{\partial w}{\partial n} + \alpha(x)w \Big|_{\partial\Omega} &= \frac{\partial z}{\partial n} + \alpha(x)z + \alpha(x)\frac{\Phi}{\alpha_0} \pm \left(\frac{\partial u}{\partial n} + \alpha(x)u \right) \Big|_{\partial\Omega} \\ &\geq \alpha(x)\frac{\Phi}{\alpha_0} \pm \varphi(x) \geq 0 \end{aligned}$$

则结合弱极值原理, 若 w 可取负最小值, 则其一定在 $\partial\Omega$ 上取到, 设最小值为 x_0 , 此时 $\frac{\partial w}{\partial n} \Big|_{x_0} \leq 0$ 则 $\left[\frac{\partial w}{\partial n} + \alpha(x)w \right] \Big|_{\partial\Omega} \leq \alpha(x_0)w(x_0) < 0$ 与上式矛盾.

则 $\forall x \in \bar{\Omega}$, 都有 $w(x) \geq 0$, 即

$$|u(x)| \leq z(x) + \frac{\Phi}{\alpha_0}.$$

则 $\max_{\bar{\Omega}} |u(x)| \leq \max_{\bar{\Omega}} z(x) + \frac{\Phi}{\alpha_0} \leq C(F + \Phi)$.

$$\text{其中 } C = \max\left\{\frac{1}{\alpha_0}, \frac{1}{2n}\left(\frac{1+d^2}{\alpha_0} + d^2\right)\right\}$$

□

针对第三边值问题, 我们有这样的关系:

定理 4.3.4. 设 $c(x) \geq 0, \alpha(x) \geq \alpha_0 > 0, u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ 是问题

$$\begin{cases} Lu = -\Delta u + c(x)u = f(x), & x \in \Omega \\ \left(\frac{\partial u}{\partial \mathbf{n}} + \alpha(x)u\right)\Big|_{\partial\Omega} = \varphi(x) \end{cases}$$

的解, 其中 $\mathbf{n}$ 表示 $\partial\Omega$ 的外法向, 如果 $\alpha(x) \equiv 0$, 则上述问题称为 *Neumann* 问题或第二边值问题。

则我们有

$$\max_{\bar{\Omega}} |u(x)| \leq C(\Phi + F)$$

其中 C 只依赖于 n, α_0, d , 而 $\Phi = \max_{\partial\Omega} |\varphi(x)|$, $F = \sup_{\Omega} |f(x)|$, $d = \sup_{x, y \in \Omega} |x - y|$.

例 4.3.3. 设 $u(x)$ 是定解问题
$$\begin{cases} -\Delta u + c(x)u = f(x), & x \in \Omega \\ \frac{\partial u}{\partial n} + \alpha(x)u\Big|_{\Gamma_1} = \varphi_1, & u|_{\Gamma_2} = \varphi_2 \end{cases}$$
 的解, 其中 $\Gamma_1 \cup \Gamma_2 = \partial\Omega, \Gamma_1 \cap \Gamma_2 = \emptyset, \Gamma_2 \neq \emptyset$. 如果 $c(x) \geq 0, \alpha(x) \geq \alpha_0 > 0$, 则有估计

$$\max_{\bar{\Omega}} |u(x)| \leq C(\sup_{\Omega} |f| + \sup_{\Gamma_1} |\varphi_1| + \sup_{\Gamma_2} |\varphi_2|).$$

解答 取 $w(x) = z(x) + \frac{\Phi_1 + \Phi_2}{\alpha_0} \pm u(x)$, 其中 $z(x) = \frac{F}{2n}\left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2\right)$, 其中 $d = \text{diam } \Omega$. 则有

$$-\Delta z + c(x)z = -\sum \frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_i} \left(\frac{F}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 - \sum x_j^2 \right) \right) \right) + c(x)z = F + c(x)z \geq F$$

我们取单位法向量 $\vec{n} = (\beta_1(x), \dots, \beta_n(x))^T$, 则有

$$\begin{aligned} ll \frac{\partial z}{\partial n} + \alpha(x)z \Big|_{\partial\Omega} &= \sum_{i=1}^n \frac{\partial z}{\partial x_i} \frac{\partial \beta_i(x)}{\partial x_i} + \alpha(x) \frac{F}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2 \right) \Big|_{\partial\Omega} \\ &= \frac{F}{2n} \left[\sum_{i=1}^n -2x_i \beta_i + \alpha(x) \left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2 \right) \right] \Big|_{\partial\Omega} \geq \frac{F}{2n} \left[-2 \sum x_i \beta_i + \alpha(x) \left(\frac{1+d^2}{\alpha_0} \right) \right] \Big|_{\partial\Omega} \\ &\geq \frac{F}{2n} \left[- \sum x_i^2 - \sum \beta_i^2(x) + 1 + d^2 \right] \geq 0 \end{aligned}$$

则

$$\begin{aligned} -\Delta w + c(x)w &= -\Delta z + c(x)z + c(x) \frac{\Phi_1 + \Phi_2}{\alpha_0} \pm (-\Delta u + c(x)u) \geq F + c(x) \frac{\Phi_1 + \Phi_2}{\alpha_0} \pm f(x) \geq 0 \\ \frac{\partial w}{\partial n} + \alpha(x)w \Big|_{\Gamma_1} &= \frac{\partial z}{\partial n} + \alpha(x)z \Big|_{\partial\Omega} + \alpha(x) \frac{\Phi_1 + \Phi_2}{\alpha_0} \pm \left(\frac{\partial u}{\partial n} + \alpha(x)u \right) \Big|_{\Gamma_1} \geq \Phi_1 + \Phi_2 \pm \varphi_1(x) \geq 0 \end{aligned}$$

同理 $\frac{\partial w}{\partial n} + \alpha(x)w \Big|_{\Gamma_2} \geq 0$.

不妨设则结合弱极值原理, 若 w 有负最小值, 则其必在 $\partial\Omega$ 上达到, 则设 w 在 $x^0 \in \Gamma_1$ 上达到负最小值则我们有 $\frac{\partial w}{\partial n} \Big|_{x=x^0} \leq 0$, 于是就

$$\frac{\partial w}{\partial n} + \alpha(x)w \Big|_{x=x^0} \leq \alpha(x)w < 0.$$

这与上面的 $\frac{\partial w}{\partial n} + \alpha(x)w \Big|_{\Gamma_1} \geq 0$ 矛盾, 因此 w 无负最小值, 则有 $w \geq 0$

$$|u(x)| \leq \frac{F}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2 \right) + \frac{\Phi_1 + \Phi_2}{\alpha_0} \leq \frac{F}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 \right) + \frac{1}{\alpha_0} (\Phi_1 + \Phi_2)$$

则我们取 $C = \max(\frac{1}{2n}(\frac{1+d^2}{\alpha_0} + d^2), \frac{1}{\alpha_0})$ 就有 $\max_{\Omega} |u(x)| \leq C(\sup_{\Omega} |f| + \sup_{\Gamma_1} |\varphi_1| + \sup_{\Gamma_2} |\varphi_2|)$. □

最后是最大模估计:

定理 4.3.5. 设 $c(x) \geq c_0 > 0$ 且 $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ 是 *Dirichlet* 问题

$$\begin{cases} -\Delta u + c(x)u = f(x) & x \in \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$$

的解, 则我们有估计

$$\int_{\Omega} |\nabla u(x)|^2 dx + \frac{c_0}{2} \int_{\Omega} |u(x)|^2 dx \leq M \int_{\Omega} |f(x)|^2 dx$$

其中 M 仅依赖于 c_0 .

另外有

定理 4.3.6. 设 $c(x) \geq c_0 > 0, \alpha(x) \geq 0, u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ 是第二、三边值问题

$$\begin{cases} Lu = -\Delta u + c(x)u = f(x), & x \in \Omega \\ \left(\frac{\partial u}{\partial \mathbf{n}} + \alpha(x)u \right) \Big|_{\partial\Omega} = \varphi(x) \end{cases}$$

的解, 则有估计

$$\int_{\Omega} |\nabla u(x)|^2 dx + \frac{c_0}{2} \int_{\Omega} |u(x)|^2 dx + \int_{\partial\Omega} \alpha(x)u^2(x) dl \leq M \int_{\Omega} |f(x)|^2 dx$$

4.4 调和函数的性质

调和函数是指满足条件 $\Delta u = 0$ 的函数 u , 这里我们来介绍他的八个性质, 首先我们结合复变函数的知识知道, 解析函数的实部和虚部是调和函数。

例 4.4.1. 求圆 $B(R)$ 上满足以下边条件的调和函数 (其中 A, B 是常数) (1) $u(R, \theta) = A \cos \theta$ (2) $u(R, \theta) = A + B \sin \theta$ (3) $u(R, \theta) = A \sin^2 \theta + B \cos^2 \theta$. (核心性质: 解析函数的实部和虚部分别是调和函数)

解答 (1) 解由 $\frac{A}{R} \rho \cos \theta$ 是解析函数 $\frac{A}{R} z = \frac{A}{R} \rho (\cos \theta + i \sin \theta)$ 的实部, 且 $\rho = R$ 时 $\frac{A}{R} \rho \cos \theta = A \cos \theta$. 则

$$u(\rho, \theta) = \frac{A}{R} \rho \cos \theta$$

(2) 解由 $A + \frac{B}{R} \rho \sin \theta$ 是解析函数 $Ai + \frac{B}{R} z = Ai + \frac{B}{R} \rho (\cos \theta + i \sin \theta)$ 的虚部, 且当 $\rho = R$ 时 $A + \frac{B}{R} \rho \sin \theta = A + B \sin \theta$, 则有

$$u(\rho, \theta) = A + \frac{B}{R} \rho \sin \theta$$

(3) 解由

$$u(R, \theta) = A \sin^2 \theta + B \cos^2 \theta = \frac{A+B}{2} + \frac{B-A}{2} \cos 2\theta$$

且 $\frac{A+B}{2} + \frac{B-A}{2R^2} \rho^2 \cos 2\theta$ 是解析函数 $\frac{A+B}{2} + \frac{B-A}{2R^2} z^2 = \frac{A+B}{2} + \frac{B-A}{2R^2} \rho^2 (\cos 2\theta + i \sin 2\theta)$ 的实部且 $\rho = R$ 时 $\frac{A+B}{2} + \frac{B-A}{2R^2} \rho^2 \cos 2\theta = \frac{A+B}{2} + \frac{B-A}{2} \cos 2\theta$, 则有

$$u(\rho, \theta) = \frac{A+B}{2} + \frac{B-A}{2R^2} \rho^2 \cos 2\theta$$

□

首先我们介绍平均值公式, 但是其证明依赖散度定理:

定理 4.4.1 (散度定理). 设 $\Omega \subset \mathbb{R}^n$ 是一个有界闭区域, 其边界 $\partial\Omega$ 是分片光滑曲面 (在 n 维空间中 $n-1$ 维流形), 设 F 是定义在 Ω 上的一个连续可微的向量场 (C^1 向量场), 则有如下等式成立:

$$\int_{\Omega} \nabla \cdot F dx = \int_{\partial\Omega} F \cdot n d\sigma$$

定理 4.4.2 (平均值公式). 设 Ω 是一个 $\mathbb{R}^n$ 中的开集, 函数 $u \in C^2(\Omega)$ 是 Ω 上的调和函数, 则对 $x_0 \in \Omega$, $R > 0$, 只要闭球 $\overline{B_R(x_0)}$ 完全包含在 Ω 内部, 则有

$$u(x_0) = \frac{1}{|\partial B_R(x_0)|} \int_{\partial B_R(x_0)} u(y) dy = \frac{1}{|B_R(x_0)|} \int_{B_R(x_0)} u(y) dy$$

解答 设 $\phi(r) = \frac{1}{|\partial B_r(x_0)|} \int_{\partial B_r(x_0)} u(y) dS(y)$, 则我们取 $y = x_0 + rw$, 则 $w \in B_1(0)$.

$$\phi(r) = \frac{1}{|\partial B_1(0)| r^{n-1}} \int_{\partial B_1(0)} u(x_0 + rw) r^{n-1} dS(w) = \frac{1}{|\partial B_1(0)|} \int_{\partial B_1(0)} u(x_0 + rw) dS(w).$$

则

$$\phi'(r) = \frac{1}{|\partial B_1(0)|} \int_{\partial B_1(0)} \frac{\partial}{\partial r} (u(x_0 + rw)) dS(w)$$

其中

$$\frac{\partial}{\partial r}u(x_0 + rw) = \nabla u(x_0 + rw) \cdot \frac{\partial(x_0 + rw)}{\partial r} = \nabla u(x_0 + rw) \cdot w = \frac{\partial u}{\partial \nu} \quad (w \text{ 即 } y \text{ 处外法向量})$$

则

$$\begin{aligned} \phi'(r) &= \frac{1}{|\partial B_1(0)|} \int_{\partial B_1(0)} \frac{\partial u}{\partial r} dS(w) = \frac{1}{|\partial B_R(x_0)| r^{1-n}} \int_{\partial B_r(x_0)} \frac{\partial u}{\partial r}(y) r^{-(n-1)} dS(y) \\ &= \frac{1}{|\partial B_r(x_0)|} \int_{\partial B_r(x_0)} \frac{\partial u}{\partial r}(y) dS(y) = \frac{1}{|\partial B_r(x_0)|} \int_{B_r(x_0)} \Delta u dx = 0 \quad (\text{散度定理}) \end{aligned}$$

故有 $\phi(r) = u(x_0) \quad \forall r > 0$ 成立. □

例 4.4.2. 设 $u(x, y)$ 在以原点为中心 R 半径的圆 $B(R)$ 内调和, 在 $\overline{B(R)}$ 上连续, 又设 $M = \iint_{B(R)} u^2(x, y) dx dy$, 试证 (1)

$$|u(0, 0)| \leq \frac{1}{\sqrt{\pi}R} M^{\frac{1}{2}} \quad (2) \quad |u(x, y)| \leq \frac{1}{\sqrt{\pi}(R-r)} M^{\frac{1}{2}}, \quad \text{其中 } r = \sqrt{x^2 + y^2}$$

解答 (1) 解结合平均值公式

$$\begin{aligned} |u(0, 0)| &= \left| \frac{1}{\pi R^2} \iint_{B_R(0)} u(x, y) dx dy \right| \leq \frac{1}{\pi R^2} \iint_{B_R(0)} |u(x, y)| dx dy \\ &\leq \frac{1}{\pi R^2} \left(\iint_{B_R(0)} dx dy \right)^{\frac{1}{2}} \left(\iint_{B_R(0)} u^2(x, y) dx dy \right)^{\frac{1}{2}} \\ &= \frac{1}{\pi R^2} (\pi R^2)^{\frac{1}{2}} \left(\iint_{B_R(0)} u^2(x, y) dx dy \right)^{\frac{1}{2}} = \frac{1}{\sqrt{\pi}R} M^{\frac{1}{2}} \end{aligned}$$

(2) 同 (1) 求解 $R-r$ 保证在 $B(R)$ 内. □

例 4.4.3. 记 $\mathbb{R}_+^2 = \{(x, y) \mid -\infty < x < \infty, y > 0\}$, 证明定解问题

$$\begin{cases} -\Delta u = f(x, y), & (x, y) \in \mathbb{R}_+^2 \\ u|_{y=0} = \varphi(x), & -\infty < x < \infty \end{cases}$$

属于 $C(\overline{\mathbb{R}_+^2}) \cap C^2(\mathbb{R}_+^2)$ 的有界解是唯一的.

解答 我们来考虑 $f = \varphi = 0 \quad \forall x \in \mathbb{R}_+^2$ 的情形, 结合 u 是 $C(\bar{\mathbb{R}}_+^2) \cap C^2(\mathbb{R}_+^2)$ 的有界解, 我们取 $L > 0$ s.t. $\varepsilon \log(L^2) > \sup |u(x)|$, 并取 $\Omega = \{(x, y) \in \mathbb{R}_+^2 : |x| < L, 0 < y < L\}$ 令 $w(x) = \pm u(x) + v(x)$, 其中 $v(x) = \varepsilon \log(x^2 + (y+1)^2)$

$$\frac{\partial v}{\partial x} = \varepsilon \frac{2x}{x^2 + (y+1)^2} \quad \frac{\partial^2 v}{\partial x^2} = \varepsilon \frac{2(x^2 + (y+1)^2) - 2x \cdot 2x}{(x^2 + (y+1)^2)^2} \quad \frac{\partial^2 v}{\partial y^2} = \varepsilon \frac{2(x^2 + (y+1)^2) - 2(y+1) \cdot 2(y+1)}{(x^2 + (y+1)^2)^2}$$

则 $\Delta v = 0$, 故有 $\Delta w = \Delta v \pm \Delta u = 0$. 结合书本附注, w 在 Ω 上的非负最大值和非正最大值必在 $\partial\Omega$ 上达到则对 $y = 0$

$$w(x, y) = \pm \varphi(x) + \varepsilon \log x^2 > \sup |u(x)| \pm \varphi(x) \geq 0$$

成立. 对 $y = L$,

$$w(x, y) = \pm u(x, L) + \varepsilon \log(x^2 + (L+1)^2) > \sup |u(x)| \pm u(x, L) \geq 0$$

对 $x = \pm L$

$$w(x, y) = \pm u(\pm L, y) + \varepsilon \log(L^2 + (y+1)^2) > \sup |u(x)| \pm u(x, L) \geq 0$$

则 $\forall (x, y) \in \Omega$, 有 $w(x, y) \geq 0$ 成立. 则有

$$u(x, y) \leq \varepsilon \log(x^2 + (y+1)^2). \quad \text{对 } (x, y) \in \Omega \text{ 成立.}$$

取 $L \rightarrow \infty$, 有 $u(x, y) \leq \varepsilon \log(x^2 + (y+1)^2)$. 取 $\varepsilon \rightarrow 0$. 有

$$u(x, y) = 0 \quad \forall x \in \mathbb{R}_+^2 \text{ 成立. } \square$$

□

定理 4.4.3 (调和函数的强极值原理). 设 $\Omega \subset \mathbb{R}^n$ 是一个连通的开集, 设 $u \in C^2(\Omega) \cap C(\bar{\Omega})$ 是 Ω 上的调和函数, 若存在内部的一点 $x_0 \in \Omega$, 使得 $u(x_0)$ 是 u 在整个 Ω 上的最大值, 则 u 在 Ω 上必须是常数.

解答 设 $\Omega \subset \mathbb{R}^n$ 是一个连通的开集, 设 $u \in C^2(\Omega) \cap C(\bar{\Omega})$ 是 Ω 上的调和函数, 若存在内部一点 $x_0 \in \Omega$, 使得 $u(x_0)$ 是 u 在整个 Ω 上的最大值, 则 u 在 Ω 上必须是常数.

设 $M = \sup_{x \in \Omega} u(x)$, 并取 $S = \{x \in \Omega \mid u(x) = M\}$, 现假设 S 非空, 则对 S 中点列 $\{x_n\}, x_n \rightarrow x$. 结合 $u \in C^2(\Omega) \cap C(\bar{\Omega})$, 有 $\lim_{n \rightarrow \infty} u(x_n) = u(\lim_{n \rightarrow \infty} x_n) = M = u(x)$ 则 S 是一个闭集. 对 $x_0 \in S$, 我们取其一个邻域 $B \subset S$ 有

$$u(x_0) = \frac{1}{|B|} \int_B u(x) dx$$

则为了保证上述等式成立, 我们应保证 $u(x) = u(x_0) \quad \forall x \in B$ 成立. 则 S 是一个开集. 则由 S 既开又闭, $S = \Omega$, 则 $\forall x \in \Omega, u(x) = M$. u 在 Ω 上须是常数. $\square$

例 4.4.4. 设 Ω 是三维有界区域, $\Omega = \mathbb{R}^3 \setminus \bar{\Omega}_0$, 又设 $u \in C^2(\Omega) \cap C(\bar{\Omega})$ 是外部问题

$$\begin{cases} -\Delta u + c(x)u = 0, & x \in \Omega \\ u|_{\partial\Omega} = \varphi \\ \lim_{|x| \rightarrow \infty} u(x) = L \end{cases}$$

的解, 其中 $c(x) \geq 0$, 且在 $\bar{\Omega}$ 上局部有界. 则有估计

$$\sup_{\Omega} |u(x)| \leq \max\{|L|, \max_{\partial\Omega} |\varphi(x)|\}$$

解答 结合 $\lim_{|x| \rightarrow \infty} u(x) = L$, 则 $\forall \varepsilon > 0, \exists R > 0$ 当 $|x| > R$ 时, 有 $|u(x) - L| < \varepsilon$. 则

$$|u(x)| \leq |u(x) - L| + |L| < |L| + \varepsilon \leq \max\{|L| + \varepsilon, \sup_{\partial\Omega} |\varphi(x)|\}$$

当 $|x| \leq R$ 时, 结合 u 在区域 $\{x \in \Omega : |x| \leq R\}$ 中取到非负最小值或非正最大值, 则若 $u(x_0) < 0$, 则

$$-u(x_0) < \sup_{\partial\Omega} |u(x)| = \sup_{\partial\Omega} |\varphi(x)| \leq \max\{|L|, \sup_{\partial\Omega} |\varphi(x)|\}$$

若 $u(x_0) \geq 0$, 则

$$u(x_0) > \sup_{\partial\Omega} |u(x)| = \sup_{\partial\Omega} |\varphi(x)| \leq \max\{|L|, \sup_{\partial\Omega} |\varphi(x)|\}$$

故有 $|u(x)| \leq \max\{|L|, \sup_{\partial\Omega} |\varphi(x)|\}$ 取 $\varepsilon \rightarrow 0$, 并对上式取上确界, 有

$$\sup_{\Omega} |u(x)| \leq \max\{|L|, \sup_{\partial\Omega} |\varphi(x)|\}$$

□

例 4.4.5. 设 Ω 为有界区域, $p_0 \in \partial\Omega$, 函数 $u \in C(\bar{\Omega} \setminus \{p_0\}) \cap C^2(\Omega)$ 满足

$$\begin{cases} -\Delta u = 0, & x \in \Omega \\ u|_{\partial\Omega \setminus \{p_0\}} = \varphi \\ \lim_{x \rightarrow p_0} |u(x)| \leq M_0 \end{cases}$$

则 $\sup_{\Omega} |u(x)| \leq \max\{M_0, \sup_{\partial\Omega} |\varphi(x)|\}$

解答 结合 $\lim_{x \rightarrow p_0} |u(x)| \leq M_0$, 则 $\forall \varepsilon > 0, \exists \delta > 0$, 当 $x \in \Omega \cap B_\delta(p_0)$ 时, 有 $|u(x)| < M_0 + \varepsilon$ 则对 $\forall x \in \Omega \setminus B_\delta(p_0)$, 取 $\partial\Omega_1 = \partial(\Omega \setminus B_\delta(p_0)) \cup (\partial\Omega \cap B_\delta(p_0))$ (这里画个图看一下), 我们都有

$$\begin{cases} -\Delta u = 0, & x \in \Omega \\ u|_{\partial\Omega_1} = \varphi \end{cases}$$

则结合弱极值原理, $\forall x \in \Omega \setminus B_\delta(p_0)$, 其非负最小值和非正最大值在 $\partial\Omega_1$ 上取到当 $u(x) < 0$ 时

$$-u(x) < \sup_{\partial\Omega_1} |u(x)| = \sup_{\partial\Omega_1} |\varphi(x)| \leq \sup_{\partial\Omega_1} |\varphi(x)| \leq \max\{\sup_{\partial\Omega} |\varphi(x)|, M_0\}$$

当 $u(x) \geq 0$ 时

$$u(x) = \sup_{\partial\Omega_1} |u(x)| = \sup_{\partial\Omega_1} |\varphi(x)| = \sup_{\partial\Omega_1} |\varphi(x)| \leq \max\{\sup_{\partial\Omega} |\varphi(x)|, M_0\}$$

结合 ε 的任意性, 我们有 $\forall x \in \Omega$

$$\sup_{x \in \Omega} |u(x)| \leq \max\{M_0, \sup_{\partial\Omega} |\varphi(x)|\} \quad \square$$

□

下面我们提出, 如果某个函数满足平均值公式, 那么我们有该函数是调和函数:

定理 4.4.4 (Koebe). 设 $\Omega \subset \mathbb{R}^n$ 是开集, 且 $u \in C(\Omega)$ 是 Ω 上的连续函数, 如果在 Ω 中任意一点 x 及其在充分小的邻域内, $u(x)$ 都等于其在球面上的平均值, 即:

$$u(x_0) = \frac{1}{|\partial B_R(x_0)|} \int_{\partial B_R(x_0)} u(y) dy$$

对于所有满足 $B_r(x) \subset \Omega$ 的球都成立, 则 $u \in C^\infty(\Omega)$ 且 $\Delta u = 0$ 。

解答 对某个球 $B_r(x)$, 我们来说 u 在该球中是调和函数, 则 $\exists$ 调和函数 h , s.t. $\forall y \in \partial B_r(x), h(y) = u(y)$. 则取 $w(y) = u(y) - h(y)$, 结合 u, h 均满足平均值公式, 我们有 w 仍满足平均值公式, 且 $w(y) = 0 \quad \forall y \in \partial B_r(x)$ 成立. 结合最值原理, w 的最大/小值必在 ∂B 上达到. 结合上述结论, 我们有 $w(y) = 0 \quad \forall y \in B_r(x)$ (即平均值性质) 则 $\forall y \in B_r(x)$, 我们有 $h(y) = u(y)$. $\square$

例 4.4.6 (Harnack 定理). 设 $\{u_n\} \in C(\bar{\Omega}) \cap C^2(\Omega)$ 是 Ω 内的调和函数列, 如果 $\{u_n\}$ 在 $\partial\Omega$ 上一致收敛, 则其在 $\bar{\Omega}$ 上也一致收敛, 且收敛于一调和函数.

解答 由 $\{u_n\}$ 是 Ω 内的调和函数, 其在 $\partial\Omega$ 上一致收敛, 则 $\forall \varepsilon > 0, \exists N > 0$, 当 $m, n > N$ 时, 有

$$|u_m - u_n| < \varepsilon$$

结合极值原理, $u_m - u_n$ 的最大值不在 Ω 内部取到, 则 $\forall x \in \Omega$, 有 $|u_m(x) - u_n(x)| < \varepsilon$. 则 $\{u_n\}$ 在 $\bar{\Omega}$ 中一致收敛于 u , 结合

$$u_N(x) = \int_{B_R(x)} \frac{1}{|B_R(x)|} u_N(y) dy$$

令 $\lim_{N \rightarrow \infty} u_N(x) = u(x)$, 对两侧 $N \rightarrow \infty$, 有

$$u(x) = \int_{B_R(x)} \frac{1}{|B_R(x)|} u(y) dy$$

结合 Koebe 定理, 有 u 为调和函数. $\square$

定理 4.4.5 (Harnack 不等式). 设 u 是区域 $\Omega \subset \mathbb{R}^n$ 上的非负调和函数, $u \geq 0$, 设 $V \subset\subset \Omega$ 是 Ω 内部的一个连通紧子集 (V 完全包含在 Ω 内部, 离边界有一段距离), 则存在一个只依赖于区域 Ω 和子集 V 的常数 C , 使得

$$\sup_V u \leq C \inf_V u$$

解答 证法一：平均值公式，结合 $V \subset\subset \Omega$ ，则我们有

$$d = \text{dist}(V, \partial\Omega) = \min_{x \in V, y \in \partial\Omega} \|x - y\| > 0$$

则我们取 $r < \frac{d}{4}$. 我们可以保证 $B_r(x) \forall x \in V$ ，都有 $B_r(x) \subset \Omega$. 对每 $x \in V$ ， $\bigcup_{x \in V} B_r(x)$ 构成 V 的一个开覆盖，且结合 V 是一个连通紧集，其存在一个有限子覆盖 $B_1 \dots B_k$ ，满足 $B_i = B_r(x_i)$. 取 $B_i = B_R(0)$ (WLOG) 结合 Poisson 公式

$$u(x) = \frac{1}{|\partial B_R(x_0)|} \int_{\partial B_R(x_0)} u(y) dS(y) = \frac{R^2 - |x|^2}{\omega_n R} \int_{\partial B_R(0)} \frac{u(y)}{|y - x|^n} dy$$

和

$$\frac{R^2 - |x|^2}{(R + |x|)^n} \leq \frac{R^2 - |x|^2}{|x - \xi|^n} \leq \frac{R^2 - |x|^2}{(R - |x|)^n}$$

即

$$\frac{R - |x|}{(R + |x|)^{n-1}} \leq \frac{R^2 - |x|^2}{|x - \xi|^n} \leq \frac{R + |x|}{(R - |x|)^{n-1}}. \quad \text{即有}$$

$$\frac{R - |x|}{(R + |x|)^{n-1}} \frac{1}{\omega_n R} \int_{\partial B_R} u(\xi) d\xi \leq u(x) \leq \frac{R + |x|}{(R - |x|)^{n-1}} \frac{1}{\omega_n R} \int_{\partial B_R} u(\xi) d\xi.$$

又由 $u(0) = \frac{1}{\omega_n R^{n-1}} \int_{\partial B_R} u(\xi) d\xi$. 我们有

$$u(0) R^{n-2} \frac{R - |x|}{(R + |x|)^{n-1}} \leq u(x) \leq u(0) R^{n-2} \frac{R + |x|}{(R - |x|)^{n-1}}$$

则 $\forall y \in B_i$.

$$\begin{aligned} u(y) &\leq R^{n-2} \frac{R + |x|}{(R - |x|)^{n-1}} u(0) \leq R^{n-2} \frac{R + |x|}{(R - |x|)^{n-1}} R^{2-n} \frac{(R + |x|)^{n-1}}{R - |x|} u(x) = \frac{(R + |x|)^n}{(R - |x|)^n} u(x) \\ u(y) &\geq R^{n-2} \frac{R - |x|}{(R + |x|)^{n-1}} u(0) \geq R^{n-2} \frac{R - |x|}{(R + |x|)^{n-1}} R^{2-n} \frac{(R - |x|)^{n-1}}{R + |x|} u(x) = \frac{(R - |x|)^n}{(R + |x|)^n} u(x) \end{aligned}$$

即在同一个球中 $\exists C > 0$ s.t.

$$\sup_{B_i} u(x) \leq C \inf_{B_i} u(x).$$

而 $\forall x, y \in V$, 结合前面的讨论, 我们可以找到 Harnack 链 $B_{i_1} \dots B_{i_k}$ s.t. $x \in B_{i_1}, y \in B_{i_k}$ 相邻的 $B_{i_j}, B_{i_{j+1}}$ 有交集, 则 $\exists x_1 \in B_{i_1} \cap B_{i_2}$ s.t. $u(x) \leq Cu(x_1)$, 结合 $x_1 \in B_{i_2} \exists x_2 \in B_{i_2} \cap B_{i_3}$ s.t. $u(x_1) \leq Cu(x_2)$, ... 结合 $x_{k-1} \in B_{i_k}$ 有 $u(x_{k-1}) \leq Cu(y)$. 则 $u(x) \leq Cu(x_1) \leq C^2u(x_2) \leq \dots \leq C^{k-1}u(x_{k-1}) \leq C^ku(y)$ 原命题成立

证法二: 对数梯度估计

前置 1 (Bochner 公式) $\Delta(|\nabla v|^2) = 2\nabla v \cdot \nabla(\Delta v) + 2|D^2v|^2$

前置 2 (Cauchy-Schwarz 不等式) $|D^2v|^2 \geq \frac{1}{n}(\Delta v)^2$

我们取 $v = \log u$, 则有

$$\nabla v = \frac{\nabla u}{u} \quad \Delta v = \frac{\Delta u}{u} - \frac{|\nabla u|^2}{u^2} = -|\nabla v|^2 \quad (*)$$

我们取 $\varphi(x) = \eta(x)|\nabla v(x)|^2 = (R^2 - |x|^2)^2|\nabla v(x)|^2$, 由 $\varphi(x) = 0 \quad \forall x \in \partial B_R(y)$ 成立, 其中 $R = d(x, \partial\Omega)$ 且 $\varphi(x) \geq 0$ 恒成立, 则 $\varphi(x)$ 在 $B_R(y)$ 内部取到最大值, 设最大值点为 x_0 , 则有

$$\nabla \varphi(x_0) = 0 \quad \Delta \varphi(x_0) \leq 0$$

记 $w = |\nabla v(x)|^2$, 则 $\varphi = \eta w$. 在 x_0 点处

$$\nabla \varphi = 2w\nabla \eta + \eta \nabla w = 0 \implies \nabla w = -2\frac{w}{\eta} \nabla \eta \quad (\#)$$

$$\begin{aligned} \Delta \varphi &= w\Delta \eta + \eta \Delta w + 2\nabla \eta \cdot \nabla w = w(\Delta \eta + 2\Delta \eta \cdot \eta^2) + \eta \Delta w + 4\eta \nabla \eta \cdot (-2\frac{w}{\eta} \nabla \eta) \\ &= 2w\eta \Delta \eta + \eta \Delta w - 6w|\nabla \eta|^2 \end{aligned}$$

结合

$$\Delta w = \Delta(|\nabla v|^2) = 2\nabla v \nabla(\Delta v) + 2|D^2v|^2 \stackrel{(*)}{=} -2\nabla v \nabla w + 2|D^2v|^2 \geq \frac{2}{n}w^2 - 2\nabla v \cdot \nabla w \text{ (利用 } w = |\nabla v|^2 \text{)}$$

则

$$\begin{aligned} 0 &\geq \Delta \varphi = 2w\eta \Delta \eta + \eta^2 w - 6w|\nabla \eta|^2 \geq 2w\eta \Delta \eta + \eta(\frac{2}{n}w^2 - 2\nabla v \nabla w) - 6w|\nabla \eta|^2 \\ &= 2w\eta \Delta \eta + \frac{2}{n}\eta^2 w^2 - 2\eta \nabla v \cdot \nabla w - 6w|\nabla \eta|^2 \end{aligned}$$

则结合 (#) 式, 我们有

$$2w\eta \Delta \eta + \frac{2}{n}\eta^2 w^2 + 4\eta \nabla v \cdot \nabla \eta \frac{w}{\eta} - 6w|\nabla \eta|^2 \leq 0$$

结合 $\eta(x) = (R^2 - |x|^2)^2$, $\nabla\eta = -2x$, $\Delta\eta = -2n$. 代入有

$$-4nw\eta + \frac{2}{n}\eta^2w^2 - 8\eta w\nabla v \cdot x - 24|x|^2w \leq 0$$

对两侧同除 $w = |\nabla v|^2$ (假设 $w \neq 0$)

$$\frac{2}{n}\eta^2w \leq 4n\eta + 8\eta|x \cdot \nabla v| + 24|x|^2$$

$$\frac{2}{n}\varphi \leq 4n\eta + 8\eta|x||\nabla v| + 24|x|^2 \leq 4n\eta + 8\eta|x|\sqrt{w} + 24|x|^2$$

则有

$$\frac{2}{n}\varphi \leq 4n\eta + 8R\sqrt{\varphi} + 24R^2 = 4n\eta + 8|x|\sqrt{\varphi} + 24|x|^2$$

取 $Y = \sqrt{\varphi(x_0)}$, 则有 $\frac{2}{n}Y^2 - 8RY - C_1R^2 \leq 0$. 则必有 $\varphi(x_0) \leq C^2R^2 (\exists C > 0)$

则 $\varphi(0) \leq \varphi(x_0) \leq C^2R^2$ 即 $R^4|\nabla v(0)|^2 \leq C^2R^2$, 另有 $|\nabla v(0)| \leq \frac{C}{R}$. 即 $\frac{|\nabla u(0)|}{u(0)} \leq \frac{C}{R}$.

则利用方法一思路, 有 $\forall x \in \Omega$, 都有 $|\nabla \log u(x)| = \frac{|\nabla u(x)|}{u(x)} \leq \frac{C}{R}$. 则 $\log u(x_2) - \log u(x_1) = \int_{\gamma} \nabla(\log u) d\vec{s} \leq \int_{\gamma} \frac{C}{R} d\vec{s} = \frac{CL}{R}$. 即 $\log\left(\frac{u(x_2)}{u(x_1)}\right) \leq \frac{CL}{R}$ $u(x_2) \leq \exp\{\frac{CL}{R}\}u(x_1)$. $\forall x_1, x_2 \in V$ 成立. □

他有这样的推论:

推论 4.4.1. 设 u 是 $B_R(x_0)$ 上的非负调和函数, 则

$$|Du(x_0)| \leq \frac{n}{R}u(x_0)$$

根据这个性质, 我们可以得到:

定理 4.4.6 (Liouville 定理). 在全平面上有下界 (或有上界) 的调和函数必为常数。

定理 4.4.7 (调和函数导数的局部估计). 对函数 u 满足 $\Delta u = 0$, 则对 $\forall B_R(x_0) \subset \Omega$, 有

$$\sup_{B_{R/2}(x_0)} |\nabla u| \leq \frac{C}{R} \|u\|_{L^\infty(B_R(x_0))}$$

解答 我们先来证 $|\nabla u(x_0)| \leq \frac{n}{R} \|u\|_{L^\infty(B_R(x_0))}$. 结合调和函数偏导数仍是调和函数, 取 $v_k = \frac{\partial u}{\partial x_k}$, 对其使用平均值公式

$$\frac{\partial u}{\partial x_k} = \frac{1}{|B_R|} \int_{B_R(x_0)} \frac{\partial u}{\partial x_k}(y) dy$$

结合 Gauss 公式 $\int_{B_R(x_0)} \frac{\partial u}{\partial x_k} dy = \int_{\partial B_R(x_0)} u \cdot \nu_k dS$ 其中 ν_k 为第 k 分量单位法向量. 则有 $\|\nu_k\| = 1$

$$\left| \frac{\partial u}{\partial x_k} \right| = \frac{1}{|B_R|} \left| \int_{B_R(x_0)} \frac{\partial u}{\partial x_k}(y) dy \right| = \frac{1}{|B_R|} \left| \int_{\partial B_R(x_0)} u \cdot \nu_k dy \right| \leq \frac{1}{|B_R|} \int_{\partial B_R(x_0)} \|u\|_{L^\infty(\partial B_R(x_0))} \|\nu_k\| dy = \frac{|\partial B_R|}{|B_R|} \|u\|_{L^\infty(\partial B_R(x_0))}$$

结合 $|B_R| = \omega_n R^n$ $|\partial B_R| = n\omega_n R^{n-1}$, 则 $\left| \frac{\partial u}{\partial x_k} \right| \leq \frac{n}{R}$. 结合 $|\nabla u| = \sqrt{\sum (\frac{\partial u}{\partial x_k})^2}$, 则有

$$|\nabla u(x_0)| \leq \frac{n\sqrt{n}}{R} \|u\|_{L^\infty(B_R(x_0))} \quad (*)$$

则对 $r = \frac{R}{2}$, 有 $B_r(x_0) \subset B_R(x_0)$, $\forall x \in B_{R/2}(x_0)$ 有 $B_r(x) \subset B_R(x)$, 对其利用 (*), 有

$$|\nabla u(x)| \leq \frac{n\sqrt{n}}{r} \|u\|_{L^\infty(B_r(x_0))} \leq \frac{2n\sqrt{n}}{R} \|u\|_{L^\infty(B_R(x_0))}$$

结合 x 选取的任意性, 有

$$\sup_{B_{R/2}(x_0)} |\nabla u| \leq \frac{C}{R} \|u\|_{L^\infty(B_R(x_0))}$$

□

在介绍最后一个性质前, 我们来明确什么是磨光, 针对某些非光滑函数, 我们希望通过一些方式将其变成光滑函数, 一般我们选择某个特定的光滑函数做卷积, 我们将标准磨光子定义为

1. 选取一个径向对称、紧支集在单位球 $B_1(0)$ 内的非负光滑函数 $\eta(|x|)$ 使得 $\int \eta dx = 1$.

2. 做缩放 $\eta_\varepsilon(x) = \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right)$.

3. 定义抛光后的函数:

$$f^\varepsilon = f * \eta_\varepsilon = \int_{\Omega} f(y) \eta_\varepsilon(x-y) dy$$

抛光后的函数具有如下两个性质:

1. $f^\varepsilon \in C^\infty(\Omega)$.
2. 若 $f \in C(\Omega)$, 则 $\{f^\varepsilon\}$ 内闭一致收敛于 f .

解答 1) 对 $\forall x \in \Omega$, 我们只要证 u_ε 在任意方向导数存在. 对任意方向 e_i , 我们有

$$\frac{u_\varepsilon(x + he_i) - u_\varepsilon(x)}{h} = \int_{\Omega} u(y) \frac{\eta_\varepsilon(x + he_i - y) - \eta_\varepsilon(x - y)}{h} dy$$

结合 $\eta_\varepsilon \in C_0^\infty(\Omega)$, 则

$$\lim_{h \rightarrow 0} \frac{u_\varepsilon(x + he_i) - u_\varepsilon(x)}{h} = \int_{\Omega} u(y) \lim_{h \rightarrow 0} \frac{\eta_\varepsilon(x + he_i - y) - \eta_\varepsilon(x - y)}{h} dy = \int_{\Omega} u(y) \frac{\partial}{\partial x_i} \eta_\varepsilon(x - y) dy$$

则有 $\frac{\partial}{\partial x_i} u_\varepsilon(x)$ 存在, 则 $u_\varepsilon \in C^1$, 以此类推, 即 $u_\varepsilon \in C^\infty(\Omega)$.

2) 对 Ω 的任意紧集 K , 结合 $u \in C(\Omega)$, 则 u 在 K 上一致连续, 则 $\forall \sigma > 0, \exists \delta > 0$, 当 $\|x - y\| < \delta$ 时

$$|u(x) - u(y)| < \sigma.$$

则结合 $u_\varepsilon(x) = u(x) \cdot 1 = u(x) \int_{B_\varepsilon(0)} \eta_\varepsilon(y) dy = \int_{B_\varepsilon(0)} \eta_\varepsilon(y) u(x) dy$. (结合 η_ε 的定义 $\Omega \setminus B_\varepsilon(0)$ 外为 0) 则

$$|u_\varepsilon(x) - u(x)| = \left| \int_{B_\varepsilon(0)} \eta_\varepsilon(y) (u(x-y) - u(x)) dy \right| \leq \int_{B_\varepsilon(0)} \eta_\varepsilon(y) |u(x-y) - u(x)| dy \leq \sigma \int_{B_\varepsilon(0)} \eta_\varepsilon(y) dy = \sigma.$$

则 $\{u_\varepsilon\}$ 内闭一致收敛于 u . □

定理 4.4.8. 若 $u \in C^2(\Omega)$, $\Delta u = 0$, 则有 $u \in C^\infty(\Omega)$.

解答 由 $u_\varepsilon(x) = \int_{B_\varepsilon(0)} u(x-y)\eta_\varepsilon(y)dy$, 结合 $\eta_\varepsilon(y)$ 的旋转对称性, 我们可取 $\eta_\varepsilon(y) = \varphi(r)$ 且 $dy = r^{n-1}drd\sigma$, 其中 σ 是 $B_1(0)$ 的单位元, 则

$$\begin{aligned} u_\varepsilon(x) &= \int_{B_\varepsilon(0)} u(x-y)\eta_\varepsilon(y)dy = \int_{B_\varepsilon(0)} u(x-rw)\varphi(r)r^{n-1}drd\sigma(w) \\ &= \int_0^\varepsilon \int_{\partial B_r} u(x-rw)\varphi(r)r^{n-1}drd\sigma(w) \end{aligned}$$

结合 $\frac{1}{|\partial B_1|} \int_{\partial B_1} u(x-rw)d\sigma(w) = u(x)$, 则 $\int_{\partial B_1} u(x-rw)d\sigma(w) = |\partial B_1|u(x) = n\omega_n u(x)$ 则

$$\begin{aligned} u_\varepsilon(x) &= \int_0^\varepsilon n\omega_n u(x)\varphi(r)r^{n-1}dr = u(x) \int_0^\varepsilon n\omega_n \varphi(r)r^{n-1}dr \\ &= u(x) \int_{B_\varepsilon(0)} \eta(y)dy = u(x) \end{aligned}$$

结合 $u_\varepsilon \in C^\infty$, 我们有 $u \in C^\infty$ □

例 4.4.7. 设 $f \in L^2(\Omega)$, 且满足 $\int_\Omega f\varphi dx = 0$, 其中 $\varphi \in C_0^\infty(\Omega)$. 证明 $f = 0$ a.e. $x \in \Omega$.

解答 首先证明 $C_0^\infty(\Omega)$ 在 $L^2(\Omega)$ 上稠密, 则对 $f \in L^2(\Omega)$. 取 $\chi_n = \{x \in \Omega, |x| < n, \text{dist}(x, \partial\Omega) > \frac{1}{n}\}$

$$f_n(x) = f(x)\chi_n(x).$$

则 f_n 具有紧支集, $|f_n(x) - f(x)| < |f(x)|$ $\lim_{n \rightarrow \infty} f_n(x) = f(x)$, 结合控制收敛定理有

$$\lim_{n \rightarrow \infty} \|f_n - f\|_{L^2(\Omega)} = 0$$

其中 f_n 是 $L^2(\Omega)$ 中具有紧支集的函数, 则对这类函数 $g \in L^2(\Omega)$. 我们取磨光子 η_ε s.t.

$$g_\varepsilon \triangleq g * \eta_\varepsilon = \int_\Omega g(y)\eta_\varepsilon(x-y)dy, \quad \text{则有 } g_\varepsilon \in C_0^\infty \text{ 且 } \text{supp}(g_\varepsilon) = \text{supp}(g) \cup \text{supp}(\eta_\varepsilon)$$

则 $g_\varepsilon \in C_0^\infty(\Omega)$, 且 $\forall \delta > 0, \exists \varepsilon > 0$, s.t. $|g_\varepsilon(x) - g(x)| < \delta/2 \quad \forall x \in \Omega$ 成立. 则对

$$\|f - f_n^\varepsilon\|_{L^2} \leq \|f - f_n\|_{L^2} + \|f_n - f_n^\varepsilon\| = \frac{\delta}{2} + \frac{\delta}{2} = \delta \quad \text{其中 } f_n^\varepsilon \in C_0^\infty(\Omega)$$

则 $\overline{C_0^\infty(\Omega)} = L^2(\Omega)$,

$\forall f \in L^2(\Omega) \quad \exists \{\varphi_k\} \subset C_0^\infty(\Omega)$ s.t. $\|f - \varphi_k\|_{L^2} \rightarrow 0$ 当 $k \rightarrow \infty$ 时.

$$\int_{\Omega} f^2 dx = \int_{\Omega} f(f - \varphi_k + \varphi_k) dx = \int_{\Omega} f(f - \varphi_k) dx + \int_{\Omega} f\varphi_k dx = \int_{\Omega} f(f - \varphi_k) dx \leq \left(\int_{\Omega} f^2 dx\right)^{\frac{1}{2}} \left(\int_{\Omega} (f - \varphi_k)^2 dx\right)^{\frac{1}{2}}$$

则 $\int_{\Omega} f^2 dx \leq \int_{\Omega} (f - \varphi_k)^2 dx < \varepsilon$ 结合 ε 的任意性, 有 $\int_{\Omega} f^2 dx = 0 \implies f = 0 \quad a.e. \ x \in \Omega$. □

例 4.4.8. 设 $u(x, y)$ 是定解问题 $\begin{cases} \Delta u = 0, & (x, y) \in B(R) \\ u|_{\partial B(R)} = \varphi(x, y) \end{cases}$ 的解, 又设 $v(x, y)$ 是定解问题 $\begin{cases} \Delta v = 0 & (x, y) \in B(R) \setminus \{0\} \\ v|_{\partial B(R)} = \varphi(x, y) \end{cases}$

的有界解, 其中 0 表示坐标原点, 且 $0 \in B(R)$, 试证 0 点是 $v(x, y)$ 的可去奇点, 即

$$u(x, y) = v(x, y) \quad \forall (x, y) \in B(R) \setminus \{0\}$$